Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Hanoi, October 2026 –

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Change Log

Date	A* M, D	In charge	Change Description
30/9/2026	A	Nguyen Viet Hoang	Add Context Diagram
30/9/2026	A	Nguyen Thanh Luan	Add User Requirements
30/9/2026	A	Nguyen Thanh Luan	Add Use Case Diagrams
30/9/2026	A	Nguyen Thanh Luan	Add Permission Matrix
30/9/2026	A	Nguyen Chi Tung Duong	Add Business Rules
30/9/2026	A	Nguyen Chi Tung Duong	Add System Messages

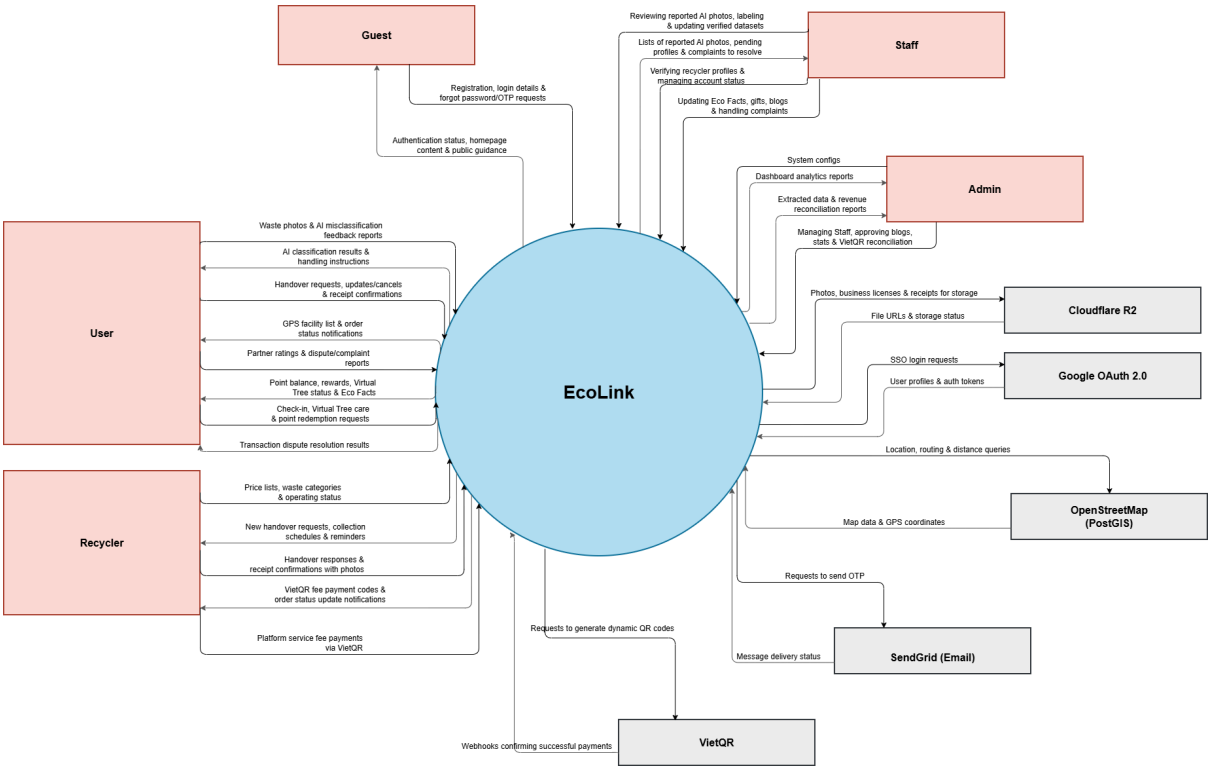
*A - Added M - Modified D - Deleted

Definition and Acronyms

Acronym	Full Name / Definition
AI	Artificial Intelligence
GPS	Global Positioning System
QR	Quick Response Code
B2C	Business-to-Consumer
B2B	Business-to-Business
EPR	Extended Producer Responsibility
UI/UX	User Interface / User Experience
MVP	Minimum Viable Product
GAP	Identified Gap in Existing Solutions
API	Application Programming Interface
CNN	Convolutional Neural Network
YOLO	You Only Look Once
OOD	Out-of-Distribution
PostGIS	PostgreSQL Geographic Extension
REST	Representational State Transfer
FE	Feature
LI	Limitation / Exclusion
UC	Use Case
BR	Business Rule
MSG	System Message
SRS	Software Requirement Specification
SDS	Software Design Specification
PMP	Project Management Plan

I. Overall Requirements

1. Context Diagram



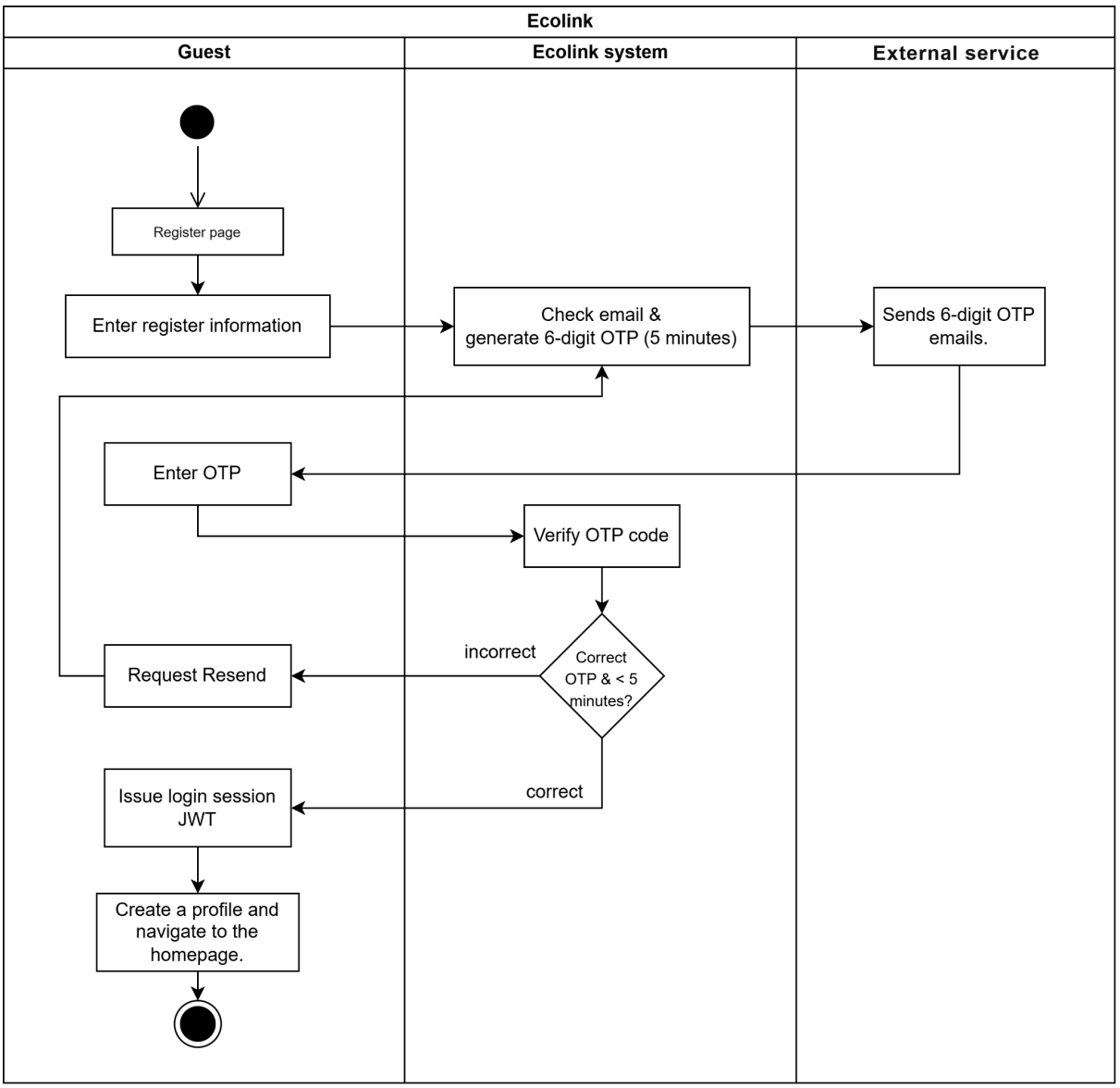
[Pink = internal entity, grey = external entity]

2. Main Business Flow

2.1 Guest Registration & Verification Flow(BF-01)

Trigger: Guest accesses the register page to create a new account.

End condition:Guest successfully verifies their email, creates a profile, receives a login session, and navigates to the homepage.



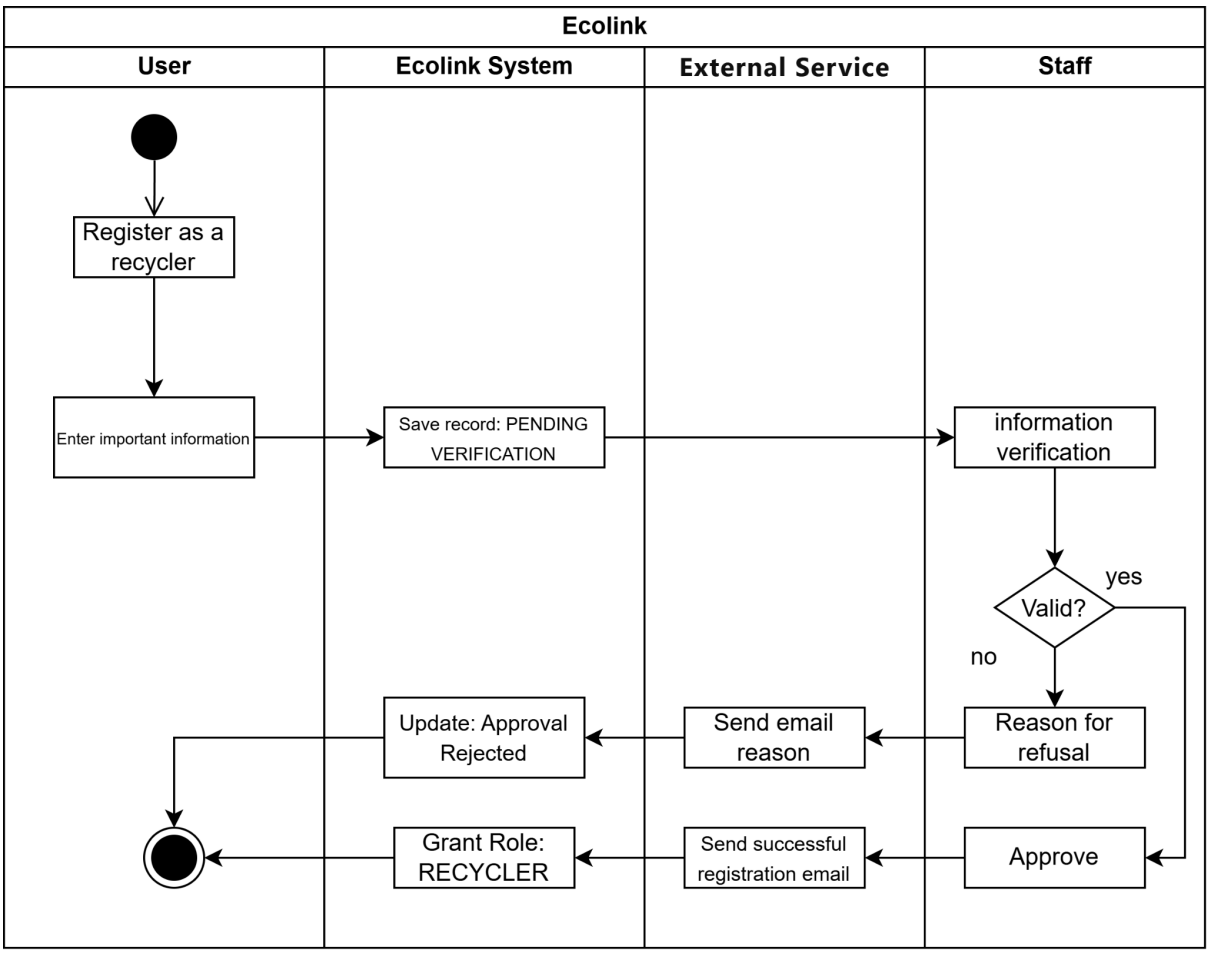
Explain:

Step 1: Enter Registration Details (Guest) The Guest opens the Register page and enters their registration information. **Step 2: Email Check & OTP Generation (Ecolink System)** The System checks the email validity and generates a 6-digit OTP valid for 5 minutes. **Step 3: OTP Dispatch (External Service)** The External Service sends the 6-digit OTP email to the Guest's registered email address. **Step 4: Enter OTP (Guest)** The Guest receives the email and enters the OTP into the system. **Step 5: OTP Verification & Evaluation (Ecolink System)** The System verifies the entered OTP code against the generated one and checks the time limit.

- **If incorrect (Wrong OTP or > 5 minutes):** The Guest must request a resend. The flow loops back to Step 2 where a new OTP is generated and sent.
- **If correct:** The System issues a login session (JWT). The Guest then creates a profile and navigates to the homepage, and the flow ends.

2.2 Recycler Registration Flow(BF-02)

Trigger: User initiates the registration process to become a Recycler on the EcoLink platform.
End condition: The User is officially granted the "RECYCLER" role, OR the application is rejected.



Explain:

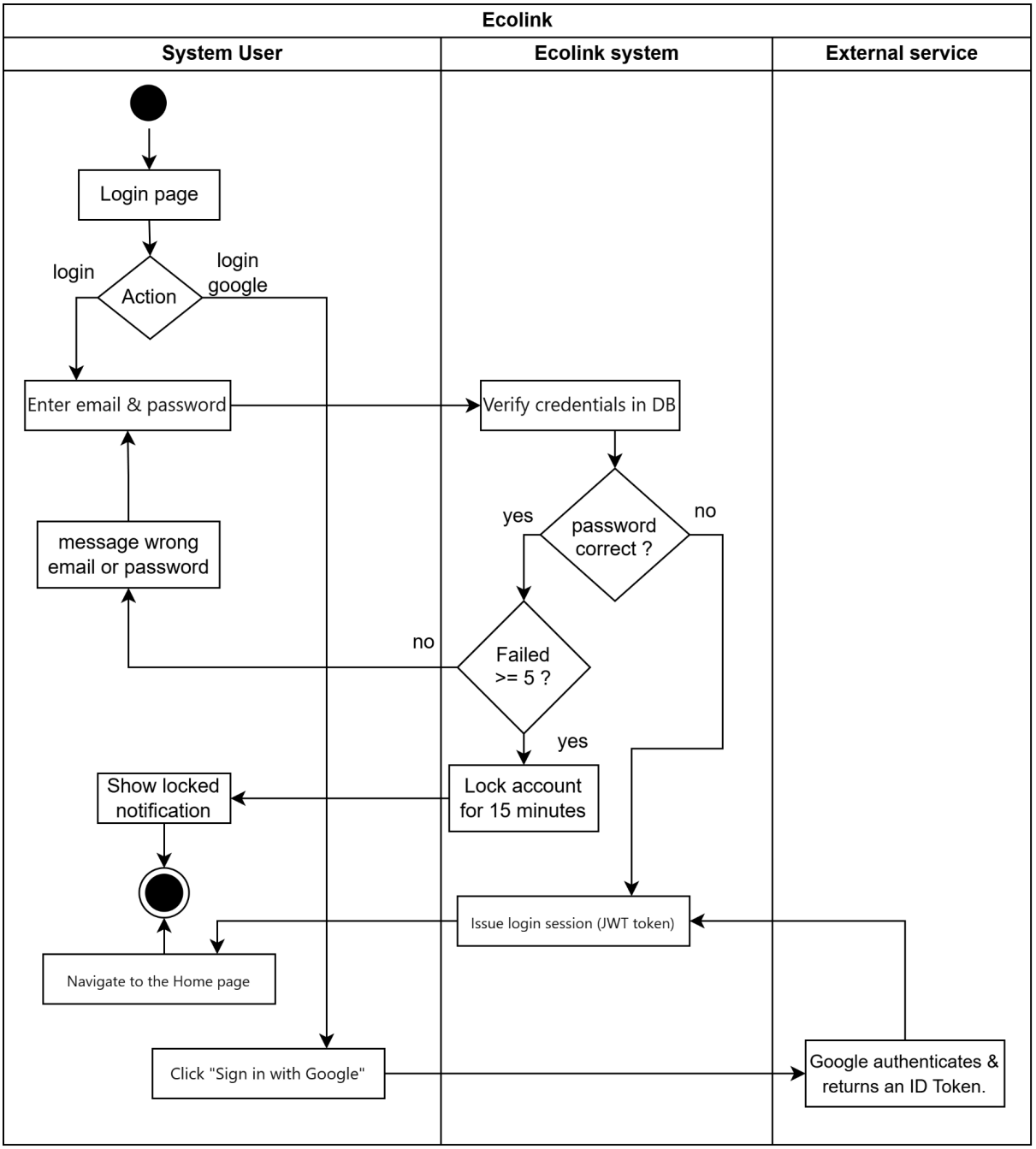
Step 1: Enter Registration Information (User) The User opens the register page and enters important information required for becoming a recycler. **Step 2: Save Pending Record (Ecolink System)** The System receives the data and saves the record with a "PENDING VERIFICATION" status. **Step 3: Information Verification (Staff)** Staff accesses the pending record from the system and performs information verification. **Step 4: Validity Evaluation (Staff & Ecolink System)** Staff evaluates whether the submitted information is valid.

- **If yes (Valid):** Staff approves the application. The System then sends a successful registration email to the User and grants them the "RECYCLER" role, and the flow ends.
- **If no (Invalid):** Staff provides a reason for refusal. The System then sends an email with the refusal reason to the User and updates the record status to "Approval Rejected", and the flow ends.

2.3 User Login & Authentication Flow (BF-03)

Trigger: System User accesses the login page to authenticate into the system.

End condition: System User successfully logs in and navigates to the Home page, OR the account is temporarily locked due to multiple failed attempts.



Explain:

Step 1: Choose Login Action (System User) The System User opens the Login page and chooses an action: manual login or login with Google.

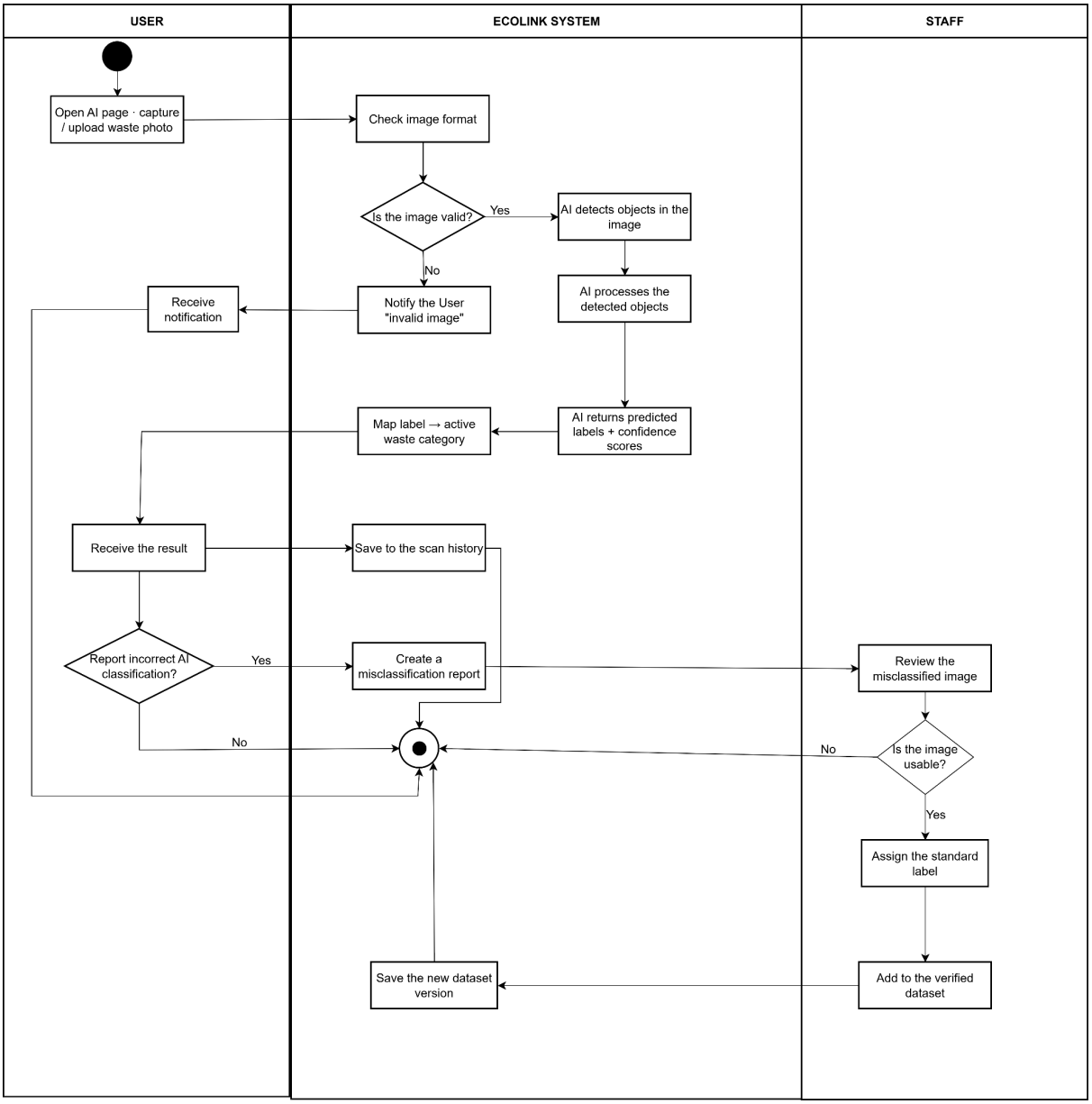
- **If login google:** The System User clicks "Sign in with Google". The External Service (Google) authenticates the user and returns an ID Token. The Ecolink System issues a login session (JWT token). The System User navigates to the Home page, and the flow ends.
- **If login (Manual):** The System User enters their email & password. The flow proceeds to Step 2. **Step 2: Credential Verification (Ecolink System)** The System receives the manual login input and verifies the credentials in the database. **Step 3: Password Evaluation (Ecolink System)** The System checks if the entered password is correct.
- **If yes (password correct):** The System issues a login session (JWT token). The System User navigates to the Home page, and the flow ends.

- **If no (password incorrect):** The System proceeds to evaluate the failure count in the next step. **Step 4: Failed Attempt Evaluation (Ecolink System)** The System checks if the number of failed login attempts is greater than or equal to 5.
- **If no (Failed < 5):** The System shows a "message wrong email or password" to the System User. The flow loops back to the "Enter email & password" step.
- **If yes (Failed >= 5):** The System locks the account for 15 minutes. The System User is shown a locked notification, and the flow terminates.

2.4 End-to-End Recyclable Waste Sorting(BF-04)

Trigger: User opens the AI page and captures or uploads a photo of recyclable waste into the EcoLink System.

End condition: The waste image is validated and classified (with the scan record saved to history), or an invalid image notification is delivered to the User, or a reported misclassification is processed by Staff and added to update the verified dataset version.



Explain:

Step 1: Upload / Capture Waste Photo (User) The User opens the AI scan page on the application and either captures a new photo of the waste item using the camera or uploads an existing image from their device.
Step 2: Image Format Validation (EcoLink System) The System receives the photo and checks the image format and file constraints.

- **If No (Invalid Image):** The System sends a notification *"invalid image"* to the User. The User receives the notification, and the flow ends.
- **If Yes (Valid Image):** The System proceeds to object detection.

Step 3: Object Detection & Feature Processing (EcoLink System) The System performs object detection on the validated image to identify waste items and processes the detected object features using the computer vision model.
Step 4: Classification & Category Mapping (EcoLink System) The System predicts the category labels along with confidence scores, and maps the top predicted label to an active waste category defined in the master catalog.
Step 5: Result Delivery & Scan History Storage (User & EcoLink System) The User receives the classification result on the screen. Simultaneously, the System automatically saves the scan record into the User's scan history.
Step 6: Misclassification Evaluation (User) The User checks the returned classification result to verify if the AI correctly identified the waste item.

- **If No:** The User accepts the classification result, and the flow ends at the completion node.
- **If Yes:** The User selects the option to report an incorrect AI classification.

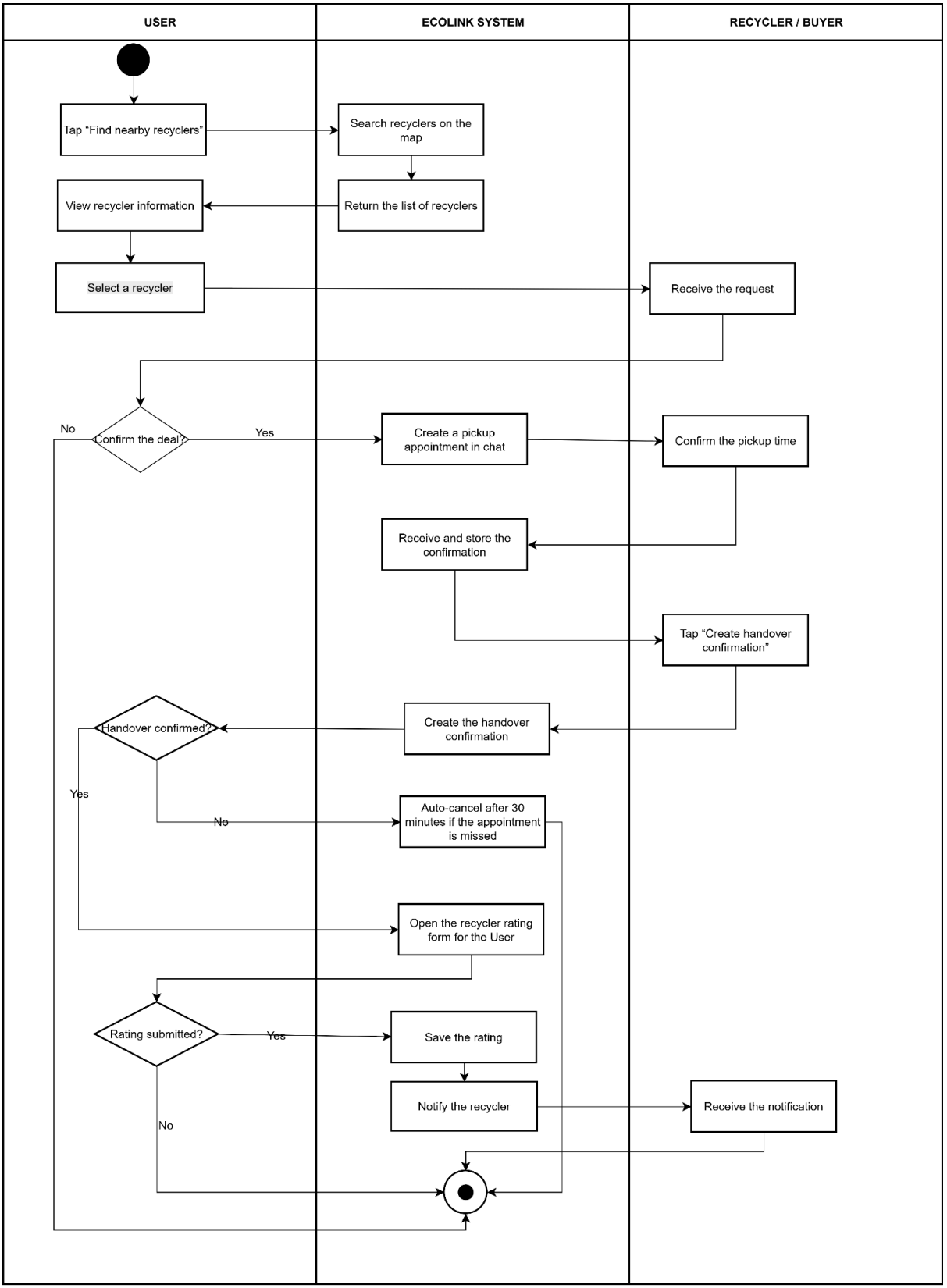
Step 7: Misclassification Report Creation (EcoLink System) The System generates a misclassification report linking the original photo, predicted label, and user-suggested correct label, then routes it to the Staff review queue.
Step 8: Image Usability Review (Staff) Staff opens the review queue and inspects the reported image to evaluate its usability for dataset retraining.

- **If No (Unusable Image):** Staff rejects or discards the unusable image, and the flow terminates.
- **If Yes (Usable Image):** Staff assigns the standard correct label to the image.

Step 9: Dataset Update & Versioning (Staff & EcoLink System) Staff adds the newly labeled image into the verified dataset. The System saves and updates the new dataset version for future model retraining, and the flow ends.

2.5 Recycler Discovery & Trade Coordination(BF-05)

Trigger: User taps "Find nearby recyclers" on the EcoLink System.
End condition: The waste handover transaction is finalized (confirmed by the user and optionally rated, or auto-cancelled after 30 minutes if missed/unconfirmed), and the recycler receives final notification.



Explain:

Step 1: Initiate Recycler Search (User) The User taps the *"Find nearby recyclers"* button on the application to discover local scrap collectors and recycling facilities.

Step 2: Spatial Search & Recycler Discovery (EcoLink System & User) The System queries active recyclers on the map based on GPS location and returns the list of matching recyclers. The User views the recycler information and profile details.
Step 3: Recycler Selection & Trade Request (User & Recycler / Buyer) The User selects a specific recycler and sends a trade request. The Recycler / Buyer receives the incoming request.
Step 4: Deal Confirmation Evaluation (Recycler / Buyer) The Recycler / Buyer reviews the trade request details to decide whether to accept the deal.

- **If No:** The Recycler / Buyer declines the deal, and the flow terminates.
- **If Yes:** The flow proceeds to appointment scheduling.

Step 5: Appointment Creation & Time Confirmation (EcoLink System & Recycler / Buyer) The System creates a pickup appointment directly within the in-app chat. The Recycler / Buyer confirms the pickup time, and the System receives and stores the confirmation record.
Step 6: Handover Confirmation Initiation (Recycler / Buyer & EcoLink System) Upon completing the physical waste collection on-site, the Recycler / Buyer taps *"Create handover confirmation"*. The System creates the handover confirmation record and dispatches it to the User for verification.
Step 7: Handover Verification Evaluation (User & EcoLink System) The User evaluates whether the handover occurred as agreed.

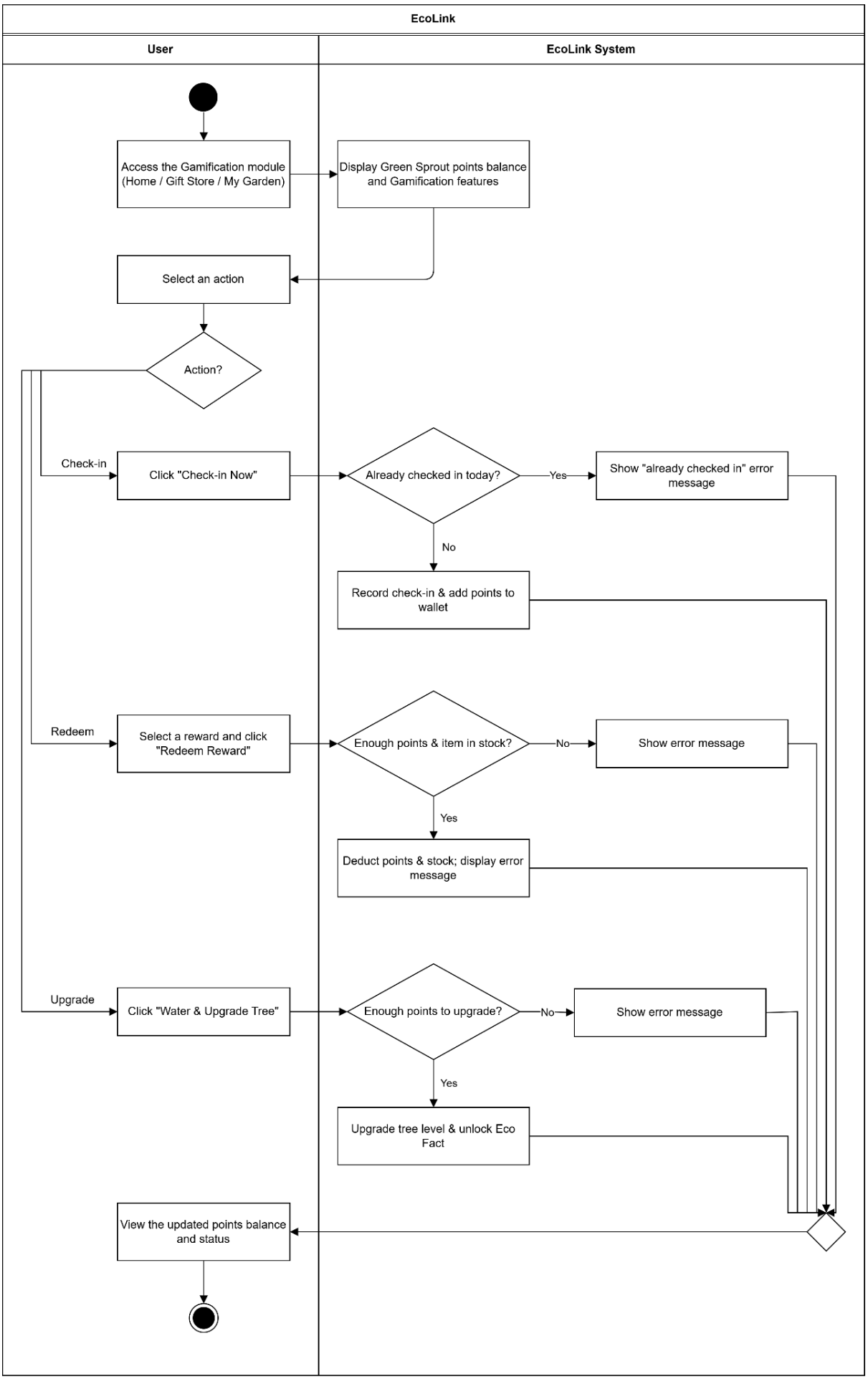
- **If No:** The User does not confirm. If the appointment is missed or unconfirmed within 30 minutes, the System automatically cancels the appointment, and the flow terminates.
- **If Yes:** The User confirms the handover, and the System opens the recycler rating form for the User.

Step 8: Rating Submission Evaluation (User, EcoLink System & Recycler / Buyer) The User decides whether to submit a rating for the Recycler.

- **If No:** The User skips submitting a rating, and the flow ends.
- **If Yes:** The User submits a star rating and comment. The System saves the rating, updates the recycler's average score and review count, and sends a notification. The Recycler / Buyer receives the notification, and the flow terminates.

2.6 Gamification & Reward Management Flow(BF-06)

Trigger: The User accesses the Gamification module (via Home, Gift Store, or My Garden) on the EcoLink System.
End condition: The User completes a gamification activity (Check-in, Reward Redemption, or Virtual Tree Upgrade) or receives an error notification, views their updated points balance and status, and the flow terminates.


EcoLink
User
EcoLink System
Access the Gamification module (Home / Gift Store / My Garden)
Display Green Sprout points balance and Gamification features
Select an action
Action?
Check-in
Click "Check-in Now"
Already checked in today?
Yes
Show "already checked in" error message
No
Record check-in & add points to wallet
Redeem
Select a reward and click "Redeem Reward"
Enough points & item in stock?
No
Show error message
Yes
Deduct points & stock; display error message
Upgrade
Click "Water & Upgrade Tree"
Enough points to upgrade?
No
Show error message
Yes
Upgrade tree level & unlock Eco Fact
View the updated points balance and status

Explain:

Step 1: Access Gamification Module (User) The User opens the application and navigates to the Gamification module from the Home screen, Gift Store, or My Garden section.

Step 2: Display Gamification Dashboard (EcoLink System) The System retrieves the User's account data and displays their current Green Sprout points balance along with available Gamification features.

Step 3: Select Gamification Action (User) The User chooses a specific action to perform on the Gamification interface.

- **If Check-in:** The User clicks "Check-in Now". The flow proceeds to Step 4.
- **If Redeem:** The User selects a desired reward and clicks "Redeem Reward". The flow proceeds to Step 5.
- **If Upgrade:** The User clicks "Water & Upgrade Tree". The flow proceeds to Step 6.

Step 4: Check-in Evaluation (EcoLink System) The System checks whether the User has already checked in on the current calendar day (Already checked in today?).

- **If Yes:** The System displays an *"already checked in"* error message. The flow proceeds to Step 7.
- **If No:** The System records the daily check-in, adds Green Sprout points to the User's wallet, and proceeds to Step 7.

Step 5: Reward Redemption Evaluation (EcoLink System) The System verifies whether the User has sufficient points and whether the selected reward is currently in stock (Enough points & item in stock?).

- **If No:** The System displays an error message (e.g., insufficient points or item out of stock). The flow proceeds to Step 7.
- **If Yes:** The System deducts the required points from the User's wallet, decrements the item stock count, displays the confirmation message, and proceeds to Step 7.

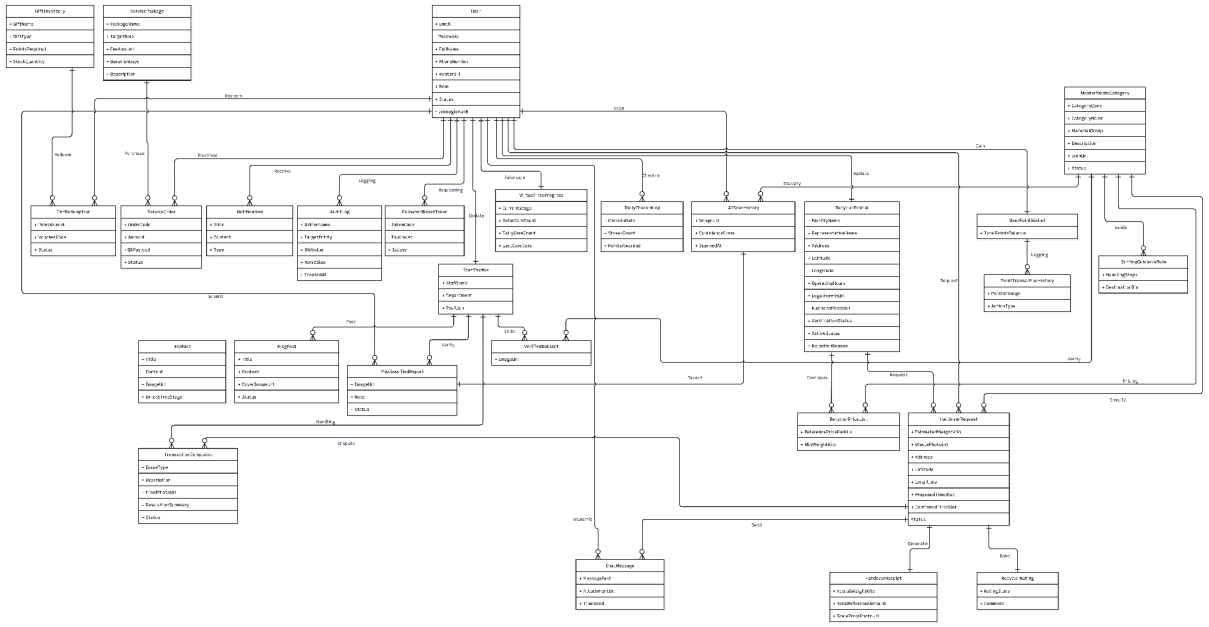
Step 6: Virtual Tree Upgrade Evaluation (EcoLink System) The System checks whether the User has accumulated enough Green Sprout points to upgrade the Virtual Tree level (Enough points to upgrade?).

- **If No:** The System displays an error message indicating insufficient points. The flow proceeds to Step 7.
- **If Yes:** The System upgrades the Virtual Tree level, unlocks a new environmental Eco Fact card, and proceeds to Step 7.

Step 7: View Updated Status (User) The User views the updated Green Sprout points balance, tree level, or voucher status on the screen, and the flow ends.

3. Conceptual Data Model

3.1 Entity Relationship Diagram



3.2 Entity Descriptions

3.2.1 User Groups & Accounts

Entity: User

Field	Value
Purpose	Represents any registered individual using the system, including regular users, recyclers, and staff.
Key business attributes	FullName, Email, PhoneNumber, Role, Status, IsGoogleAuth
Business identity	Email must be unique across the entire system.
Status / lifecycle	Active (normal) · Inactive (deactivated/banned) · Unverified (pending OTP/email confirmation)

Entity: RecyclerProfile

Field	Value
Purpose	Stores the business and operational details for users acting as scrap buyers or recycling facilities.
Key business attributes	FacilityName, RepresentativeName, Address, OperatingHours, LegalPermitUrl, VerificationStatus, ActiveStatus
Business identity	Linked 1:1 with a User account. A user can have at most one RecyclerProfile.
Status / lifecycle	Pending Verification → Verified (or Rejected) → Active / Inactive (business temporarily closed)

Entity: StaffProfile

Field	Value
Purpose	Represents system administrators or staff members who manage operations and resolve disputes.
Key business attributes	StaffCode, Department, Position
Business identity	StaffCode must be unique. Linked 1:1 with a User account.
Status / lifecycle	Active (currently employed) · Inactive (resigned/suspended)

Entity: PasswordResetToken

Field	Value
Purpose	Manages temporary security tokens generated when a user requests to reset a forgotten password.
Key business attributes	TokenCode, ExpiredAt, IsUsed
Business identity	TokenCode is unique and linked to a specific User.
Status / lifecycle	Valid → Used (terminal) or Expired (terminal)

3.2.2 Waste Sorting & AI Group

Entity: MasterWasteCategory

Field	Value
Purpose	Defines the standard categories and material groups of waste (e.g., Plastic, Paper) used for classification and pricing.
Key business attributes	CategoryCode, CategoryName, MaterialGroup, Description, IconUrl, Status
Business identity	CategoryCode must be unique across the system.
Status / lifecycle	Active (available for scanning/pricing) · Inactive (deprecated)

Entity: SortingGuidanceRule

Field	Value
Purpose	Provides step-by-step instructions on how users should properly sort and handle specific types of waste.
Key business attributes	HandlingSteps, DestinationBin
Business identity	Linked 1:1 to a specific MasterWasteCategory.

Status / lifecycle	Published → Updated (when rules change)

Entity: AIScanHistory

Field	Value
Purpose	Records the history of user-uploaded images scanned by the AI for waste classification.
Key business attributes	ImageUrl, ConfidenceScore, ScannedAt
Business identity	Identified by a system-generated ID, linked to the scanning User.
Status / lifecycle	Scanned (permanent record, no status changes)

Entity: MisclassifiedReport

Field	Value
Purpose	Reports submitted by users when the AI incorrectly classifies a scanned waste item.
Key business attributes	ImageUrl, Note, Status
Business identity	Linked to a specific AIScanHistory record.
Status / lifecycle	Pending Review → Reviewed/Approved (added to dataset) or Rejected

Entity: VerifiedDataset

Field	Value
Purpose	Stores accurately labeled waste images used to retrain and improve the computer vision model.
Key business attributes	ImageUrl
Business identity	Linked to a specific MasterWasteCategory (the correct label).
Status / lifecycle	Active (used in current training set) · Archived

3.2.3 Scrap Collection & Trading Group

Entity: HandoverRequest

Field	Value
Purpose	A request created by a user to hand over classified scrap to a specific recycler at a designated location and time.
Key business attributes	estimated_weight_kg, waste_photo_url, address, proposed_time_slot, confirmed_time_slot, status

Business identity	System-generated ID.
Status / lifecycle	Pending (awaiting recycler) → Confirmed (accepted) → Completed (terminal) or Cancelled (terminal)

Entity: HandoverReceipt

Field	Value
Purpose	Acts as the final proof of a completed scrap transaction, recording the actual weight and total payment amount.
Key business attributes	actual_weight_kg, total_reference_amount, scale_proof_photo_url
Business identity	Linked 1:1 with a completed HandoverRequest.
Status / lifecycle	Draft → Confirmed (agreed by both parties)

Entity: RecyclerPriceList

Field	Value
Purpose	Manages the purchasing prices set by individual recyclers for different categories of waste materials.
Key business attributes	reference_price_per_kg, min_weight_kg, is_active
Business identity	The combination of RecyclerProfile and MasterWasteCategory must be unique.
Status / lifecycle	Active (current price) · Inactive (historical/disabled price)

Entity: TransactionComplaint

Field	Value
Purpose	Logs disputes raised by users regarding a specific scrap handover transaction.
Key business attributes	IssueType, Description, ProofPhotoUrl, ResolutionSummary, Status
Business identity	A specific HandoverReceipt can have at most one active TransactionComplaint.
Status / lifecycle	Open → Investigating (by Staff) → Resolved (terminal) or Dismissed (terminal)

Entity: RecyclerRating

Field	Value
Purpose	Stores reviews and star ratings given by users to evaluate their experience with a specific recycler.
Key business attributes	rating_stars, comment

Business identity	A User can leave at most one rating per completed HandoverRequest.
Status / lifecycle	Published → Hidden (if violating community guidelines)

Entity: ChatMessage

Field	Value
Purpose	Represents messages exchanged between users and recyclers to coordinate handovers.
Key business attributes	message_text, attachment_url, sent_at
Business identity	System-generated ID, linked to a specific HandoverRequest or User-Recycler thread.
Status / lifecycle	Sent → Read

3.2.4 Services & Payments Team

Entity: ServicePackage

Field	Value
Purpose	Defines subscription plans or premium services offered within the platform (e.g., recycler visibility boost).
Key business attributes	PackageName, TargetRole, FeeAmount, DurationDays, Description
Business identity	PackageName must be unique.
Status / lifecycle	Active (available for purchase) · Inactive (legacy package)

Entity: ServiceOrder

Field	Value
Purpose	Records the financial transactions when a user or recycler purchases a ServicePackage.
Key business attributes	OrderCode, Amount, QRPayload, Status
Business identity	OrderCode must be unique across the entire system.
Status / lifecycle	Pending Payment → Paid (terminal) or Expired/Failed (terminal)

3.2.5 Rewards, Gifts & Gamification Group

Entity: UserPointWallet

Field	Value

Purpose	Stores the current balance of reward points a user has earned through eco-friendly actions.
Key business attributes	TotalPointsBalance
Business identity	Linked 1:1 with a User account.
Status / lifecycle	Active (balance updates dynamically based on PointTransactionHistory)

Entity: PointTransactionHistory

Field	Value
Purpose	Logs all additions and deductions of a user's reward points.
Key business attributes	PointsChange (positive or negative), ActionType
Business identity	System-generated ID linked to a UserPointWallet.
Status / lifecycle	Logged (permanent, immutable record)

Entity: DailyCheckinLog

Field	Value
Purpose	Tracks a user's daily app login streaks to award bonus points.
Key business attributes	CheckinDate, StreakCount, PointsAwarded
Business identity	The combination of User and CheckinDate must be unique.
Status / lifecycle	Completed (permanent record)

Entity: VirtualTreeProgress

Field	Value
Purpose	Represents a gamification feature tracking the growth stages of a user's virtual plant.
Key business attributes	CurrentStage, TotalCareCount, DailyCareCount, LastCareDate
Business identity	Linked 1:1 with a User account.
Status / lifecycle	Seed → Sprout → Sapling → Mature Tree (cycles repeat or upgrade)

Entity: GiftInventory

Field	Value

Purpose	Catalogs the available physical or digital rewards that users can exchange points for.
Key business attributes	GiftName, GiftType, PointsRequired, StockQuantity
Business identity	System-generated ID or unique Gift SKU.
Status / lifecycle	In Stock → Out of Stock → Discontinued

Entity: GiftRedemption

Field	Value
Purpose	Records the transactions where a user spends points to redeem a specific gift.
Key business attributes	PointsSpent, VoucherCode, Status
Business identity	VoucherCode (if applicable) must be unique.
Status / lifecycle	Processing → Issued/Delivered (terminal) or Failed/Refunded (terminal)

Entity: EcoFact

Field	Value
Purpose	Educational snippets and environmental facts used to inform users and unlock tree stages.
Key business attributes	Title, Content, ImageUrl, UnlockTreeStage
Business identity	System-generated ID.
Status / lifecycle	Published · Draft

3.2.6 News, Announcements & System Group

Entity: BlogPost

Field	Value
Purpose	Represents news articles, announcements, or community posts published on the platform.
Key business attributes	Title, Content, CoverImageUrl, Status
Business identity	System-generated ID.
Status / lifecycle	Draft → Published → Archived

Entity: Notification

Field	Value
Purpose	Manages system alerts, transaction updates, and messages sent to users.
Key business attributes	Title, Content, Type
Business identity	System-generated ID linked to a target User.
Status / lifecycle	Unread → Read

Entity: AuditLog

Field	Value
Purpose	Records system-level actions and data changes for security auditing.
Key business attributes	ActionName, TargetEntity, OldValue, NewValue, CreatedAt
Business identity	System-generated ID.
Status / lifecycle	Logged (append-only, permanent, immutable)

3.3 Data Business Rules

Rule	Description	Enforced On
Unique Email & Phone	No two users may share the same email address or phone number, regardless of their role.	Account creation and profile update
Single Active Profile	A User can only be linked to one RecyclerProfile or one StaffProfile. A single account cannot act as both a Recycler and Staff.	Profile creation
Recycler Verification Lock	A Recycler cannot accept HandoverRequests, appear in user searches, or activate items in RecyclerPriceList until their VerificationStatus is 'Verified'.	Recycler operations
Soft Deletion Policy	User accounts, Recycler profiles, and Staff profiles are only deactivated (Status = Inactive), never permanently deleted, to preserve relational history (logs, receipts, chats).	Account deactivation
Minimum Weight Requirement	The EstimatedWeightKilo in a HandoverRequest must be greater than or equal to the MinWeightKilo defined by the chosen recycler in their RecyclerPriceList.	Handover request creation

Future Time Slot	The ProposedTimeSlot in a HandoverRequest must be a future datetime (e.g., at least 2 hours from the current time).	Handover request creation
Transaction Price Locking	The reference price used to calculate TotalReferenceAmount in HandoverReceipt is locked based on the RecyclerPriceList at the exact moment the HandoverRequest is Confirmed, preventing price manipulation during transit.	Receipt generation
Rating Eligibility	A User can only create a RecyclerRating if there is a corresponding HandoverReceipt (meaning the transaction was successfully completed).	Rating submission
Complaint Window & Limit	A TransactionComplaint can only be submitted within a predefined time window (e.g., 48 hours) after the HandoverReceipt is generated. Only one active complaint is allowed per receipt.	Complaint submission

4. User Requirements

4.1 Actors

#	Actor	Description
1	**Guest**	An unauthenticated visitor to the system. Guests can view public homepage content and project information, register for a new account (a User)
2	**System User**	An abstract base actor representing any account holder or system identity interacting with shared authentication and common account management services (Login, Logout, Password Management, Profile, Notifications). All specialized roles (Citizen User, Recycler, Collector, Staff, Admin) inherit from System User.
3	**User**	An ordinary individual or household seeking to classify and sell recyclable waste. Uses AI photo scanning to identify waste, looks up rule-based sorting guidance, discovers nearby recyclers via GPS, creates handover requests (including preferred collection time slots), negotiates via chatbox, configures payout bank accounts, rates recyclers, submits trade complaints, checks in daily, redeems points for vouchers, and nurtures Virtual Trees (Gamification).
4	**Recycler / Buyer**	An individual scrap collector, scrap yard, or recycling facility. Responsible for managing business profiles, setting accepted waste categories and purchasing price lists, toggling active/online status for GPS discovery, accepting or rejecting incoming handover requests, negotiating via chatbox, creating actual handover receipts, and scanning VietQR codes to settle transaction amounts and platform fees.
5	**Staff (Operations Staff)**	Back-office operational staff responsible for administrative maintenance and support. Key tasks include reviewing misclassified AI images, labeling data to curate verified datasets, verifying and approving Recycler registration applications, handling trade disputes and complaints, viewing User/Buyer

		accounts (locking/deleting violating accounts), managing Gift Inventory and Eco Fact content, conducting staff check-ins, and drafting blog posts for Admin approval.
6	**Administrator (Admin)**	The highest-privileged system actor responsible for platform governance and configuration. Manages system analytics dashboards, exports data reports, manages and assigns permissions for Staff accounts, configures master waste categories and Rule Engine policies, sets Gamification rules and leaderboard cycles, approves or rejects blog posts, configures platform transaction fee percentages, and oversees financial reconciliation (Pay-in/Pay-out tracking).
7	**VietQR (External Payment Gateway)**	An external payment service integrated via API/Webhook. Receives dynamic QR payment generation requests from the system and sends inbound payment webhooks to confirm successful transfers from Recyclers (Pay-in).
8	**Google (External OAuth Provider)**	An external third-party authentication service enabling single sign-on (SSO) and quick registration via Google OAuth 2.0, providing verified user profile information (email_verified).
9	**Gmail / SMTP (External Mail Service)**	An external email delivery infrastructure used by the system to dispatch automated account verification emails, password reset OTPs/links, and critical transaction or dispute notifications to users.

4.2 Use Cases (UC)

ID	Use Case	Feature	Use Case Description
G-01	View Homepage	Guest Services	Allows unauthenticated visitors to view project introduction, major features, public guides, and landing page content.
G-02	Register as User	User Authentication	Allows household citizens to create a new User account by providing basic personal details or via Google OAuth.
SU-01	Login System	User Authentication	Authenticates registered users (User, Recycler, Staff, Admin) using email and password credentials or Google Single Sign-On and establishes an active authenticated session.
SU-02	Change Password	Profile Management	Allows logged-in users to update their current password by verifying their old password.
SU-03	View & Update Profile	Profile Management	Allows authenticated system users to view and update their personal profile information and avatar.

SU-04	View Notifications	Notification System	Displays in-app system notifications, transaction updates, and alerts to the logged-in user.
SU-05	Logout System	User Authentication	Terminates the active user session and redirects the user to the public landing page.
SU-06	Forgot Password	User Authentication	Enables system users who forgot their passwords to request a secure password reset link or OTP via their registered email address.
U-01	Identify Waste via Photo	AI Classification & Rule Engine	Allows users to capture or upload a waste photo for real-time classification by the AI Vision Model.
U-02	Get Sorting Guidance	AI Classification & Rule Engine	Retrieves localized step-by-step handling and disposal instructions from the Rule Engine based on identified waste categories.
U-03	Search & View Waste Details	AI Classification & Rule Engine	Allows users to manually search for waste items by keyword and view detailed classification and handling information.
U-04	Report Incorrect AI Classification	AI Classification & Rule Engine	Enables users to submit feedback and correct labels when the AI model misidentifies a waste image.
U-05	Find Nearby Recyclers	Location-Based Recycler Discovery	Utilizes PostGIS spatial querying to locate, filter, and display active recycling facilities near the user's GPS location.
U-06	Create Handover Request	Recycling Trade Request	Allows users to submit a waste handover request specifying estimated weight, photo, and preferred collection time slot.
U-07	Update / Cancel Handover Request	Recycling Trade Request	Enables users to modify preferred time slots or cancel a pending handover request before recycler acceptance.
U-08	Negotiate via Chatbox	In-App Messaging & Negotiation	Facilitates direct real-time communication between User and Recycler to discuss collection schedules and pricing.
U-09	Confirm Handover Receipt	Recycling Trade Request	Allows the user to review and confirm actual receipt details (weight, price, photo) submitted by the Recycler to complete the trade.
U-10	Rate Recycler	Rating & Review	Enables users to submit star ratings and comments for a Recycler upon completing a handover transaction.

U-11	Submit Transaction Complaint	Dispute Resolution	Allows users to submit a dispute report regarding transaction discrepancies, incorrect pricing, or misconduct.
U-12	View Transaction History	Payment & Disbursement	Displays a historical log of completed trade transactions, received payouts, and earned platform reward points.
U-13	Daily Check-in	Gamification & Rewards	Rewards users with daily Green Points for logging into the platform consistently.
U-14	Redeem Points for Gifts	Gamification & Rewards	Allows users to exchange earned Green Points for physical gifts or merchant discount vouchers in the platform store.
U-15	Upgrade Virtual Tree	Gamification & Rewards	Enables users to spend Green Sprouts to grow their Virtual Tree and unlock educational environmental facts (Eco Facts).
U-16	Purchase In-System Services via VietQR	In-System Services & Payments	Allows users to purchase and pay for internal system service packages (Ad-Free experience, EcoPlus Green Membership, verified badges) via dynamic VietQR codes.
U-17	Apply for Recycler Upgrade	Recycler Onboarding	Allow users to submit an application to upgrade to Recycler status.
R-01	Manage Accepted Waste & Pricing	Recycler Profile & Purchasing Info	Allows recyclers to configure accepted waste categories, minimum quantities, reference price lists, and business hours.
R-02	Toggle Active Status	Recycler Profile & Purchasing Info	Enables recyclers to switch their operational status (Online/Offline) to control visibility on the public GPS map.
R-03	Accept / Reject Handover Request	Recycling Trade Request	Allows recyclers to review incoming handover requests with preferred time slots and accept or reject them based on capacity.
R-04	Cancel Collection Transaction	Recycling Trade Request	Enables recyclers to cancel an accepted handover transaction prior to physical collection with a mandatory reason.
R-05	Negotiate via Chatbox	In-App Messaging & Negotiation	Facilitates direct real-time communication between Recycler and User to discuss collection schedules and pricing.
R-06	Create Handover Receipt	Recycling Trade Request	Allows recyclers to input actual weight/pricing, upload proof photo, and generate a digital handover receipt for user

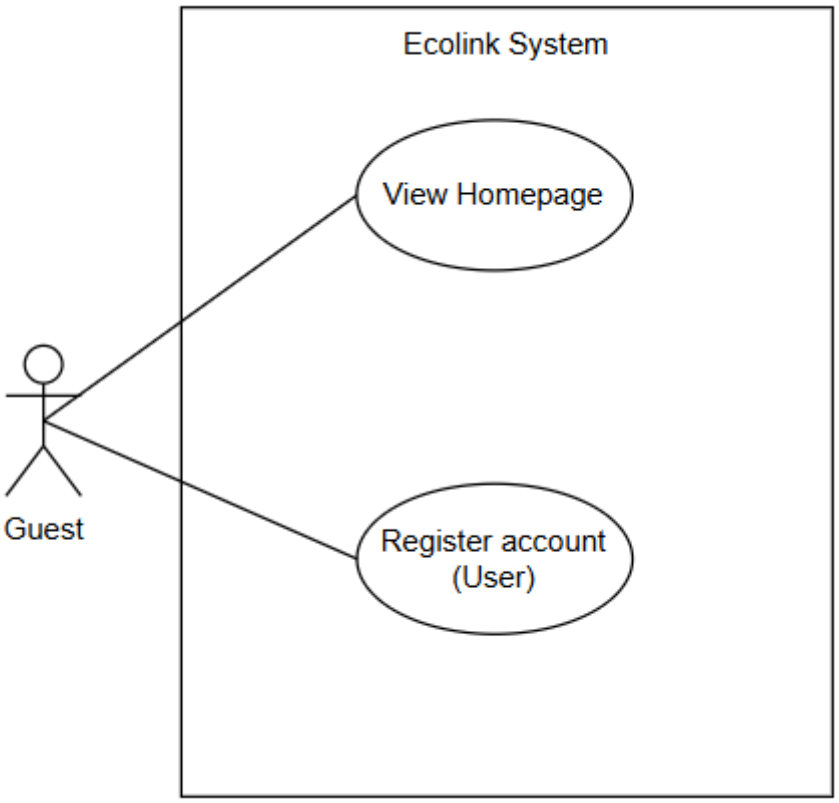
			confirmation. Trade settlement is handled directly outside the system.
R-07	Purchase In-System Services via VietQR	In-System Services & Payments	Allows recyclers to purchase and pay for in-system service packages (Verified Green Tick badge, priority map placement) via dynamic VietQR code scan.
S-01	Review Misclassified Images	AI Model Lifecycle & Data Curation	Enables Staff to inspect user-reported misclassified AI images and verify user-suggested correct labels.
S-02	Label Image Data	AI Model Lifecycle & Data Curation	Allows Staff to annotate and apply standardized labels to raw waste images during inspection (<> from S-01).
S-03	Update Verified Dataset	AI Model Lifecycle & Data Curation	Allows Staff to curate verified clean images into the official test and retraining dataset versions.
S-04	Verify Recycler Application	Recycler Verification	Enables Staff to review submitted business licenses and verify applicant authenticity before approving public Recycler profiles.
S-05	Handle Complaints	Dispute Resolution	Allows Staff to investigate reported trade disputes, review chat history/receipts, and issue binding resolution decisions.
S-06	View Users & Recyclers Accounts	User Account Management	Enables Staff to view account details and transaction histories of User and Recycler accounts.
S-07	Lock / Delete Accounts	User Account Management	Allows Staff to temporarily lock or permanently disable user/recycler accounts violating platform terms.
S-08	Create Blog Post	Content & Blog Management	Enables Staff to draft environmental educational articles and news posts for Admin approval.
S-09	Manage Eco Fact Content	Gamification & Knowledge Management	Allows Staff to perform CRUD operations on Eco Fact knowledge entries and environmental trivia.
S-10	Manage Gift Inventory	Gamification & Knowledge Management	Enables Staff to manage gift catalog items, stock quantities, and voucher codes available for point redemption.
S-11	Staff Daily Check-in	Staff Operations	Tracks daily work attendance and activity logs for internal operational staff members.

A-01	View Dashboard & Analytics	System Administration & Analytics	Displays high-level platform metrics, total recycled tonnage, AI accuracy rates, active users, and financial transaction summaries.
A-02	Export Data Reports	System Administration & Analytics	Allows Admin to export system analytics, transaction logs, and environmental impact data into Excel/CSV files.
A-03	Manage Staff Accounts	Staff Management	Enables Admin to create, update, and manage accounts for internal operational Staff members.
A-04	Manage Staff Permissions	Staff Management	Allows Admin to assign or revoke specific operational permissions and role access levels for Staff accounts.
A-05	Manage Master Waste Categories	Master Data Management	Enables Admin to define, update, or deprecate master waste categories and material groupings across the system.
A-06	Configure Sorting Policy	Master Data Management	Allows Admin to configure localized sorting rules and guidelines within the Rule Engine.
A-07	Manage Gamification Config	Gamification Management	Enables Admin to configure point conversion ratios, daily point caps, bonus multipliers, and Virtual Tree progression thresholds.
A-08	Configure Leaderboard Cycles	Gamification Management	Allows Admin to set evaluation periods (weekly/monthly/quarterly) and automated reset schedules for contribution leaderboards.
A-09	Approve / Reject Blog Post	Content & Blog Management	Enables Admin to review, publish, or reject blog article drafts submitted by Staff.
A-10	Configure In-System Service Fees	Financial Management	Allows Admin to configure pricing, fee schedules, and durations for in-system upgrade service packages (Ad-Free experience, Green Tick verified badges, account upgrades) via VietQR.
A-11	Manage Financial Reconciliation	Financial Management	Allows Admin to audit and reconcile in-system service payments collected via dynamic VietQR and monitor internal revenue records.

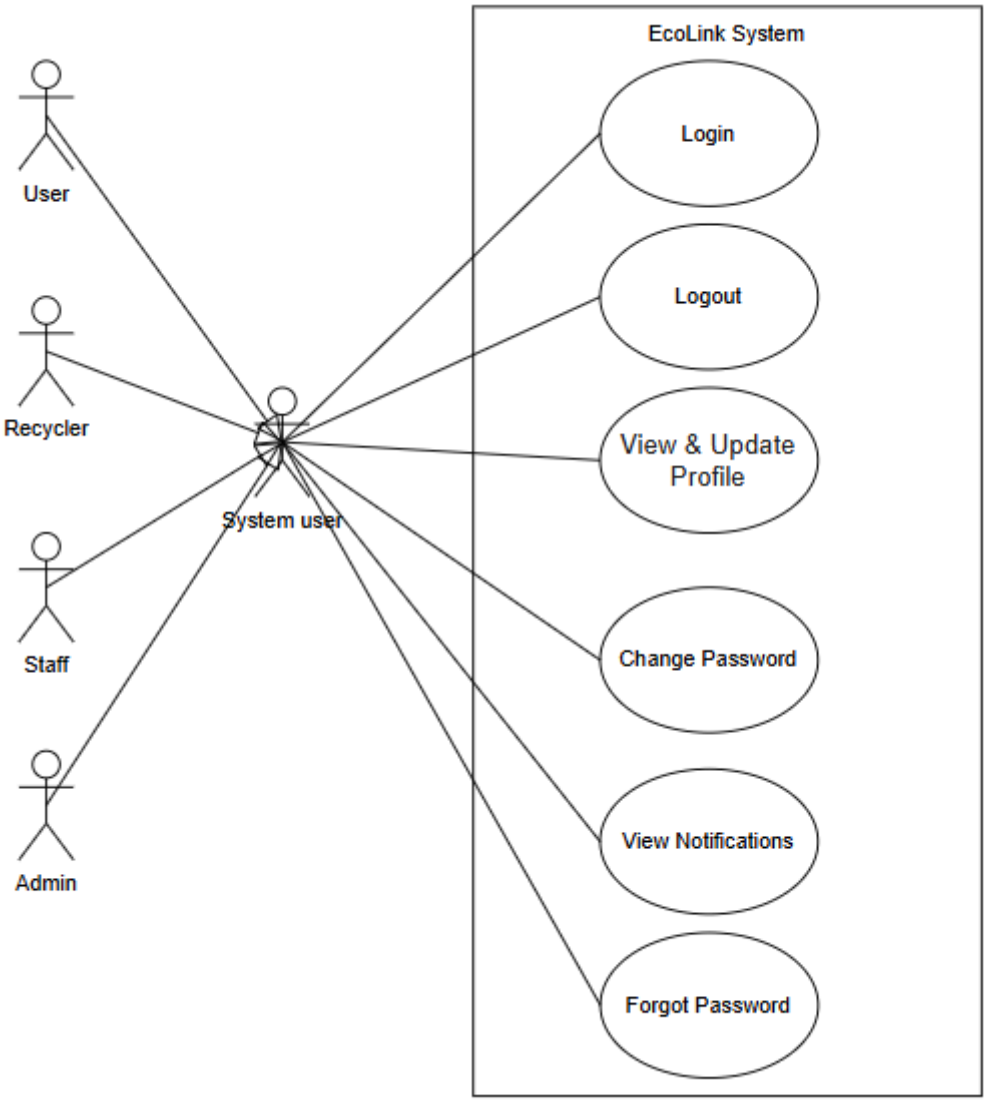
4.3 Use Case Diagrams

This section provides the Use Case Diagrams illustrating the functional boundaries, actor interactions, and inter-use case relationships (<<include>> and <<extend>>) within the EcoLink platform. To ensure structural clarity and maintainability, separate diagrams are provided for each primary actor, concluded by a system-wide unified diagram.

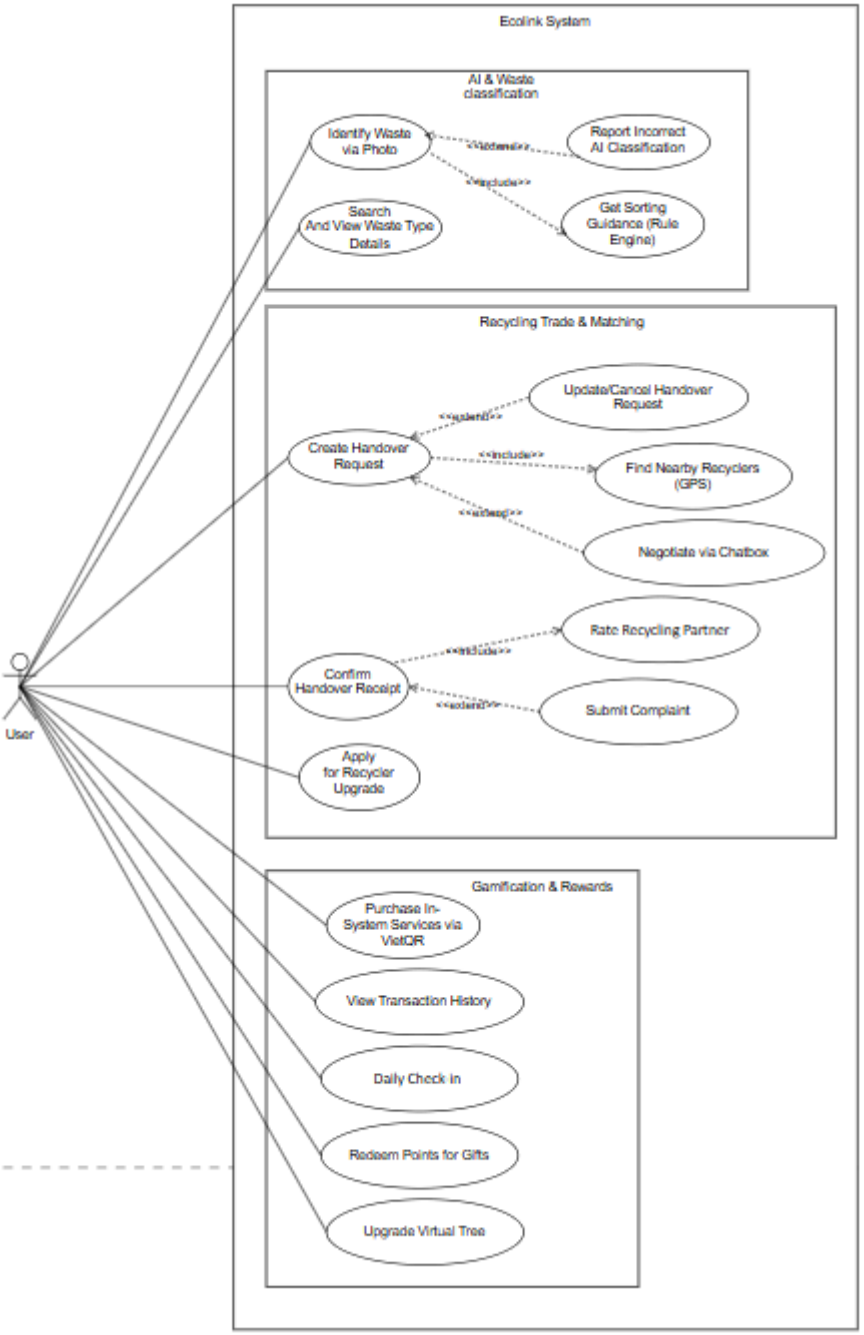
4.3.1 UCs for Guest



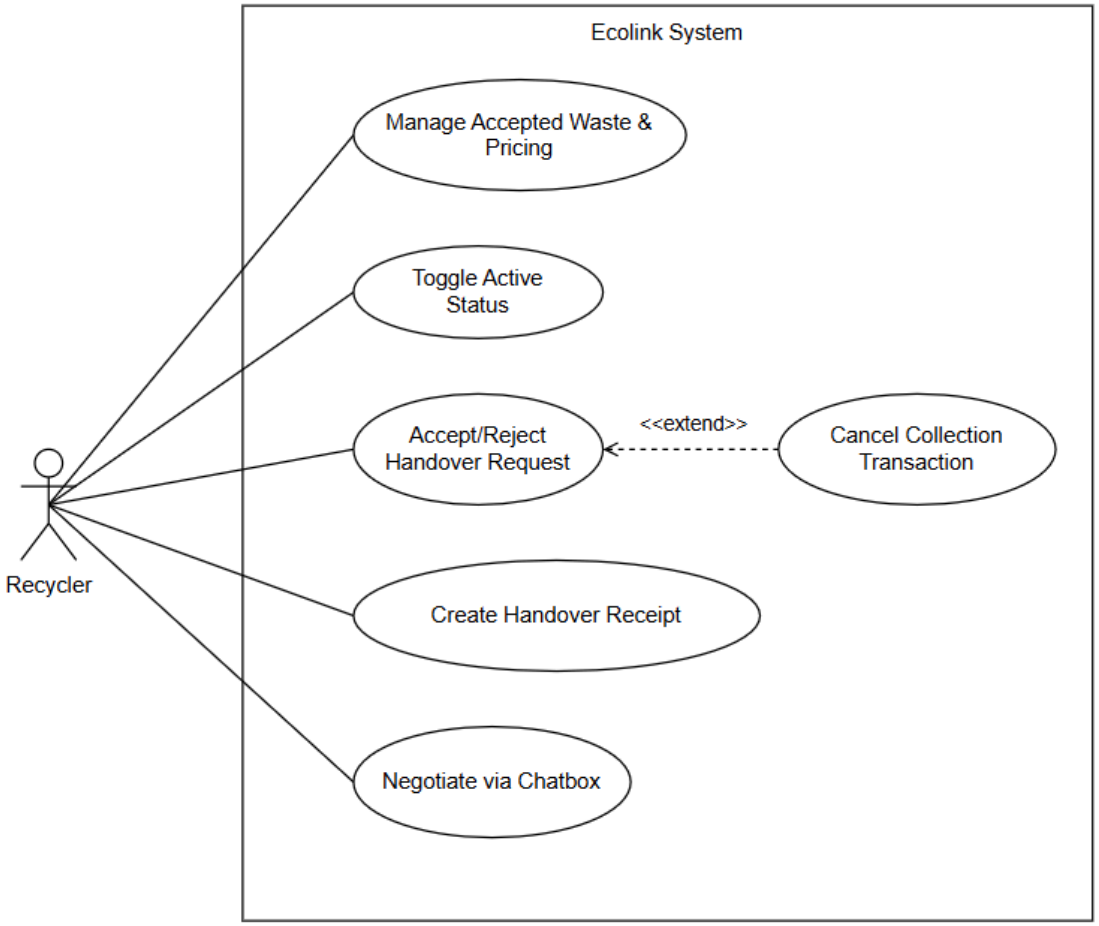
4.3.2 UCs for System user



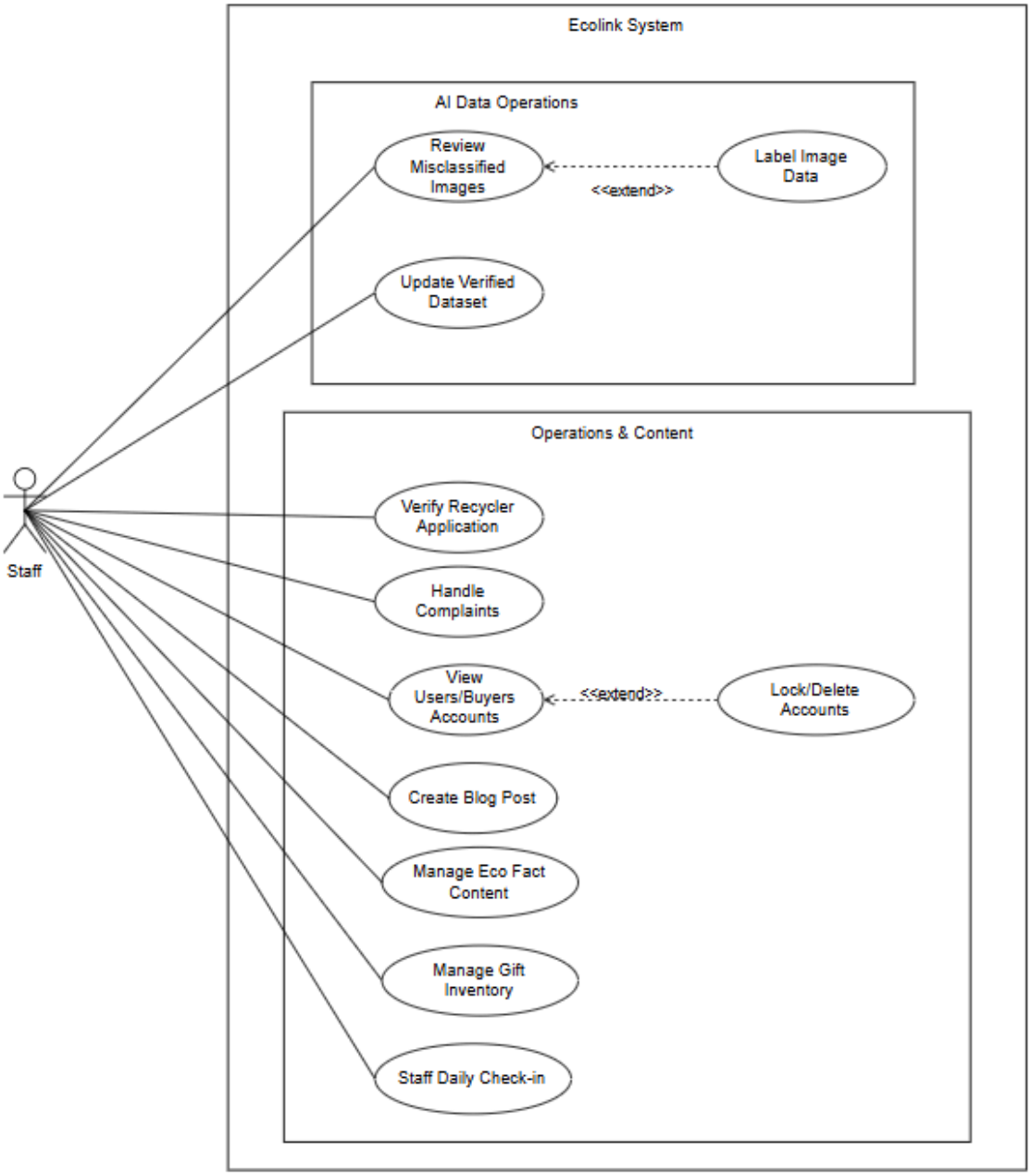
4.3.3 UCs for User



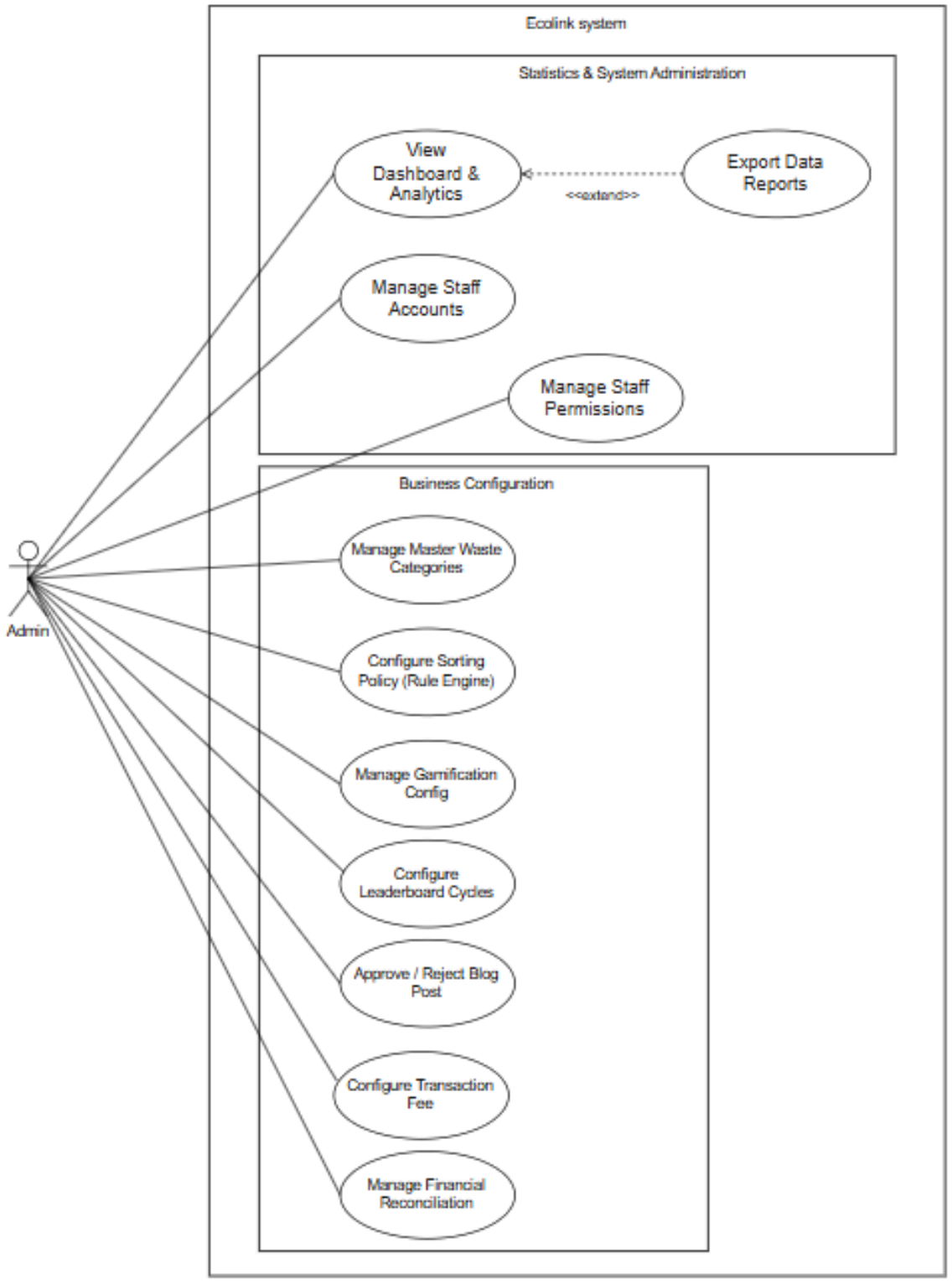
4.3.4 UCs for Recycler



4.3.5 UCs for Staff



4.3.6 UCs for Admin



4.4 Permission Matrix

Full = unrestricted · Restricted = conditions apply (see footnotes) · No = denied

Entity /	Action	Guest	User	Recycler	Staff	Admin

Business Domain						
Citizen User Accounts	Register	Full	No	No	No	No
	Edit / Update Own Profile	No	Full	Full	Full	Full
	View Accounts List & Details	No	No	No	Full	Full
	Lock / Delete Accounts	No	No	No	Restricted	Full
Staff Accounts & Permissions	Create / Edit / Delete Staff	No	No	No	No	Full
	Assign / Revoke Permissions	No	No	No	No	Full
Recycler Verification Application	Submit Application	No	Full	No	No	No
	Verify / Approve / Reject	No	No	No	Full	Full
AI Waste Scan & Classification	Capture / Upload Photo for AI Scan	No	Full	No	No	No
	Report Misclassified AI Image	No	Full	No	No	No
	Review & Label Misclassified Data	No	No	No	Full	Full
	Manage Verified Dataset (Retrain)	No	No	No	Full	Full
Master Waste Category & Rules	View Active Categories & Guidance	Full	Full	Full	Full	Full
	Create / Edit / Deprecate Category	No	No	No	No	Full
	Configure Sorting Policy (Rule Engine)	No	No	No	No	Full
Recycler	Manage Purchasing	No	No	Full	No	No

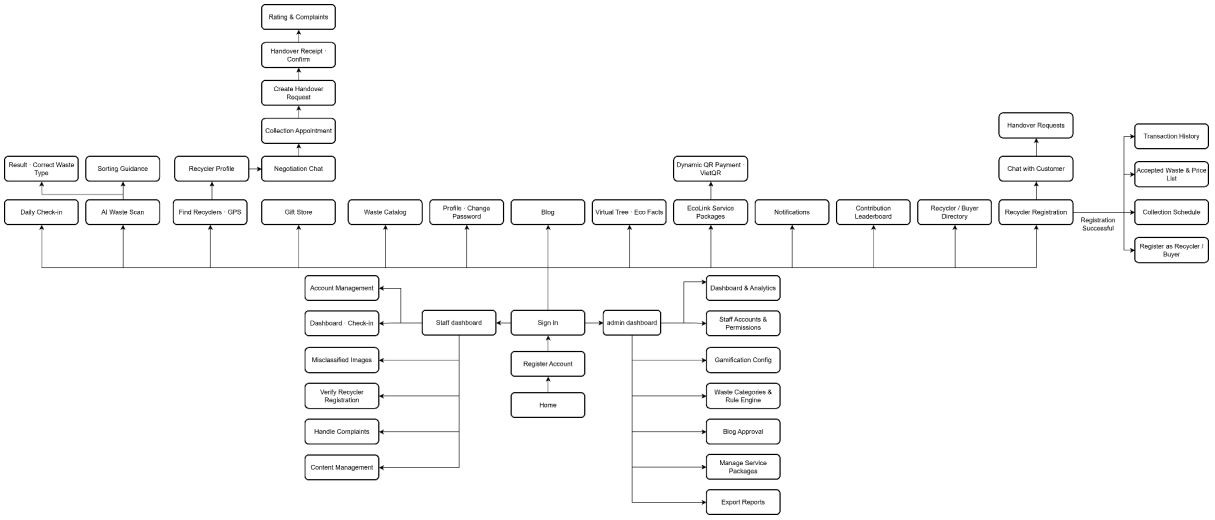
Pricing & GPS Visibility	Price List					
	Toggle Active Status (GPS Map)	No	No	Full	No	No
	Search Nearby Recyclers (GPS)	No	Full	No	No	No
Handover Request & Trade	Create Handover Request	No	Full	No	No	No
	Update / Cancel Handover Request	No	Restricted	No	No	No
	Accept / Reject Handover Request	No	No	Restricted	No	No
	View Handover Details & History	No	Restricted	Restricted	Full	Full
Chatbox Negotiation	Send / View Messages	No	Restricted	Restricted	Restricted	No
Dispute & Complaints	Submit Complaint	No	Restricted	Restricted	No	No
	Resolve Complaint & Issue Outcome	No	No	No	Full	Full
Gamification & Rewards	Daily Check-in	No	Full	No	Full	No
	Redeem Points for Gifts / Voucher	No	Full	No	No	No
	Nurture Virtual Tree & Unlock Fact	No	Full	No	No	No
	Manage Gift Inventory & Eco Fact	No	No	No	Full	Full
	Configure Gamification Rules	No	No	No	No	Full
Blog Posts	View Published Blog Posts	Full	Full	Full	Full	Full
	Create / Edit Blog Drafts	No	No	No	Full	Full

	Approve / Reject / Publish Blog	No	No	No	No	Full
Financial & System Analytics	View Dashboard Analytics	No	No	Noa	No	Full
	Export Data Reports	No	No	No	No	Full
	Configure Transaction Fee & Payout	No	No	No	No	Full
	Manage Financial Reconciliation	No	No	No	No	Full

5. System Functionalities

5.1 Screen Inventory

a. Screen Flow



b. Screen List

#	Screen Name	Feature	Description
1	Sign In	Authentication	Allows any user to authenticate and reach their role-specific area of the system
2	User Register	Authentication	Allows a new User to create their own account
3	Home	Public Portal	Serves as the entry page of EcoLink, introducing the platform and guiding visitors to registration and public information
4	Waste Catalog	Public Portal	Displays all recyclable waste categories with description, image and preparation guideline for users

5	Recycler / Buyer Directory	Public Portal	Shows the list of registered recyclers and large buyers for visitors to browse before signing in
6	Blog	Content	Displays recycling knowledge and green-living articles published by EcoLink
7	Notifications	Notifications	Collects in-app alerts for requests, appointments, approvals and complaints of the signed-in account
8	Profile · Change Password	Account	Lets the signed-in user view and update personal information and change the account password
9	Recycler Registration	Registration	Allows a signed-in user to submit a recycler profile and supporting documents for staff verification
10	Register as Recycler / Buyer	Registration	Provides the registration form and acceptance criteria for becoming a Recycler or a large Buyer
11	Registration Successful	Registration	Confirms that the registration request has been submitted and is waiting for staff review
12	AI Waste Scan	AI Waste Scan	Allows the user to photograph or upload waste images for the AI engine to recognise and classify objects
13	Result · Correct Waste Type	AI Waste Scan	Shows the recognised waste types, score and estimated weight, and lets the user correct any wrong classification
14	Sorting Guidance	AI Waste Scan	Provides step-by-step guidance on how to sort each recognised waste type before handing it over
15	Daily Check-in	Gamification	Allows the user to check in every day to collect eco points and keep the daily streak alive
16	Gift Store	Gamification	Allows the user to exchange accumulated eco points for real gift items
17	Virtual Tree · Eco Facts	Gamification	Grows a virtual tree from eco points and unlocks eco facts and rewards as the tree grows
18	Contribution Leaderboard	Gamification	Ranks users by total eco points and recycling contribution to motivate continuous participation

19	Find Recyclers · GPS	Waste Handover	Locates nearby recyclers and large buyers on the map using the GPS location of the user
20	Recycler Profile	Waste Handover	Displays the profile, accepted waste list, price list and rating of a selected recycler
21	Create Handover Request	Waste Handover	Creates a handover order that selects the recycler, waste type, weight and pickup address
22	Negotiation Chat	Waste Handover	Provides real-time chat where the user and the recycler agree on price, time and handover details
23	Collection Appointment	Waste Handover	Allows both parties to fix the date and time the recycler comes to collect the waste
24	Handover Receipt · Confirm	Waste Handover	Records the handover and lets the user confirm that the waste has been received as agreed
25	Rating & Complaints	Transaction Management	Allows the user to rate the recycler and to raise or track a complaint about a completed transaction
26	EcoLink Service Packages	Service & Payment	Displays the available EcoLink service packages with their benefits, period and platform fee
27	Dynamic QR Payment · VietQR	Service & Payment	Generates a dynamic VietQR code so the recycler pays the EcoLink platform fee after a handover
28	Accepted Waste & Price List	Recycler	Shows the waste types a recycler accepts together with the current price per unit or per kilogram
29	Handover Requests	Recycler	Lists the incoming handover requests from customers for the recycler to review and respond to
30	Chat with Customer	Recycler	Provides real-time chat between the recycler and the customer for price and schedule negotiation
31	Collection Schedule	Recycler	Shows the pickup schedule of the recycler and allows the recycler to confirm or reschedule a visit
32	Transaction History	Recycler	Lists all past handovers of the recycler with weight, value and platform fee status

33	Account Management	Staff	Allows staff to view, lock and manage the accounts of users, recyclers and buyers
34	Dashboard · Check-in	Staff	Gives staff an overview of today's activity with the daily check-in statistics of the platform
35	Misclassified Images	Staff	Shows images that the AI classified with low confidence so staff can correct the classification
36	Verify Recycler Registration	Staff	Allows staff to review submitted profiles and approve or reject recycler registrations
37	Handle Complaints	Staff	Presents the open complaints and allows staff to investigate and resolve them
38	Content Management	Staff	Allows staff to create, edit and schedule blog posts and public content
39	Dashboard & Analytics	Admin	Shows key metrics of the platform such as user growth, recycling volume, revenue and ESG statistics
40	Staff Accounts & Permissions	Admin	Allows the administrator to create staff accounts and assign their permissions
41	Gamification Config	Admin	Configures eco point rules, levels, rewards and the gamification parameters of the platform
42	Waste Categories & Rule Engine	Admin	Maintains the waste category master data and the classification rules of the AI engine
43	Blog Approval	Admin	Allows the administrator to review and approve blog articles drafted by staff
44	Manage Service Packages	Admin	Allows the administrator to create, edit and price the EcoLink service packages
45	Export Reports	Admin	Exports operational, financial and recycling reports to Excel or PDF for reporting

5.2 External API Inventory

#	API Name	Calling Path	Description
1	AI Waste Classification Interface (EXT-01)	EcoLink Backend → AI Inference Service (FastAPI / YOLOv8)	Submits scrap waste photos uploaded by citizens to the computer vision model and retrieves predicted classification labels, confidence scores, and sorting instructions.

2	Cloudflare R2 Cloud Object Storage Interface (EXT-02)	EcoLink Web App / Backend ↔ Cloudflare R2 Storage (S3 API)	Issues pre-signed URLs and uploads persistent image assets, including captured waste photos, scrap scale proof images, and recycler verification licenses.
3	Google OAuth 2.0 Identity Interface (EXT-03)	EcoLink Web Frontend / Backend ↔ Google Identity Platform	Authenticates citizen users and scrap collectors via Single Sign-On (SSO) using verified Google email accounts.
4	VietQR Payment Gateway Interface (EXT-04)	EcoLink Backend ↔ VietQR API & Partner Bank IPN Webhook	Generates dynamic NAPAS 247 banking QR codes with embedded order references and receives real-time IPN payment webhooks for in-system service packages (Collector Verification Badges, Ad-Free subscriptions).
5	OpenStreetMap & Nominatim Geocoding Interface (EXT-05)	EcoLink Web Frontend / Backend ↔ Nominatim OSM Geocoding Service	Converts human-readable street addresses into geographic coordinates (Latitude/Longitude) and provides open-source raster map tiles for recycler discovery.
6	Firebase Cloud Messaging (FCM) Push Service (EXT-06)	EcoLink Backend → Google FCM Push Service → Client Devices	Delivers instant push notification alerts to web and mobile browsers regarding trade offers, scheduled collection reminders, and dispute updates.
7	Twilio SendGrid Transactional Email Interface (EXT-07)	EcoLink Backend → Twilio SendGrid Transactional Email API	Transmits critical security emails, single-use password reset tokens (OTPs), account activation links, and formal dispute arbitrations.

5.3 Background Job Inventory

#	Job Name	Group	Description
1	Auto-Complete Handover Receipt (JOB-01)	Scheduled	Every 60 minutes
2	Reset Daily Check-in & Validate Consecutive Streak (JOB-02)	Scheduled	Daily at 00:00 (ICT / GMT+7)
3	Expire Pending Handover Requests (JOB-03)	Scheduled	Every 30 minutes
4	Flag Outdated Scrap Price Lists (JOB-04)	Scheduled	Daily at 01:00 (ICT / GMT+7)
5	Close Leaderboard Cycle & Reset Top Rankings (JOB-05)	Scheduled	Periodic (Weekly / Monthly / Quarterly configured by Admin)
6	Export Curated Scrap Images for AI Retraining (JOB-06)	Scheduled	Weekly on Sunday at 02:00 (ICT)
7	Purge Stale Unverified Accounts (JOB-07)	Scheduled	Daily at 03:00 (ICT / GMT+7)
8	Enforce Dispute Window Expiration (JOB-08)	Scheduled	Every 60 minutes
9	Send Handover Appointment Reminders (JOB-09)	Scheduled	Triggered {X} hours prior to scheduled appointment
10	Cancel Expired In-System VietQR Payment Orders (JOB-10)	Scheduled	Every 5 minutes

11	Auto-Unlock Temporarily Locked Accounts (JOB-11)	Scheduled	Every 5 minutes
12	Expire Unused Reward Vouchers (JOB-12)	Scheduled	Daily at 00:05 (ICT / GMT+7)

II. Use Case Specifications

1. AI Waste Classification & Sorting Guidance

1.1 Identify Waste via Photo

Primary Actor(s):	User	**Secondary Actor(s):**	AI Classification Service
Description:	Allows an authenticated User to capture or upload a photo of a waste item in order to identify its waste type, retrieve step-by-step sorting instructions from the Rule Engine, and view the estimated scrap market value. The system uses an AI classification model to predict the waste category and asks the User to confirm the result before it is recorded.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The AI Classification Service is operational and accessible. **3.** The User is on the Waste Sorting / AI Scan screen.		
Postconditions:	**- Success:** The classification result card (predicted category, confidence score, recyclable badge, cleaning/disposal instructions, and estimated price range) is displayed. After the User confirms the category, the classification is recorded in the User's scan history. **- Failure:** If any operation fails, the system data remains unchanged, any uploaded temporary files are discarded, no scan record is persisted, and an informative error message is displayed to the User.		
Normal Flow:	**NF-01:** The User selects "Scan Waste via AI" on the home dashboard. **NF-02:** The system opens the camera viewfinder with a capture guide overlay and an option to select an image from the device gallery. **NF-03:** The User captures a photo or selects an existing image, and then taps "Analyze". **NF-04:** The system validates the file format and size, uploads the image to Cloudflare R2 storage, and then invokes the AI Classification Service. **NF-05:** The AI Classification Service processes the image and returns the predicted category labels together with their confidence scores. **NF-06:** The system maps the top prediction to an active category in the Rule Engine, then displays the result card (waste category, recyclable status, confidence percentage, disposal instructions, scrap price range) and MSG-CLS-03. **NF-07:** The User reviews the result and taps "Confirm", after which the system records the classification in the User's scan history.		
Alternative Flow:	**AF-01:** At NF-06, if the AI Classification Service returns two or three competing categories with close confidence scores, the system displays the top-3 predictions and the User selects the matching category, then the flow continues at NF-07. **AF-02:** At NF-07, if the User disagrees with the suggested category, the User selects another category manually and the system records the adjusted category.		

Extension Flow:	**ETF-01:** At NF-03, if the User taps "Cancel" or dismisses the camera/gallery picker without choosing an image, the system returns to the Waste Sorting home screen without invoking the AI Classification Service.
Exceptions Flow:	**EF-01:** At NF-04, if the file is not in a supported format (JPG/PNG), the system displays MSG-CLS-01 and halts processing. **EF-02:** At NF-04, if the file exceeds {X} MB, the system displays MSG-CLS-02 and halts processing. **EF-03:** At NF-05, if the connection to the AI Classification Service times out (> 15 seconds) or the service is unavailable, the system displays MSG-CLS-05 and discards the uploaded temporary file. **EF-04:** At NF-06, if the highest confidence score is below the Admin-configured threshold, the system flags the result as uncertain, displays MSG-CLS-04, and requires the User to confirm or select the category manually. **EF-05:** At NF-06, if the top prediction cannot be mapped to an active waste category, the system does not display the prediction and prompts the User to retry or search the waste catalog manually. **EF-06:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-CLS-01:** Only images in supported formats (JPG/PNG) and within {X} MB are accepted. Invalid files are rejected before AI processing. **BR-CLS-02:** An AI prediction is displayed only if it maps to an active category in the waste catalog. **BR-CLS-03:** If model confidence is below the threshold configured by Admin, the result is flagged "uncertain" and the user must confirm or choose the category manually. **BR-CLS-04:** A user must confirm or correct the suggested category before the classification is recorded as final. **BR-CLS-08:** Waste images are accessible only to the owner, the Recycler in an active related handover, Staff reviewing feedback, and Admin.

1.2 Get Sorting Guidance

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to view step-by-step cleaning, pre-treatment, and bin disposal instructions for a waste category in order to prepare and dispose of the item correctly. The guidance is retrieved from the active policy in the Rule Engine after an AI scan confirmation or a manual catalog search.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** An active sorting policy exists for the target waste category in the Rule Engine. **3.** The User is on the classification result screen or the waste details screen, having confirmed an AI-classified category or selected a category from the search results.		
Postconditions:	**- Success:** The sorting guidance card (numbered pre-treatment steps, designated bin type, recyclable badge, estimated scrap price) is displayed. This is a read-only operation and no data is modified. **- Failure:** If any operation fails, the system data remains unchanged and a fallback message is displayed to the User.		

Normal Flow:	**NF-01:** The User confirms the AI-classified category (see Identify Waste via Photo) or selects a category from the search results (see Search & View Waste Details). **NF-02:** The system queries the Rule Engine for the single active sorting policy version that matches the waste category. **NF-03:** The system displays the Sorting Guidance Card with the step-by-step handling procedure (e.g., empty, rinse, flatten), designated bin type, recyclability indicator, and scrap price range. **NF-04:** The User reviews the guidance.
Alternative Flow:	**AF-01:** At NF-04, the User taps "Share" to copy the guidance link or share it to an external application. **AF-02:** At NF-04, the User taps "Find Nearby Recyclers" and the system navigates to the Find Nearby Recyclers (GPS) screen.
Extension Flow:	**ETF-01:** At NF-04, if the User taps "Back", the system returns to the previous screen without modifying any data.
Exceptions Flow:	**EF-01:** At NF-02, if no active sorting policy is configured for the selected category, the system displays a message that sorting guidance for this category is currently unavailable. **EF-02:** At NF-02, if the Rule Engine is unreachable, the system displays an error message and offers a retry option. **EF-03:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-SRT-01:** Sorting guidance is served only from active rules in the Rule Engine, and each waste category has exactly one active guide version at a time. **BR-SRT-02:** Only Admin can create, edit, or retire catalog entries and sorting policies. Changes apply to new lookups only. **BR-SRT-03:** A sorting guide must contain at least one ordered handling step and a destination before it can be published.

1.3 Search & View Waste Details

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to look up waste categories by keyword or material type (plastic, paper, metal, glass, e-waste) and view their recyclability information and scrap market price range without using the camera scan.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** At least one active waste category exists in the master waste catalog. **3.** The User is on the Waste Catalog / Search screen.		
Postconditions:	**- Success:** The matching categories are listed and the detailed properties of the selected category are displayed. **- Failure:** If any operation fails or no category matches, the system data remains unchanged and an informative message is displayed to the User.		

Normal Flow:	**NF-01:** The User taps the search input field on the Waste Catalog / Search screen. **NF-02:** The User enters a keyword (e.g., "plastic bottle"), and then the system displays real-time autocomplete suggestions after 2 characters are typed. **NF-03:** The User selects a suggestion or taps "Search", after which the system displays the list of matching active categories. **NF-04:** The User taps a category card, and then the system displays the category details (material family, recyclability badge, pre-treatment overview, scrap price range). **NF-05:** The User taps "View Full Guidance" to continue to Get Sorting Guidance.
Alternative Flow:	**AF-01:** At NF-01, the User filters directly by a material chip (e.g., Plastic, Metal, Paper, Glass) instead of entering text. **AF-02:** At NF-05, the User taps "Find Recyclers" and the system navigates to the Find Nearby Recyclers (GPS) screen.
Extension Flow:	**ETF-01:** At NF-04, if the User taps "Back", the system returns to the search result list without modifying any data.
Exceptions Flow:	**EF-01:** At NF-03, if no category matches the search term, the system displays MSG-SRT-01 and suggests common search terms. **EF-02:** At NF-03, if the search service is unreachable, the system displays a network connection error message. **EF-03:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-SRT-02:** Only Admin can create, edit, or retire catalog entries and sorting policies. Changes apply to new lookups only. **BR-SRT-04:** A category referenced by classification results, Recycler profiles, or transactions can only be deactivated, never deleted.

1.4 Report Incorrect AI Classification

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to flag an incorrect AI classification by submitting the original photo together with the category the User believes is correct. The report enters the Staff review queue in order to verify the result and improve the AI training data.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** A scan record exists in the system for the classification being reported. **3.** The User is viewing the AI scan result screen.		
Postconditions:	**- Success:** A misclassification report is persisted, linked to the scan record, the original image, the AI-predicted category, and the User-submitted category, and is added to the Staff review queue. MSG-CLS-06 is displayed. **- Failure:** If any operation fails, the system data remains unchanged, no report is created, and an informative error message is displayed to the User.		

Normal Flow:	**NF-01:** The User taps "Report Incorrect Classification" on the AI scan result screen. **NF-02:** The system displays the feedback form with the AI-predicted category and a category dropdown. **NF-03:** The User selects the correct waste category, optionally enters an explanatory note, and then taps "Submit Feedback". **NF-04:** The system validates the input, creates the report linked to the source image, adds it to the Staff review queue, and finally displays MSG-CLS-06.
Alternative Flow:	**AF-01:** At NF-03, the User searches or filters the category dropdown by material type to find the correct category.
Extension Flow:	**ETF-01:** At NF-02, if the User taps "Cancel", the system closes the form and returns to the scan result screen without recording any feedback.
Exceptions Flow:	**EF-01:** At NF-04, if no correct category is selected, the system highlights the required field and prevents submission. **EF-02:** At NF-04, if the report cannot be saved because of a system error, the system displays an error message and suggests retrying. **EF-03:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-CLS-05:** Every misclassification report is linked to the original image, the predicted category, and the user-suggested category, and enters the Staff review queue. **BR-CLS-06:** Only Staff-reviewed and labeled images may be added to training/validation datasets. **BR-CLS-07:** An image must not appear in both the training and validation datasets. Validation dataset changes are made by Staff only. **BR-CLS-08:** Waste images are accessible only to the owner, the Recycler in an active related handover, Staff reviewing feedback, and Admin.

1.5 Review Misclassified Images

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to review images reported by Users as misclassified by the AI in order to determine the actual waste category and decide whether each image moves on to labeling.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** At least one misclassification report is waiting in the review queue. **3.** The Staff is on the AI Data Management screen.		
Postconditions:	**- Success:** The report status is updated (confirmed for labeling or rejected), the decision is recorded in the audit log, and confirmed images are ready for labeling. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.		

Normal Flow:	**NF-01:** The Staff selects "Misclassification Report Queue". **NF-02:** The system displays the queued reports (thumbnail, AI prediction, confidence, User-proposed category, report date, status). **NF-03:** The Staff selects a report row. **NF-04:** The system displays the full-size image with zoom tools, the AI prediction, the User-proposed category, and any notes. **NF-05:** The Staff inspects the image → clicks "Confirm for Labeling" or "Reject Report". **NF-06:** The system updates the report status → records the decision in the audit log → displays a success notification → loads the next report in the queue.
Alternative Flow:	**AF-01:** At NF-02, the Staff filters the queue by confidence range or category. **AF-02:** At NF-05, if the image is too blurry or unusable, the Staff selects "Discard Image" and the system marks the report as rejected (unusable) and loads the next report.
Extension Flow:	**ETF-01:** At NF-04, if the Staff clicks "Skip" or "Back to Queue", the system leaves the report unchanged and loads the next report or returns to the queue.
Exceptions Flow:	**EF-01:** At NF-04, if the image cannot be retrieved from storage, the system displays an error message and allows the Staff to skip the report. **EF-02:** At NF-05, if the report has already been processed by another Staff member, the system notifies the Staff and refreshes the queue. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-CLS-05:** Every misclassification report is linked to the original image, the predicted category, and the user-suggested category, and enters the Staff review queue. **BR-CLS-06:** Only Staff-reviewed and labeled images may be added to training/validation datasets. **BR-CLS-08:** Waste images are accessible only to the owner, the Recycler in an active related handover, Staff reviewing feedback, and Admin. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

1.6 Label Image Data

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to assign an accurate standard label to a reviewed image in order to build reliable training data for AI retraining.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** The Staff is viewing a reviewed image that was confirmed for labeling. **3.** The standard waste category list is active.		
Postconditions:	**- Success:** The image is assigned a standard label and its status is updated to Labeled, ready for the Verified Dataset. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.		

Normal Flow:	**NF-01:** The Staff clicks "Assign Standard Label". **NF-02:** The system displays the list of active standard categories. **NF-03:** The Staff selects the correct label and then clicks "Save Label". **NF-04:** The system validates the selection, records the label, updates the image status, logs the action, and then displays a completion notice.
Alternative Flow:	**AF-01:** At NF-02, the Staff searches or filters the category list by material group to find the correct label.
Extension Flow:	**ETF-01:** At NF-02, if the Staff clicks "Cancel", the system closes the form without any change.
Exceptions Flow:	**EF-01:** At NF-03, if no label is selected, the system displays MSG-AUTH-02 and prevents saving. **EF-02:** At NF-04, if the selected category has been deactivated in the meantime, the system rejects the label, refreshes the category list, and asks the Staff to choose another category. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-CLS-06:** Only Staff-reviewed and labeled images may be added to training/validation datasets. **BR-CLS-07:** An image must not appear in both the training and validation datasets. Validation dataset changes are made by Staff only. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

1.7 Update Verified Dataset

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to compile labeled images into the Verified Dataset in order to prepare reliable data for AI retraining and accuracy evaluation.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** At least one labeled image is pending inclusion in the Verified Dataset. **3.** The Staff is on the Verified Dataset screen of Dataset Management.		
Postconditions:	**- Success:** The selected labeled images are added to the verified dataset, a new dataset version is created, and the version history is logged. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.		
Normal Flow:	**NF-01:** The Staff selects "Verified Dataset". **NF-02:** The system displays the current version information, dataset statistics, and the pending labeled images. **NF-03:** The Staff clicks "Update Dataset". **NF-04:** The system displays a confirmation dialog with the update statistics. **NF-05:** The Staff enters batch notes → clicks "Confirm Update". **NF-06:** The system adds the images to the official dataset, creates a new version log, and then displays a success notification.		
Alternative Flow:	**AF-01:** At NF-02, the Staff exports the dataset statistics report and the system displays MSG-ADM-02.		

Extension Flow:	**ETF-01:** At NF-04, if the Staff clicks "Cancel", the system aborts the update and keeps the dataset unchanged.
Exceptions Flow:	**EF-01:** At NF-05, if the batch notes are missing, the system displays MSG-AUTH-02 and prevents the update. **EF-02:** At NF-06, if an image already belongs to the training or validation dataset on the other side, the system blocks its inclusion and lists the conflicting images. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-CLS-06:** Only Staff-reviewed and labeled images may be added to training/validation datasets. **BR-CLS-07:** An image must not appear in both the training and validation datasets. Validation dataset changes are made by Staff only. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

1.8 Manage Master Waste Categories

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to manage the master catalog of recyclable waste categories in order to keep waste classification and sorting guidance accurate. The Admin can view the category list, create new categories, update category attributes, and deactivate categories that are no longer in use.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has catalog management privileges. **3.** The Admin is on the Master Waste Categories screen.		
Postconditions:	**- View:** The system displays the latest list of master waste categories. **- Create:** A new category is created in Active status and is available for subsequent lookups. **- Update:** The category attributes are updated and apply to new lookups only. **- Deactivate:** The category is set to Inactive and hidden from new lookups, while historical records are preserved. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Admin.		

Normal Flow:	**View Waste Categories** **NF-01:** The Admin navigates to the "Master Waste Categories" section. **NF-02:** The system displays the category list (Category Code, Name, Material Group, Base Unit, Status, Linked Rules Count). **Create Category** **NF-03:** The Admin clicks "Add Category". **NF-04:** The system displays the category creation form. **NF-05:** The Admin enters the Category Code, Name, Description, and Standard Unit (kg/piece), uploads an icon, and then clicks "Save". **NF-06:** The system validates code uniqueness and data completeness, creates the category in Active status, writes an audit log entry, displays MSG-SRT-03, and then refreshes the category table. **Update Category** **NF-03 (from NF-02):** The Admin clicks "Edit" on a category row. **NF-04:** The system displays the edit form pre-populated with the current data. **NF-05:** The Admin modifies the attributes and then clicks "Save Changes". **NF-06:** The system validates the input, saves the changes (applied to new lookups only), writes an audit log entry, displays MSG-SRT-03, and then updates the table row. **Deactivate Category** **NF-03 (from NF-02):** The Admin clicks "Deactivate" on a category row. **NF-04:** The system displays a confirmation dialog. **NF-05:** The Admin confirms the action. **NF-06:** The system sets the category status to Inactive, writes an audit log entry, displays a success notification, and then refreshes the table.
Alternative Flow:	**AF-01:** At NF-02, the Admin searches by keyword or filters the list by material group or status, and the system updates the displayed categories.
Extension Flow:	**ETF-01:** At NF-05 (Create/Update), if the Admin clicks "Cancel" or closes the form, the system discards all entered data and returns to the category list. **ETF-02:** At NF-05 (Deactivate), if the Admin clicks "Cancel" on the confirmation dialog, the category status remains unchanged.
Exceptions Flow:	**EF-01:** At NF-06 (Create/Update), if required fields are empty, the system highlights them, displays MSG-AUTH-02, and prevents saving. **EF-02:** At NF-06 (Create), if the Category Code already exists, the system displays a duplicate code error message and keeps the form data. **EF-03:** At NF-03 (Deactivate), if the Admin requests permanent deletion of a category referenced by classification results, Recycler profiles, or transactions, the system blocks the deletion and only allows deactivation. **EF-04:** At NF-03, if a non-Admin account attempts to change the catalog, the system denies the request and displays MSG-ADM-04. **EF-05:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.

Business Rule	**BR-SRT-02:** Only Admin can create, edit, or retire catalog entries and sorting policies. Changes apply to new lookups only. **BR-SRT-04:** A category referenced by classification results, Recycler profiles, or transactions can only be deactivated, never deleted. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

1.9 Configure Sorting Policy (Rule Engine)

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to configure the Rule Engine in order to define the mandatory ordered handling steps and the designated destination for each waste category. The Admin can save a policy as a draft or publish it as the single active guidance version for the category.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has policy configuration privileges and the target waste category is in Active status. **3.** The Admin is on the Sorting Policy & Rule Engine screen.		
Postconditions:	**- Success:** When published, the new policy becomes the sole active guidance version for the category and previous versions are archived. When saved as a draft, the policy is stored and is not visible to Users. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the active sorting guidance is not affected, and an informative error message is displayed to the Admin.		
Normal Flow:	**NF-01:** The Admin navigates to the "Sorting Policy & Rule Engine" screen. **NF-02:** The system displays the waste categories with their active policy version and status. **NF-03:** The Admin selects a waste category to create or revise its sorting policy. **NF-04:** The system displays the policy editor with the current policy (if any). **NF-05:** The Admin configures the ordered handling steps (e.g., empty liquids, rinse and dry, remove label, flatten), assigns the designated destination (Recycle Bin / Recycler Depot), adds conditional notes such as cleanliness requirements or contamination warnings, and then clicks "Publish Policy". **NF-06:** The system validates that at least one ordered step and one destination exist, sets the new policy as the sole active version, archives previous versions, writes an audit log entry, and then displays MSG-SRT-04.		
Alternative Flow:	**AF-01:** At NF-05, if the Admin clicks "Save as Draft", the system stores the policy draft without publishing it to Users.		
Extension Flow:	**ETF-01:** At NF-05, if the Admin clicks "Cancel", the system discards the changes and returns to the policy overview.		

Exceptions Flow:	**EF-01:** At NF-06, if the policy lacks an ordered handling step or a destination, the system blocks publishing and prompts for the missing components. **EF-02:** At NF-06, if the category was deactivated while the policy was being edited, the system blocks publishing and informs the Admin. **EF-03:** At NF-03, if a non-Admin account attempts to modify Rule Engine policies, the system denies the request and displays MSG-ADM-04. **EF-04:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-SRT-01:** Sorting guidance is served only from active rules in the Rule Engine, and each waste category has exactly one active guide version at a time. **BR-SRT-02:** Only Admin can create, edit, or retire catalog entries and sorting policies. Changes apply to new lookups only. **BR-SRT-03:** A sorting guide must contain at least one ordered handling step and a destination before it can be published. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

2. Recycler Profile & Matching

2.1 Find Nearby Recyclers (GPS)

Primary Actor(s):	User	**Secondary Actor(s):**	Map Provider (OpenStreetMap / Google Maps)
Description:	Allows an authenticated User to view verified scrap collectors and recycling facilities on an interactive map within a configurable radius of the User's GPS location in order to choose a suitable recycler, based on its accepted materials, prices, and operating schedule.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The User is on the home screen or a screen that offers access to "Find Recyclers".		
Postconditions:	**- Success:** The interactive map with recycler markers within the radius is displayed, together with a summary list sorted by ascending distance. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the User.		
Normal Flow:	**NF-01:** The User taps "Find Recyclers" on the main navigation bar. **NF-02:** The system requests the device's current GPS coordinates. **NF-03:** The system queries active, approved recyclers within the default radius (5 km), and then plots the markers on the interactive map. **NF-04:** The system displays a scrollable recycler summary list below the map, ordered by ascending distance. **NF-05:** The User taps a map marker or a recycler card, after which the system displays the Recycler Profile Card (business name, general area address, accepted materials, price per kg, operating hours, average rating). **NF-06:** The User reviews the profile and then taps "Request Handover" to continue to Create Handover Request.		

Alternative Flow:	**AF-01:** At NF-03, the User adjusts the search radius slider (2 km – 20 km) and the system refreshes the map markers. **AF-02:** At NF-04, the User filters by accepted material type (e.g., only recyclers accepting "Metal") and the map and list update accordingly.
Extension Flow:	**ETF-01:** At NF-05, if the User taps "Close" or taps outside the profile card, the system collapses the card and returns to the map overview.
Exceptions Flow:	**EF-01:** At NF-02, if GPS access is denied or disabled, the system displays MSG-MTH-02 and offers a manual location search bar. **EF-02:** At NF-03, if no recycler operates within the radius, the system displays MSG-MTH-01 and prompts the User to expand the radius. **EF-03:** At NF-03, if the map service is unreachable, the system displays a network connection error message. **EF-04:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-MTH-01:** A Recycler must be approved by Staff before appearing in search results or receiving handover requests. **BR-MTH-03:** A Recycler profile can be published only with at least one accepted waste category, collection conditions, and a valid location. **BR-MTH-04:** Search returns only approved, active Recyclers within the user's search radius that accept the identified waste category, sorted by distance by default. **BR-MTH-05:** A user's GPS coordinates are used only to compute distance and never shown to Recyclers. The exact address is shared only after a request is accepted. **BR-MTH-06:** A suspended Recycler is hidden from search immediately, but existing active transactions remain visible to both parties.

2.2 Manage Accepted Waste & Pricing

Primary Actor(s):	Recycler	**Secondary Actor(s):**	
Description:	Allows an authenticated Recycler to manage the list of accepted scrap types, public reference prices per kilogram (VND/kg), and minimum collection weights in order to keep the facility profile accurate and attractive to Users.		
Preconditions:	**1.** The Recycler is authenticated and has an active session with role Recycler, and the Recycler account has been approved by Staff. **2.** The Recycler is on the Pricing & Categories screen.		
Postconditions:	**- View:** The system displays the facility's current accepted scrap types and prices. **- Add:** The new scrap type is saved to the facility profile and displayed publicly to Users. **- Update:** The updated price, minimum weight, or purchasing status is saved and displayed publicly to Users. **- Failure:** If any operation fails, the system data remains unchanged, the previous pricing data is retained, and an informative error message is displayed to the Recycler.		

Normal Flow:	**View Categories & Pricing** **NF-01:** The Recycler selects "Scrap Pricing" on the navigation bar. **NF-02:** The system displays the list of scrap types accepted by the facility (Waste Name, Material Group, Reference Price in VND/kg, Minimum Collection Weight, Status). **Add Scrap Type** **NF-03:** The Recycler clicks "Add Scrap Type". **NF-04:** The system displays a form with a standard waste type dropdown, a price input (VND/kg), and a minimum weight input (kg). **NF-05:** The Recycler selects a scrap type and enters the price and minimum weight, and then clicks "Save". **NF-06:** The system validates the data, saves the scrap type to the facility profile, and then refreshes the list. **Update Pricing or Suspend Purchasing** **NF-03 (from NF-02):** The Recycler clicks the "Edit" icon on a scrap row. **NF-04:** The system displays a dialog to change the price, minimum weight, or toggle the suspension of purchasing. **NF-05:** The Recycler changes the information and then clicks "Update". **NF-06:** The system validates the data, records the new information, and then updates the category list.
Alternative Flow:	**AF-01:** At NF-02, the Recycler uses the quick search box or filters by material group, and the system filters the displayed rows.
Extension Flow:	**ETF-01:** At NF-05 (Add/Update), if the Recycler clicks "Cancel" or clicks outside the dialog, the system closes the form and retains all previous information.
Exceptions Flow:	**EF-01:** At NF-06, if mandatory fields are left blank, the system displays MSG-AUTH-02 and prevents saving. **EF-02:** At NF-06, if the price or minimum weight is not a positive number, the system highlights the invalid field and prevents saving. **EF-03:** At NF-06, if the change would leave the profile without any accepted waste type or a valid location, the system displays MSG-MTH-06 and keeps the profile unchanged. **EF-04:** At any point, if the Recycler's session has expired, the system displays MSG-NTF-01 and redirects the Recycler to the Login page.
Business Rule	**BR-MTH-01:** A Recycler must be approved by Staff before appearing in search results or receiving handover requests. **BR-MTH-03:** A Recycler profile can be published only with at least one accepted waste category, collection conditions, and a valid location. **BR-MTH-07:** Removing an accepted category from a Recycler profile does not affect handover requests already accepted.

2.3 Toggle Active Status

Primary Actor(s):	Recycler	**Secondary Actor(s):**	Map Provider (OpenStreetMap / Google Maps)

Description:	Allows an authenticated Recycler to switch the facility's readiness status between Active and Resting in order to control whether the facility appears on the map for nearby Users and receives new handover requests.
Preconditions:	**1.** The Recycler is authenticated and has an active session with role Recycler, and the Recycler account has been approved by Staff. **2.** The Recycler is on the Dashboard screen.
Postconditions:	**- Success:** When Active, the facility is displayed as ready on the map for Users within the radius and can receive new requests. When Resting, the facility is hidden from the map and stops receiving new requests, while existing transactions remain unaffected. **- Failure:** If any operation fails, the toggle state remains unchanged and an informative error message is displayed to the Recycler.
Normal Flow:	**NF-01:** The Recycler views the status toggle switch on the Dashboard. **NF-02:** The Recycler toggles the switch between Resting and Active. **NF-03:** The system displays a confirmation dialog for the status change. **NF-04:** The Recycler clicks "Agree". **NF-05:** The system records the new status, updates the facility's visibility on the map, and then displays the updated status.
Alternative Flow:	**AF-01:** N/A
Extension Flow:	**ETF-01:** At NF-04, if the Recycler clicks "Cancel", the system closes the dialog and retains the previous toggle position.
Exceptions Flow:	**EF-01:** At NF-02, if the facility has no accepted waste type configured or no valid location, the system displays MSG-MTH-06 and prevents activation. **EF-02:** At NF-05, if the account is deactivated or suspended, the system displays MSG-AUTH-04 and terminates the session. **EF-03:** At NF-05, if the map visibility update fails, the system keeps the previous status and displays an error message. **EF-04:** At any point, if the Recycler's session has expired, the system displays MSG-NTF-01 and redirects the Recycler to the Login page.
Business Rule	**BR-MTH-01:** A Recycler must be approved by Staff before appearing in search results or receiving handover requests. **BR-MTH-03:** A Recycler profile can be published only with at least one accepted waste category, collection conditions, and a valid location. **BR-MTH-06:** A suspended Recycler is hidden from search immediately, but existing active transactions remain visible to both parties. **BR-AUTH-05:** A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately.

2.4 Verify Recycler Application

Primary Actor(s):	Staff	**Secondary Actor(s):**	Recycler, Gmail
Description:	Allows an authenticated Staff member to verify the documents and licenses of a recycling facility in order to approve or reject its account before it appears in search results.		

Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** At least one Recycler application is in Pending Verification status. **3.** The Staff is on the Recycler Management screen.
Postconditions:	**- Approve:** The Recycler account is set to Verified and becomes visible to Users, and MSG-MTH-04 is sent to the Recycler. **- Reject:** The Recycler account is set to Rejected with a recorded reason, MSG-MTH-05 is sent to the Recycler, and the Recycler may resubmit a corrected profile. The decision is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.
Normal Flow:	**Approve Application** **NF-01:** The Staff selects "Pending Applications". **NF-02:** The system displays the list of pending applications. **NF-03:** The Staff selects an application. **NF-04:** The system displays the ID/licenses, facility photos, and location. **NF-05:** The Staff clicks "Approve". **NF-06:** The system displays a confirmation dialog then the Staff clicks "Agree". **NF-07:** The system updates the status to Verified then writes an audit log entry and sends MSG-MTH-04 to the Recycler. **Reject Application** **NF-05 (from NF-04):** The Staff clicks "Reject". **NF-06:** The system displays the rejection reason form. **NF-07:** The Staff selects a reason or enters details, and then clicks "Send". **NF-08:** The system updates the status to Rejected, writes an audit log entry, and then sends MSG-MTH-05 to the Recycler.
Alternative Flow:	**AF-01:** At NF-02, the Staff searches or filters the applications by submission date or area.
Extension Flow:	**ETF-01:** At NF-06 (Approve) or NF-07 (Reject), if the Staff clicks "Cancel", the system aborts the action and keeps the application in Pending Verification status.
Exceptions Flow:	**EF-01:** At NF-07 (Reject), if no rejection reason is provided, the system displays MSG-AUTH-02 and prevents submission. **EF-02:** At NF-07, if the application has already been processed by another Staff member, the system notifies the Staff and refreshes the list. **EF-03:** At NF-07 (Approve) or NF-08 (Reject), if the notification email cannot be delivered, the system displays MSG-NTF-02 and keeps the decision saved. **EF-04:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-MTH-01:** A Recycler must be approved by Staff before appearing in search results or receiving handover requests. **BR-MTH-02:** A rejected verification must include a reason, and the Recycler may resubmit a corrected profile. **BR-NTF-06:** A failed email delivery is logged and retried up to {N} times, and never blocks or rolls back the business operation. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

3. Handover Request & Negotiation

3.1 Create Handover Request

Primary Actor(s):	User	**Secondary Actor(s):**	Recycler
Description:	Allows an authenticated User to submit a handover request to a selected Recycler, specifying the waste categories, estimated weights, photos, and a preferred collection time slot, in order to arrange the sale and collection of recyclable waste.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** An active, approved Recycler has been selected by the User. **3.** The User is on the Recycler Profile Card screen.		
Postconditions:	**- Success:** A new Handover Request is created in Pending status with a unique Request ID and the proposed collection time slot, and the Recycler is notified with MSG-NEG-04. MSG-HND-01 is displayed to the User. **- Failure:** If any operation fails, the system data remains unchanged, any uploaded temporary files are discarded, no request is created, and an informative error message is displayed to the User.		
Normal Flow:	**NF-01:** The User taps "Request Handover" on the Recycler Profile Card. **NF-02:** The system displays the request form pre-filled with the selected Recycler and its accepted materials. **NF-03:** The User selects the waste categories to sell and then enters the estimated weight (kg) for each category. **NF-04:** The User uploads at least one photo of the waste items, after which the system uploads the images to Cloudflare R2 storage and displays thumbnail previews. **NF-05:** The User selects a preferred collection date and time slot within the Recycler's declared schedule. **NF-06:** The User reviews the summary and then taps "Submit Request". **NF-07:** The system validates all required inputs, creates the request in Pending status, notifies the Recycler (MSG-NEG-04), displays MSG-HND-01, and finally navigates to the Request Detail screen.		
Alternative Flow:	**AF-01:** At NF-03, if the User wants to discuss the terms before submitting, the User taps "Message Recycler" and the system opens the Negotiate via Chatbox screen.		
Extension Flow:	**ETF-01:** Between NF-02 and NF-06, if the User taps "Cancel" or "Back", the system asks for confirmation; upon confirmation, the system discards all draft inputs and returns to the Recycler Profile Card.		

Exceptions Flow:	**EF-01:** At NF-07, if a mandatory field is invalid (no waste category, weight ≤ 0, no photo, or no time slot), the system highlights the invalid fields and prevents submission. **EF-02:** At NF-04, if an image is in an unsupported format or exceeds the size limit, the system displays MSG-CLS-01 or MSG-CLS-02 and rejects the image. **EF-03:** At NF-07, if the User already holds 3 active requests in Pending or Accepted status, the system blocks the creation and displays an error message stating that the maximum number of active handover requests has been reached. **EF-04:** At NF-07, if the selected Recycler is no longer approved or active, or no longer accepts a selected waste category, the system blocks the submission and prompts the User to choose another Recycler or category. **EF-05:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-HND-01:** A handover request can be sent only to an approved, active Recycler that accepts the waste category in the request. **BR-HND-02:** Handover status changes are one-way and permanent: Pending → (Negotiating) → Accepted → Scheduled → Completed, or terminal Rejected/Cancelled/Expired/Disputed. No status may revert. **BR-HND-08:** Full contact details are revealed only after a request is accepted. **BR-NEG-04:** The User proposes their desired collection time slot directly inside the Handover Request when creating it. If the slot doesn't work for the Recycler, the two parties negotiate a new time through the chat. **BR-MTH-05:** A user's GPS coordinates are used only to compute distance and never shown to Recyclers. The exact address is shared only after a request is accepted.

3.2 Update / Cancel Handover Request

Primary Actor(s):	User	**Secondary Actor(s):**	Recycler
Description:	Allows an authenticated User to modify the time slot or estimated weight of a Pending handover request, or to cancel a request before the transaction is completed, in order to keep the request consistent with the User's actual plan.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The User's handover request exists in Pending status (for editing) or has not reached Completed status (for cancellation). **3.** The User is on the My Requests screen.		
Postconditions:	**- Update:** The request attributes are updated and the Recycler is notified of the changes. **- Cancel:** The request status is set to Cancelled (terminal) and the Recycler is notified with MSG-HND-05. A late cancellation is recorded when it falls within the configured period before the confirmed appointment. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the User.		

Normal Flow:	**Update Request** **NF-01:** The User navigates to "My Requests" and then selects a request in Pending status. **NF-02:** The system displays the Request Detail screen with the available actions. **NF-03:** The User taps "Edit Request". **NF-04:** The system displays the editable fields populated with the current data. **NF-05:** The User adjusts the time slot, estimated weight, or notes, and then taps "Save Changes". **NF-06:** The system validates the inputs, updates the request, notifies the Recycler, and then displays a success confirmation. **Cancel Request** **NF-03 (from NF-02):** The User taps "Cancel Request". **NF-04:** The system displays a confirmation dialog that requires a cancellation reason. **NF-05:** The User selects or enters a reason, and then taps "Confirm Cancellation". **NF-06:** The system updates the request status to Cancelled, notifies the Recycler (MSG-HND-05), and then displays a success notification.
Alternative Flow:	**AF-01:** At NF-04 (Update), if the User wants to renegotiate terms instead of editing the request, the User taps "Message Recycler" and the system opens the Negotiate via Chatbox screen.
Extension Flow:	**ETF-01:** At NF-05 (Update), if the User taps "Cancel", the system discards the changes and returns to the Request Detail screen. **ETF-02:** At NF-05 (Cancel), if the User taps "No" or dismisses the dialog, the request remains in its current status.
Exceptions Flow:	**EF-01:** At NF-06 (Update), if the Recycler has accepted the request in the meantime, the system informs the User that the request can no longer be edited and suggests negotiating via chat or cancelling. **EF-02:** At NF-06 (Update), if a required field is invalid, the system highlights the invalid fields and prevents saving. **EF-03:** At NF-05 (Cancel), if the cancellation is within {X} hours before the confirmed appointment, the system displays MSG-HND-06 before executing it. **EF-04:** At NF-03, if the request has already reached Completed or another terminal status, the system disables the edit and cancel actions. **EF-05:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-HND-02:** Handover status changes are one-way and permanent: Pending → (Negotiating) → Accepted → Scheduled → Completed, or terminal Rejected/Cancelled/Expired/Disputed. No status may revert. **BR-HND-06:** Either party may cancel until the handover is Completed. Cancellations within {X} hours of a confirmed appointment are recorded as late cancellations. **BR-HND-07:** Once Completed, a transaction cannot be edited or deleted, only viewed.

3.3 Accept / Reject Handover Request

Primary Actor(s):	Recycler	**Secondary Actor(s):**	User
Description:	Allows an authenticated Recycler to review incoming handover requests and either accept them, which locks the collection time slot, or reject them, in order to respond to Users and plan the collection schedule.		
Preconditions:	**1.** The Recycler is authenticated and has an active session with role Recycler. **2.** At least one Pending handover request addressed to the Recycler exists. **3.** The Recycler is on the Handover Requests screen.		
Postconditions:	**- Accept:** The request status is set to Accepted, the collection time slot is locked, the User's contact details are revealed to the Recycler, and the User is notified with MSG-HND-02. **- Reject:** The request status is set to Rejected with a recorded reason and the User is notified so that another facility can be chosen. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Recycler.		
Normal Flow:	**Accept Request** **NF-01:** The Recycler selects "Handover Requests". **NF-02:** The system displays the incoming requests (sender name, distance, waste category, estimated weight, proposed time, attached photos). **NF-03:** The Recycler selects a request to view its details. **NF-04:** The Recycler clicks "Accept". **NF-05:** The system updates the status to Accepted, locks the collection time slot, reveals the User's contact details, and then notifies the User (MSG-HND-02). **Reject Request** **NF-04 (from NF-03):** The Recycler clicks "Reject". **NF-05:** The system displays a dialog to select a rejection reason. **NF-06:** The Recycler selects a reason and then clicks "Confirm Reject". **NF-07:** The system updates the status to Rejected, and then notifies the User.		
Alternative Flow:	**AF-01:** At NF-03, if the proposed time is inconvenient, the Recycler clicks "Message" to negotiate with the User before deciding (see Negotiate via Chatbox).		
Extension Flow:	**ETF-01:** At NF-05 (Reject), if the Recycler clicks "Cancel", the system closes the dialog and keeps the request in Pending status.		
Exceptions Flow:	**EF-01:** At NF-05 (Accept), if the User has cancelled the request in the meantime, the system displays MSG-HND-05 and refreshes the request list. **EF-02:** At NF-07 (Reject), if no rejection reason is selected, the system displays MSG-AUTH-02 and prevents submission. **EF-03:** At any point, if the Recycler's session has expired, the system displays MSG-NTF-01 and redirects the Recycler to the Login page.		

Business Rule	**BR-NEG-05:** The collection time slot becomes officially confirmed the moment the Recycler clicks "Accept" on the User's Handover Request — no separate confirmation step is needed. **BR-HND-02:** Handover status changes are one-way and permanent: Pending → (Negotiating) → Accepted → Scheduled → Completed, or terminal Rejected/Cancelled/Expired/Disputed. No status may revert. **BR-HND-03:** A Recycler can view and respond only to requests addressed to them. **BR-HND-08:** Full contact details are revealed only after a request is accepted.

3.4 Cancel Collection Appointment

Primary Actor(s):	Recycler	**Secondary Actor(s):**	User
Description:	Allows an authenticated Recycler to cancel a previously accepted collection appointment because of an unexpected situation, recording a clear reason, in order to inform the User promptly.		
Preconditions:	**1.** The Recycler is authenticated and has an active session with role Recycler. **2.** The handover request is in Accepted or Scheduled status and has not been completed. **3.** The Recycler is on the Collection Schedule screen.		
Postconditions:	**- Success:** The request status is set to Cancelled, the reason is recorded, and the User is notified with MSG-HND-05. A late cancellation is recorded when it falls within the configured period before the appointment. **- Failure:** If any operation fails, the appointment remains unchanged and an informative error message is displayed to the Recycler.		
Normal Flow:	**NF-01:** The Recycler goes to "Collection Schedule" and then selects the request to view its details. **NF-02:** The Recycler clicks "Cancel Appointment". **NF-03:** The system displays a warning dialog and requests a cancellation reason. **NF-04:** The Recycler selects a reason and then clicks "Confirm Cancel". **NF-05:** The system updates the status to Cancelled, notifies the User (MSG-HND-05), and then displays a success notification.		
Alternative Flow:	**AF-01:** At NF-02, if the Recycler only needs to change the collection time, the Recycler clicks "Message" to negotiate a new time (see Negotiate via Chatbox) instead of cancelling.		
Extension Flow:	**ETF-01:** At NF-04, if the Recycler clicks "Back", the system closes the dialog and the appointment remains unchanged.		
Exceptions Flow:	**EF-01:** At NF-04, if the cancellation is within {X} hours before the appointment time, the system displays MSG-HND-06 and records it as a late cancellation. **EF-02:** At NF-04, if no reason is selected, the system displays MSG-AUTH-02 and prevents submission. **EF-03:** At NF-05, if the User has cancelled or completed the request in the meantime, the system notifies the Recycler and refreshes the schedule. **EF-04:** At any point, if the Recycler's session has expired, the system displays MSG-NTF-01 and redirects the Recycler to the Login page.		

Business Rule	**BR-HND-02:** Handover status changes are one-way and permanent: Pending → (Negotiating) → Accepted → Scheduled → Completed, or terminal Rejected/Cancelled/Expired/Disputed. No status may revert. **BR-HND-06:** Either party may cancel until the handover is Completed. Cancellations within {X} hours of a confirmed appointment are recorded as late cancellations.

3.5 Negotiate via Chatbox

Primary Actor(s):	User, Recycler	**Secondary Actor(s):**	
Description:	Allows the User and the Recycler to exchange real-time messages and images, and to submit price or quantity counter-offers, within a specific handover request in order to clarify logistics, adjust the terms, or agree on an alternative collection time.		
Preconditions:	**1.** The User or Recycler is authenticated and has an active session with the corresponding role. **2.** A handover request exists between the User and the Recycler (Pending, Negotiating, Accepted, or Completed). **3.** The User or Recycler is on the Request Detail screen.		
Postconditions:	**- Success:** Messages and image attachments are delivered in real time and permanently stored in the communication log of the request. When a counter-offer is accepted, the agreed terms are stored as an immutable snapshot. **- Failure:** If any operation fails, the system data remains unchanged, undelivered messages are not persisted, and an in-line retry indicator is displayed.		
Normal Flow:	**NF-01:** The sender (User or Recycler) taps the "Chat / Message" icon on the Request Detail screen. **NF-02:** The system displays the chronological conversation history and the request summary. **NF-03:** The sender types a message or attaches an image, and then taps "Send". **NF-04:** The system validates the content, uploads the attachment to Cloudflare R2 storage, delivers the message in real time, stores it, and finally displays it with a timestamp. **NF-05:** The other party receives the message with an in-app alert (MSG-NEG-02) and can reply by repeating the same steps.		
Alternative Flow:	**AF-01:** At NF-03, the sender submits a structured price or quantity counter-offer; when the other party accepts it, the system registers the agreed terms and displays MSG-NEG-03.		
Extension Flow:	**ETF-01:** At NF-03, if the sender taps "Back", the system closes the chat view and returns to the Request Detail screen; the conversation history remains saved.		

Exceptions Flow:	**EF-01:** At NF-04, if the network connection drops, the system marks the message with a "Tap to retry" indicator and does not persist it. **EF-02:** At NF-03, if an attachment is in an unsupported format or exceeds the size limit, the system displays MSG-CLS-01 or MSG-CLS-02 and rejects the attachment. **EF-03:** At NF-03, if the message exceeds the maximum length, the system displays MSG-NTF-03 and prevents sending. **EF-04:** At AF-01, if the request is already Accepted, Rejected, Cancelled, or Expired, the system blocks the counter-offer while the chat remains available. **EF-05:** At any point, if the session has expired, the system displays MSG-NTF-01 and redirects the sender to the Login page.
Business Rule	**BR-NEG-01:** Negotiation messages exist only within a specific handover request and are visible only to its User and Recycler (and assigned Staff during a dispute). The full message history is retained and never deleted, so it can serve as evidence if a dispute arises. **BR-NEG-02:** Making or changing a price/quantity offer is allowed only while the request is Pending or Negotiating, and is locked once it is Accepted, Rejected, Cancelled, or Expired. The chat channel itself stays open for both parties even after the handover is Completed. **BR-NEG-03:** Only the latest offer is valid; each counter-offer supersedes the previous one. Upon acceptance, the agreed terms (price, quantity, time) are stored as an immutable snapshot. **BR-NEG-04:** The User proposes their desired collection time slot directly inside the Handover Request when creating it. If the slot doesn't work for the Recycler, the two parties negotiate a new time through the chat. **BR-NEG-05:** The collection time slot becomes officially confirmed the moment the Recycler clicks "Accept" on the User's Handover Request — no separate confirmation step is needed.

4. Handover Completion & Rating

4.1 Create Handover Receipt

Primary Actor(s):	Recycler	**Secondary Actor(s):**	User
Description:	Allows an authenticated Recycler to record the actual collected weight, upload proof photos, and submit a digital handover receipt after the physical collection, in order to request the User's confirmation. Payment for the waste is settled directly between the two parties outside the system.		
Preconditions:	**1.** The Recycler is authenticated and has an active session with role Recycler. **2.** The handover request is in Accepted or Scheduled status and the waste has been weighed on site. **3.** The Recycler is on the Request Detail screen.		
Postconditions:	**- Success:** The handover receipt is created with the confirmed weights and proof photos, the request status is set to Awaiting User Confirmation, and the User is notified to confirm the receipt. **- Failure:** If any operation fails, the system data remains unchanged, the request status is retained, and an informative error message is displayed to the Recycler.		

Normal Flow:	**NF-01:** The Recycler opens the request and then clicks "Create Handover Receipt". **NF-02:** The system displays the receipt form with the scrap list and weight inputs. **NF-03:** The Recycler enters the actual weight (kg) for each waste category, and then uploads the proof photos. **NF-04:** The system validates the inputs, calculates the total weight and reference value, and then displays a preview. **NF-05:** The Recycler verifies the figures with the User and then clicks "Submit Receipt". **NF-06:** The system saves the receipt, sets the request status to Awaiting User Confirmation, notifies the User, and then displays a success notification.
Alternative Flow:	**AF-01:** At NF-03, if extra scrap items that were not listed in the request are present, the Recycler adds them from the Recycler's accepted catalog.
Extension Flow:	**ETF-01:** At NF-05, if the parties disagree, the Recycler clicks "Cancel", the receipt is not submitted, and the request status is retained.
Exceptions Flow:	**EF-01:** At NF-04, if the actual weight deviates from the agreed weight by more than {X}%, the system displays MSG-HND-04. **EF-02:** At NF-04, if a required weight or proof photo is missing, the system displays MSG-AUTH-02 and prevents submission. **EF-03:** At NF-03, if a proof photo is in an unsupported format or exceeds the size limit, the system displays MSG-CLS-01 or MSG-CLS-02 and rejects the photo. **EF-04:** At any point, if the Recycler's session has expired, the system displays MSG-NTF-01 and redirects the Recycler to the Login page.
Business Rule	**BR-HND-04:** A handover is Completed only when the Recycler confirms with a handover photo and the User confirms the receipt. **BR-HND-05:** Final settlement uses the Recycler-confirmed actual quantity and the agreed unit price. If the actual quantity deviates from the agreed one by more than {X}%, the User may accept or raise a dispute before settlement. Payment of the settlement amount happens directly between the Recycler and the User (e.g., cash or bank transfer at the time of handover) — the system only records the figures, it does not process or hold the money. **BR-HND-07:** Once Completed, a transaction cannot be edited or deleted, only viewed.

4.2 Confirm Handover Receipt

Primary Actor(s):	User	**Secondary Actor(s):**	Recycler
Description:	Allows an authenticated User to review the digital receipt created by the Recycler (measured weights, scale photos, unit prices, and total amount) and confirm it after receiving payment directly from the Recycler outside the system, in order to complete the transaction and earn Eco-points.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The Recycler has submitted the handover receipt and the request is in Awaiting User Confirmation status. **3.** The User is on the Receipt Confirmation screen.		

Postconditions:	**- Success:** The request status is set to Completed and the transaction is permanently locked. Eco-points are credited to both the User and the Recycler, the Recycler is notified, and the rating form for the Recycler is opened. **- Failure:** If any operation fails or the receipt is disputed, the system data remains unchanged, the status is not changed, no points are awarded, and an informative error message is displayed to the User.
Normal Flow:	**NF-01:** The User receives a push notification or opens a request in "Awaiting Your Confirmation" status. **NF-02:** The system displays the digital receipt (actual weights, scale photos, agreed unit prices, total amount). **NF-03:** The User checks the recorded weights and photos against the goods handed over. **NF-04:** The User receives the payment directly from the Recycler (cash or direct bank transfer outside the platform). **NF-05:** The User taps "Confirm Receipt & Complete". **NF-06:** The system updates the status to Completed, credits Eco-points to the User and the Recycler (MSG-GAM-01), displays MSG-HND-03, and then opens the Rate Recycler form.
Alternative Flow:	**AF-01:** At NF-03, if a weight discrepancy is found, the User requests a re-weighing or coordinates a receipt adjustment with the Recycler via Negotiate via Chatbox before confirming. **AF-02:** At NF-03, if an unresolved disagreement exists or the payment is withheld, the User taps "Dispute Receipt" and the system opens the Submit Complaint form.
Extension Flow:	**ETF-01:** At NF-05, if the User taps "Back" without confirming, the system returns to the request list and the request remains in Awaiting User Confirmation status.
Exceptions Flow:	**EF-01:** At NF-03, if the actual weight deviates from the agreed weight by more than {X}%, the system displays MSG-HND-04. **EF-02:** At NF-06, if the completion fails because of a network or system error, the receipt remains in Awaiting User Confirmation status and no points are awarded. **EF-03:** At NF-01, if the User neither confirms nor disputes the receipt within 48 hours, the scheduled background job automatically completes the receipt in accordance with the system policy. **EF-04:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.

Business Rule	**BR-HND-04:** A handover is Completed only when the Recycler confirms with a handover photo and the User confirms the receipt. **BR-HND-05:** Final settlement uses the Recycler-confirmed actual quantity and the agreed unit price. If the actual quantity deviates from the agreed one by more than {X}%, the User may accept or raise a dispute before settlement. Payment of the settlement amount happens directly between the Recycler and the User (e.g., cash or bank transfer at the time of handover) — the system only records the figures, it does not process or hold the money. **BR-HND-07:** Once Completed, a transaction cannot be edited or deleted, only viewed. **BR-GAM-01:** Points are awarded only for verified events, such as a Completed handover (based on confirmed category and quantity) — credited to both the completing User and the completing Recycler — and only once per event per role. No points for cancelled/disputed transactions; points are reversed if a dispute outcome reverses the transaction.

4.3 Rate Recycler

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to submit a 1 to 5 star rating and an optional review for a Recycler after a Completed handover, in order to give feedback on the service quality and help other Users choose reliable facilities.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The related handover request is in Completed status. **3.** The User is on the Rate Recycler form, opened after confirming the receipt or from the completed transaction in the transaction history.		
Postconditions:	**- Success:** The review is saved, the Recycler's average rating is recalculated immediately, and MSG-RVW-01 is displayed. **- Failure:** If any operation fails, the system data remains unchanged, the review is discarded, existing ratings are not affected, and an informative error message is displayed to the User.		
Normal Flow:	**NF-01:** The system displays the "Rate Recycler" form. **NF-02:** The User selects a star rating (1 to 5). **NF-03:** The User optionally enters feedback about punctuality, honesty in weighing, or professionalism. **NF-04:** The User taps "Submit Review". **NF-05:** The system validates the input, saves the review, recalculates the Recycler's average rating, displays MSG-RVW-01, and then closes the form.		
Alternative Flow:	**AF-01:** At NF-01, if a review already exists for the transaction, the system displays the submitted review in read-only mode and does not allow a new one.		
Extension Flow:	**ETF-01:** At NF-01, if the User taps "Skip", the system closes the form without saving any review.		

Exceptions Flow:	**EF-01:** At NF-04, if no star rating is selected, the system highlights the stars and prevents submission. **EF-02:** At NF-01, if the transaction is not in Completed status, the system blocks access and displays MSG-RVW-02. **EF-03:** At NF-05, if the review cannot be saved because of a system error, the system displays an error message and keeps the form data for retry. **EF-04:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-RVW-01:** A rating/review can be submitted only by a party of a Completed transaction, once per party per transaction, within {N} days after completion. **BR-RVW-02:** A rating is 1–5 stars with an optional comment. Once submitted, it cannot be edited, but the author may delete it entirely (see BR-RVW-03). **BR-RVW-03:** The review's author (User or Recycler, whichever submitted it) may delete their own review at any time; this is permanent and cannot be restored. Exception: Staff/Admin may hide or delete a review if it has been reported and found to violate community standards (e.g., offensive language, fabricated information). Aside from the author and Staff/Admin under this exception, no other party — including the one being rated — can delete or hide a review. The displayed average rating recalculates immediately after any deletion/hiding.

5. Complaint Resolution

5.1 Submit Complaint

Primary Actor(s):	User	**Secondary Actor(s):**	Staff
Description:	Allows an authenticated User to file a formal complaint about a handover transaction (e.g., inaccurate weight, unpaid amount, misconduct) with a description and photo evidence, in order to start Staff arbitration.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The handover request is in Completed status, or the User rejected the receipt during Confirm Handover Receipt. **3.** The complaint window has not expired ({N} days after completion) and no unresolved complaint exists for the transaction. **4.** The User is on the Transaction Details screen.		
Postconditions:	**- Success:** A complaint ticket is created in Open status, the transaction is tagged "Under Dispute", and the complaint is routed to the Staff queue. MSG-DSP-01 is displayed. **- Failure:** If any operation fails, the system data remains unchanged, no complaint is created, the transaction tag remains unchanged, and an informative error message is displayed to the User.		
Normal Flow:	**NF-01:** The User taps "Submit Complaint" on the Transaction Details screen. **NF-02:** The system displays the complaint form with reason categories (Inaccurate Weight, Unpaid Amount, Misconduct, Other), a description field, and a photo evidence upload area. **NF-03:** The User selects a category, enters the incident description, attaches supporting photos, and then taps "Submit Complaint". **NF-04:** The system validates the data, creates the complaint ticket in Open status, tags the transaction "Under Dispute", routes it to the Staff queue, and then displays MSG-DSP-01.		

Alternative Flow:	**AF-01:** N/A
Extension Flow:	**ETF-01:** At NF-02, if the User taps "Cancel", the system closes the form and returns to the Transaction Details screen without creating a complaint.
Exceptions Flow:	**EF-01:** At NF-04, if the reason category or description is missing, the system highlights the required fields and prevents submission. **EF-02:** At NF-01, if the complaint window ({N} days after completion) has expired, the system blocks the action and displays MSG-DSP-03. **EF-03:** At NF-01, if an open complaint already exists for the transaction, the system displays MSG-DSP-04 and navigates to the existing complaint. **EF-04:** At NF-03, if an attached photo is in an unsupported format or exceeds the size limit, the system displays MSG-CLS-01 or MSG-CLS-02 and rejects the photo. **EF-05:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-DSP-01:** Only a User or Recycler who is a party to the transaction can raise a dispute, within {N} days after completion, with at most one open dispute per transaction. **BR-DSP-02:** A dispute must include a reason category and a description; supporting photos are optional. **BR-DSP-04:** Only Staff can resolve a dispute, and every decision must record an outcome (e.g., the User should receive a full/partial refund from the Recycler, or the dispute is dismissed) and a rationale. Since payment happens directly between the two parties outside the system, the outcome is a binding recommendation communicated to both for them to carry out themselves — the system has no mechanism to automatically transfer or refund money. **BR-DSP-06:** Dispute data (messages, photos, transaction details) is accessible only to the two parties and the assigned Staff.

5.2 Handle Transaction Complaints

Primary Actor(s):	Staff	**Secondary Actor(s):**	User, Recycler
Description:	Allows an authenticated Staff member to receive and mediate complaints about handover transactions (e.g., weight differences, no-shows), evaluate the evidence, and issue a binding recommendation, in order to resolve disputes fairly between the parties.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** At least one complaint is waiting for resolution. **3.** The Staff is on the Complaint Management screen.		
Postconditions:	**- Success:** The complaint is closed with a recorded outcome and rationale, both parties are notified with MSG-DSP-02, and the decision is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the complaint remains open, and an informative error message is displayed to the Staff.		

Normal Flow:	**NF-01:** The Staff selects "Transaction Complaints". **NF-02:** The system displays the list of complaints (ID, parties, type, time). **NF-03:** The Staff opens a complaint. **NF-04:** The system displays the evidence, the handover receipt, and the chat history. **NF-05:** The Staff evaluates the case and then selects a resolution (Accept / Reject). **NF-06:** The Staff enters the ruling summary and then clicks "Issue Conclusion". **NF-07:** The system validates the input, closes the complaint, writes an audit log entry, and then notifies both parties (MSG-DSP-02).
Alternative Flow:	**AF-01:** At NF-05, if more information is required, the Staff updates the status to "Requesting More Info" and the system notifies the parties.
Extension Flow:	**ETF-01:** At NF-04 or NF-06, if the Staff clicks "Back to List", the system returns to the complaint list without saving any decision.
Exceptions Flow:	**EF-01:** At NF-06, if the ruling summary is blank, the system displays MSG-AUTH-02 and prevents submission. **EF-02:** At NF-07, if the complaint has already been resolved by another Staff member, the system notifies the Staff and refreshes the list. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-DSP-04:** Only Staff can resolve a dispute, and every decision must record an outcome (e.g., the User should receive a full/partial refund from the Recycler, or the dispute is dismissed) and a rationale. Since payment happens directly between the two parties outside the system, the outcome is a binding recommendation communicated to both for them to carry out themselves — the system has no mechanism to automatically transfer or refund money. **BR-DSP-06:** Dispute data (messages, photos, transaction details) is accessible only to the two parties and the assigned Staff. **BR-GAM-01:** Points are awarded only for verified events, such as a Completed handover (based on confirmed category and quantity) — credited to both the completing User and the completing Recycler — and only once per event per role. No points for cancelled/disputed transactions; points are reversed if a dispute outcome reverses the transaction. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

6. Gamification & Rewards

6.1 Daily Check-in

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to check in once per day in order to earn Eco-points, maintain a consecutive-day streak, and receive streak bonuses.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The User has not checked in today (the daily check-in resets at 00:00 ICT). **3.** The User is on the Home dashboard.		

Postconditions:	**- Success:** A daily check-in record is created, the User's point balance is credited, the streak count is updated, and MSG-GAM-08 is displayed. **- Failure:** If any operation fails, the system data remains unchanged, no points are awarded, the streak is not changed, and an informative error message is displayed to the User.
Normal Flow:	**NF-01:** The User opens the "Daily Check-in" widget on the Home dashboard. **NF-02:** The system displays the 7-day check-in calendar, the current streak count, and today's reward preview. **NF-03:** The User taps "Check-in Now". **NF-04:** The system verifies the eligibility, records the check-in, calculates the points (base reward plus streak bonus), credits the points to the User's balance, updates the streak, and then displays MSG-GAM-08.
Alternative Flow:	**AF-01:** At NF-02, if the User missed the check-in yesterday, the system informs the User that the streak will restart from Day 1 before the check-in is performed.
Extension Flow:	**ETF-01:** At NF-02, if the User closes the widget without tapping "Check-in Now", the system returns to the Home dashboard without creating a check-in record.
Exceptions Flow:	**EF-01:** At NF-03, if the User has already checked in today (e.g., a duplicate tap), the system disables the check-in button and displays a message that the User has already checked in today. **EF-02:** At NF-04, if the User has reached the daily point cap for non-transaction actions, the system displays MSG-GAM-05. **EF-03:** At NF-04, if a database write error occurs, the system rolls back the point credit and displays an error message. **EF-04:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-GAM-01:** Points are awarded only for verified events, such as a Completed handover (based on confirmed category and quantity) — credited to both the completing User and the completing Recycler — and only once per event per role. No points for cancelled/disputed transactions; points are reversed if a dispute outcome reverses the transaction. **BR-GAM-02:** Daily point-earning from non-transactional actions is capped at {N} per user/recycler. **BR-GAM-09:** Each User/Staff account may check in once per day. The reward points increase with the length of the consecutive-day streak, and the streak automatically resets to Day 1 if a day is missed.

6.2 Staff Daily Check-in

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to check in and check out once per day in order to log working hours and build an attendance streak.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** The Staff has not checked in today. **3.** The Staff is on the Staff Dashboard.		

Postconditions:	**- Success:** The shift start and end times are recorded, the working hours are calculated, and the attendance streak is updated. MSG-GAM-08 is displayed at check-in. **- Failure:** If any operation fails, the system data remains unchanged, no shift record is created or changed, and an informative error message is displayed to the Staff.
Normal Flow:	**NF-01:** The Staff views the "Shift Check-in" widget on the Staff Dashboard. **NF-02:** The system displays the current time and the "Check-in" button. **NF-03:** The Staff clicks "Check-in". **NF-04:** The system records the start time, updates the status, displays MSG-GAM-08, and then switches the button to "Check-out". **NF-05:** At the end of the working day, the Staff clicks "Check-out". **NF-06:** The system records the end time, calculates the working hours, and then marks the shift as complete.
Alternative Flow:	**AF-01:** At NF-02, the Staff clicks "View History" to review the monthly attendance streaks.
Extension Flow:	**ETF-01:** N/A
Exceptions Flow:	**EF-01:** At NF-03, if the Staff has already checked in today, the system prevents a duplicate check-in and displays a message that the check-in has already been recorded. **EF-02:** At NF-05, if no check-in exists for the day, the system disables the check-out action. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-GAM-09:** Each User/Staff account may check in once per day. The reward points increase with the length of the consecutive-day streak, and the streak automatically resets to Day 1 if a day is missed.

6.3 Redeem Points for Gifts

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to redeem accumulated Eco-points for rewards in the Gift Store (e.g., e-vouchers, eco-friendly items) in order to benefit from the User's green activities.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** At least one active gift with available stock exists in the Gift Store. **3.** The User is on the Gift Store screen.		
Postconditions:	**- Success:** The required points are deducted from the User's balance and the gift stock is reduced atomically. A redemption record is created and MSG-GAM-02 is displayed. **- Failure:** If any operation fails, the system data remains unchanged, the point balance and gift stock are not changed, and an informative error message is displayed to the User.		

Normal Flow:	**NF-01:** The User navigates to the "Gift Store" tab. **NF-02:** The system displays the current point balance and the reward catalog (image, name, point cost, stock). **NF-03:** The User selects a gift and then taps "Redeem Reward". **NF-04:** The system displays a confirmation dialog with the point cost and the remaining balance. **NF-05:** The User taps "Confirm Redemption". **NF-06:** The system validates the point balance and stock, deducts the points and reduces the stock in one atomic transaction, creates the redemption record, and then displays MSG-GAM-02.
Alternative Flow:	**AF-01:** At NF-02, the User filters the rewards by type (e-vouchers or physical items) or sorts them by point cost.
Extension Flow:	**ETF-01:** At NF-04, if the User taps "Cancel", the system closes the dialog without deducting any points.
Exceptions Flow:	**EF-01:** At NF-06, if the point balance is insufficient, the system displays MSG-GAM-03 and halts the redemption. **EF-02:** At NF-06, if the gift has run out of stock in the meantime, the system displays MSG-GAM-04 and refreshes the catalog. **EF-03:** At NF-06, if the atomic transaction fails, the system rolls back all changes without deducting any points. **EF-04:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-GAM-03:** Point balance cannot be negative. A redemption deducts points and issues the voucher atomically. **BR-GAM-04:** A reward can be redeemed only while active and in stock; each redemption reduces stock atomically. **BR-GAM-05:** A voucher is single-use, bound to the redeeming account, and expires on its expiry date.

6.4 Upgrade Virtual Tree

Primary Actor(s):	User	**Secondary Actor(s):**	
Description:	Allows an authenticated User to spend Eco-points to grow the User's Virtual Tree through its development stages in order to unlock environmental Eco Facts and achievement badges.		
Preconditions:	**1.** The User is authenticated and has an active session with role User. **2.** The User's Virtual Tree has not reached the maximum level. **3.** The User is on the Virtual Tree screen.		
Postconditions:	**- Success:** The upgrade points are deducted, the tree level increases by one, the corresponding Eco Fact is permanently unlocked, and MSG-GAM-06 and MSG-GAM-07 are displayed. **- Failure:** If any operation fails, the system data remains unchanged, the point balance and tree level are not changed, and an informative error message is displayed to the User.		

Normal Flow:	**NF-01:** The User selects the "My Garden / Virtual Tree" icon on the home screen. **NF-02:** The system displays the tree at its current level, the progress toward the next stage, the required upgrade points, and the current point balance. **NF-03:** The User taps "Water & Upgrade Tree". **NF-04:** The system displays the upgrade cost confirmation dialog. **NF-05:** The User taps "Confirm Upgrade". **NF-06:** The system verifies the point balance → deducts the points → raises the tree level by one → unlocks the corresponding Eco Fact → updates the tree graphic → displays MSG-GAM-06 and MSG-GAM-07.
Alternative Flow:	**AF-01:** At NF-02, the User taps "Knowledge Library" to browse the Eco Facts that have already been unlocked without upgrading the tree.
Extension Flow:	**ETF-01:** At NF-04, if the User taps "Later", the system closes the dialog without any change.
Exceptions Flow:	**EF-01:** At NF-06, if the point balance is insufficient, the system displays MSG-GAM-03 and disables the upgrade button. **EF-02:** At NF-02, if the tree has reached the maximum level, the system hides the upgrade button and displays a message that the tree has reached full maturity. **EF-03:** At NF-06, if the level update fails because of a system error, the system preserves the User's points and displays an error message. **EF-04:** At any point, if the User's session has expired, the system displays MSG-NTF-01 and redirects the User to the Login page.
Business Rule	**BR-GAM-03:** Point balance cannot be negative. A redemption deducts points and issues the voucher atomically. **BR-GAM-06:** Virtual Tree stage progression is one-way. Care actions are limited to {N} per day; an unlocked Eco Fact stays unlocked. **BR-GAM-08:** Earning points and redeeming them for vouchers is available to both the User and Recycler roles. Virtual Tree care, Eco Facts, and the contribution leaderboard remain scoped to the User role only.

6.5 Manage Eco Fact Content

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to manage the environmental knowledge facts shown in the Virtual Tree feature and link them to the tree unlock levels, in order to keep the educational content accurate and engaging.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** The Staff is on the Gamification Content Management screen.		
Postconditions:	**- View:** The system displays the latest list of Eco Facts. **- Add:** The new Eco Fact is saved and linked to the selected tree level. **- Edit:** The updated Eco Fact content is saved and reflected immediately. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.		

Normal Flow:	**View Eco Facts** **NF-01:** The Staff selects "Eco Fact Knowledge Base". **NF-02:** The system displays the list of Eco Facts (Code, Title, Tree Level, Date, Status). **Add Eco Fact** **NF-03:** The Staff clicks "Add New Eco Fact". **NF-04:** The system displays the form (Title, Description, Image, Tree Level). **NF-05:** The Staff enters the data and selects the tree level, and then clicks "Save". **NF-06:** The system validates the input, saves the Eco Fact, and then refreshes the list. **Edit Eco Fact** **NF-03 (from NF-02):** The Staff clicks "Edit" on an Eco Fact. **NF-04:** The system displays the form pre-filled with the current data. **NF-05:** The Staff edits the content and then clicks "Save Changes". **NF-06:** The system validates the input, updates the Eco Fact, and then refreshes the list.
Alternative Flow:	**AF-01:** At NF-02, the Staff filters the Eco Facts by tree level.
Extension Flow:	**ETF-01:** At NF-05 (Add/Edit), if the Staff clicks "Cancel", the system discards the changes and returns to the list.
Exceptions Flow:	**EF-01:** At NF-06, if a required field is missing, the system displays MSG-AUTH-02 and prevents saving. **EF-02:** At NF-06, if a text field exceeds its maximum length, the system displays MSG-NTF-03 and prevents saving. **EF-03:** At NF-06, if the uploaded image is in an unsupported format or exceeds the size limit, the system displays MSG-CLS-01 or MSG-CLS-02 and rejects the image. **EF-04:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-GAM-06:** Virtual Tree stage progression is one-way. Care actions are limited to {N} per day; an unlocked Eco Fact stays unlocked. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status.

6.6 Manage Gift Inventory

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to manage the reward catalog, point costs, and stock levels, and to suspend unavailable gifts, in order to keep the Gift Store accurate for Users.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** The Staff has been granted the reward inventory permission. **3.** The Staff is on the Gift & Voucher Management screen.		

Postconditions:	**- View:** The system displays the latest gift inventory. **- Add:** The new gift is saved to the catalog. **- Restock/Suspend:** The available stock or status is updated; out-of-stock gifts cannot be redeemed (MSG-GAM-04). **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.
Normal Flow:	**View Inventory** **NF-01:** The Staff selects "Gift & Voucher Inventory". **NF-02:** The system displays the list of gifts (Name, Type, Points, Stock, Status). **Add Gift** **NF-03:** The Staff clicks "Add New Gift". **NF-04:** The system displays the gift creation form. **NF-05:** The Staff enters the gift details and point cost → clicks "Save". **NF-06:** The system validates the input → saves the gift → refreshes the list. **Restock / Suspend Gift** **NF-03 (from NF-02):** The Staff clicks "Edit" or "Restock" on a gift. **NF-04:** The system displays a dialog to add stock or toggle the gift status. **NF-05:** The Staff enters the quantity or selects the status → clicks "Update Inventory". **NF-06:** The system validates the input → updates the stock or status → refreshes the list.
Alternative Flow:	**AF-01:** At NF-02, the Staff clicks "View Redemption History" to review the redemptions of a gift.
Extension Flow:	**ETF-01:** At NF-05 (Add/Restock/Suspend), if the Staff clicks "Cancel", the system discards the changes and keeps the inventory unchanged.
Exceptions Flow:	**EF-01:** At NF-06, if required information is missing, the system displays MSG-AUTH-02 and prevents saving. **EF-02:** At NF-06, if the point cost or stock quantity is not a valid non-negative number, the system highlights the invalid field and prevents saving. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-GAM-04:** A reward can be redeemed only while active and in stock; each redemption reduces stock atomically. **BR-GAM-05:** A voucher is single-use, bound to the redeeming account, and expires on its expiry date. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status.

6.7 Manage Gamification Config

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to configure the gamification point rates per kilogram of waste, the daily non-transaction point cap, the daily check-in streak bonuses, and the Virtual Tree stage thresholds, in order to keep the reward system balanced and prevent point abuse.		

Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has system configuration privileges and the Gamification module is enabled. **3.** The Admin is on the Gamification Configuration screen.
Postconditions:	**- Success:** The updated gamification rules are saved and take effect immediately for subsequent User activities. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the gamification parameters are not changed, and an informative error message is displayed to the Admin.
Normal Flow:	**NF-01:** The Admin navigates to the "Gamification Configuration" section. **NF-02:** The system displays the current parameters (point rates per kg by waste type, daily non-transaction cap {N}, check-in streak multipliers, Virtual Tree stage threshold table). **NF-03:** The Admin adjusts the reward point values per recyclable material category. **NF-04:** The Admin configures the daily Virtual Tree care limit and maps the Eco Facts to the growth stages. **NF-05:** The Admin clicks "Save Configuration". **NF-06:** The system validates the values (points are non-negative and level thresholds are strictly increasing) → saves the configuration → writes an audit log entry → displays a success notification.
Alternative Flow:	**AF-01:** At NF-05, if the Admin clicks "Reset to Defaults", the system asks for confirmation and restores the baseline parameters.
Extension Flow:	**ETF-01:** At NF-05, if the Admin clicks "Cancel" or navigates away, the system discards the unsaved changes.
Exceptions Flow:	**EF-01:** At NF-06, if a point value is negative or the stage thresholds are not strictly increasing, the system highlights the invalid inputs and prevents saving. **EF-02:** At NF-05, if a non-Admin account attempts to change the configuration, the system denies the request and displays MSG-ADM-04. **EF-03:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.

Business Rule	**BR-GAM-01:** Points are awarded only for verified events, such as a Completed handover (based on confirmed category and quantity) — credited to both the completing User and the completing Recycler — and only once per event per role. No points for cancelled/disputed transactions; points are reversed if a dispute outcome reverses the transaction. **BR-GAM-02:** Daily point-earning from non-transactional actions is capped at {N} per user/recycler. **BR-GAM-06:** Virtual Tree stage progression is one-way. Care actions are limited to {N} per day; an unlocked Eco Fact stays unlocked. **BR-GAM-08:** Earning points and redeeming them for vouchers is available to both the User and Recycler roles. Virtual Tree care, Eco Facts, and the contribution leaderboard remain scoped to the User role only. **BR-GAM-09:** Each User/Staff account may check in once per day. The reward points increase with the length of the consecutive-day streak, and the streak automatically resets to Day 1 if a day is missed. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

6.8 Configure Leaderboard Cycles

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to configure the leaderboard calculation cycle (Weekly, Monthly, or Quarterly) and the ranking criteria, in order to run a fair and consistent contribution ranking.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has leaderboard governance privileges. **3.** The Admin is on the Leaderboard Configuration screen.		
Postconditions:	**- Success:** The leaderboard cycle and ranking criteria are updated and the scheduled aggregation job is synchronized. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the leaderboard configuration is not changed, and an informative error message is displayed to the Admin.		
Normal Flow:	**NF-01:** The Admin navigates to the "Leaderboard Cycles Configuration" page. **NF-02:** The system displays the active cycle mode (Weekly, Monthly, Quarterly) and the current ranking metric. **NF-03:** The Admin selects the ranking metric (Total Recycled Waste Weight or Completed Handover Count without disputes). **NF-04:** The Admin chooses the evaluation cycle duration (Weekly / Monthly / Quarterly). **NF-05:** The Admin sets the top contributor spotlight option (only display names are shown). **NF-06:** The Admin clicks "Apply Settings". **NF-07:** The system validates the parameters (only Completed, non-disputed transactions are counted) → synchronizes the scheduled aggregation job → writes an audit log entry → displays a confirmation notification.		

Alternative Flow:	**AF-01:** At NF-02, if the Admin clicks "Trigger Recalculation Now", the system immediately recalculates the ranking from the completed records using the current criteria.
Extension Flow:	**ETF-01:** At NF-06, if the Admin clicks "Cancel" or closes the settings window, the system discards the unsaved changes.
Exceptions Flow:	**EF-01:** At NF-06, if a required setting (ranking metric or cycle) is not selected, the system highlights the missing field and prevents saving. **EF-02:** At NF-07, if the synchronization of the scheduled job fails, the system keeps the previous configuration and displays an error notification. **EF-03:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-GAM-07:** The leaderboard counts only Completed, non-disputed transactions per the metric Admin configures, and shows only display names. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

7. In-System Services & Payments

7.1 Purchase In-System Services via VietQR

Primary Actor(s):	User, Recycler	**Secondary Actor(s):**	VietQR Service
Description:	Allows an authenticated User or Recycler to purchase in-system service packages (e.g., ad-free experience, Green Tick badge, map priority) by scanning a dynamic VietQR code, in order to activate the purchased privileges automatically after the payment is verified.		
Preconditions:	**1.** The User or Recycler is authenticated and has an active session with the corresponding role. **2.** The VietQR Service is operational. **3.** The User or Recycler is on the Service Packages screen.		
Postconditions:	**- Success:** The payment order is verified as paid, the purchased service package is activated on the account, a confirmation is sent, and MSG-PAY-02 is displayed. **- Failure:** If any operation fails or the payment window expires, the system data remains unchanged, no service is activated, the QR code is invalidated (MSG-PAY-03), and an informative message is displayed.		

Normal Flow:	**NF-01:** The User or Recycler selects "EcoLink Services". **NF-02:** The system displays the available packages (name, benefits, duration, price). **NF-03:** The User or Recycler selects a package → clicks "Register to Pay". **NF-04:** The system creates a service order with a unique reference code → requests a dynamic VietQR code from the VietQR Service → displays the QR code with a 15-minute countdown and MSG-PAY-01. **NF-05:** The User or Recycler scans the QR code with a banking application and authorizes the transfer. **NF-06:** The system receives the payment notification → validates the signature, the order reference, and the amount. **NF-07:** The system marks the order as Paid → activates the purchased package → displays MSG-PAY-02 → sends an in-app confirmation.
Alternative Flow:	**AF-01:** At NF-04, if only one device is used, the User or Recycler taps "Save QR to Photos" or "Copy Bank Info & Memo" to complete the payment in the banking application.
Extension Flow:	**ETF-01:** At NF-04, if the User or Recycler taps "Cancel Order", the system voids the pending order and the QR code, and returns to the package list without any charge.
Exceptions Flow:	**EF-01:** At NF-04, if the 15-minute countdown expires without a payment, the system disables the QR code and displays MSG-PAY-03. **EF-02:** At NF-06, if the transferred amount does not match the package price, the system does not activate the package and flags the order for review in Manage Financial Reconciliation. **EF-03:** At NF-04, if the VietQR Service does not respond within 10 seconds, the system displays an error message that the payment gateway is temporarily unavailable and suggests retrying later. **EF-04:** At any point, if the session has expired, the system displays MSG-NTF-01 and redirects the User or Recycler to the Login page.
Business Rule	**BR-PAY-01:** Payments for in-system services (e.g., ad removal, verified badges, feature upgrades) are processed through dynamic VietQR codes linked to a unique order reference and are marked successful only after a verified payment confirmation (valid signature and matching amount). **BR-PAY-02:** In-system payment events are handled idempotently, so that each VietQR payment is credited and activated exactly once. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status.

7.2 Configure Transaction Fee

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to configure the VietQR pricing and fee schedule of the in-system upgrade services (e.g., ad removal, verified badge), in order to control the prices applied to future purchases.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has financial policy privileges. **3.** The Admin is on the Service Fee & Upgrade Pricing screen.		

Postconditions:	**- Success:** The updated service package pricing is saved and applied to future purchases. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the active pricing is not changed, and an informative error message is displayed to the Admin.
Normal Flow:	**NF-01:** The Admin navigates to the "Service Fee & Upgrade Pricing" screen. **NF-02:** The system displays the current service packages and their prices (e.g., Ad-free Package, Recycler Verified Badge, Premium Analytics Package). **NF-03:** The Admin selects a package → updates its fixed price (VND) or promotional rate. **NF-04:** The Admin sets the package validity (e.g., 30 days, 1 year, lifetime) and the effective date. **NF-05:** The Admin clicks "Save Fee Policy". **NF-06:** The system validates the values → saves the pricing → writes an audit log entry → displays a confirmation notification.
Alternative Flow:	**AF-01:** At NF-03, the Admin clicks "Add New Package", enters the package name, promotional VietQR price, and benefits, and saves the new package.
Extension Flow:	**ETF-01:** At NF-05, if the Admin clicks "Cancel", the system discards the changes and restores the current pricing values.
Exceptions Flow:	**EF-01:** At NF-06, if the price is negative or exceeds the allowed maximum, the system highlights the invalid input and prevents saving. **EF-02:** At NF-05, if a non-Admin account attempts to change the pricing, the system denies the request and displays MSG-ADM-04. **EF-03:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-PAY-01:** Payments for in-system services (e.g., ad removal, verified badges, feature upgrades) are processed through dynamic VietQR codes linked to a unique order reference and are marked successful only after a verified payment confirmation (valid signature and matching amount). **BR-PAY-02:** In-system payment events are handled idempotently, so that each VietQR payment is credited and activated exactly once. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

7.3 Manage Financial Reconciliation

Primary Actor(s):	Admin	**Secondary Actor(s):**	VietQR Service
Description:	Allows an authorized Admin to monitor the in-system service payments made through dynamic VietQR and reconcile the revenue records of each period, in order to ensure accurate financial reporting.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has financial oversight privileges and service payments have been processed through VietQR. **3.** The Admin is on the Financial Reconciliation & Revenue screen.		

Postconditions:	**- Success:** The reconciliation period is marked as Reconciled and its financial records are locked, and a reconciliation report is generated. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the period remains open for audit, and an informative error message is displayed to the Admin.
Normal Flow:	**NF-01:** The Admin navigates to the "Financial Reconciliation & Revenue" section. **NF-02:** The system displays the monthly reconciliation periods with the total VietQR transactions, successful payments, total service revenue, and pending transactions. **NF-03:** The Admin selects a reconciliation period. **NF-04:** The system displays the payment orders of the period (Order ID, account, purchased package, transaction reference, amount in VND, payment time, match status). **NF-05:** The Admin checks the transaction ledger against the bank statement → clicks "Confirm Reconciliation Period". **NF-06:** The system marks the period as Reconciled → locks its records against modification → writes an audit log entry → displays a confirmation message.
Alternative Flow:	**AF-01:** At NF-04, if a transaction has a discrepancy (e.g., delayed notification or reference mismatch), the Admin clicks "Investigate Transaction" to view the payment notification log and manually re-verify the match. **AF-02:** At NF-04, the Admin clicks "Export Reconciliation Sheet" to download an Excel spreadsheet of the period.
Extension Flow:	**ETF-01:** At NF-05, if the Admin leaves the reconciliation screen without confirming, the period remains open and no data is changed.
Exceptions Flow:	**EF-01:** At NF-05, if unmatched or pending payment orders remain in the period, the system displays a warning and requires them to be resolved before closing the period. **EF-02:** At NF-04, if the Admin attempts to modify or delete the records of a Reconciled period, the system blocks the action. **EF-03:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-PAY-01:** Payments for in-system services (e.g., ad removal, verified badges, feature upgrades) are processed through dynamic VietQR codes linked to a unique order reference and are marked successful only after a verified payment confirmation (valid signature and matching amount). **BR-PAY-02:** In-system payment events are handled idempotently, so that each VietQR payment is credited and activated exactly once. **BR-ADM-04:** Accounts with transaction history can only be deactivated, never deleted. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

8. Account & Access Management

8.1 View Homepage

Primary Actor(s):	Guest	**Secondary Actor(s):**	N/A
Description:	Allows an unauthenticated visitor to access the public homepage, view platform highlights, browse feature overviews, read environmental articles, access public guides, and navigate to registration or login.		
Preconditions:	**1.** The visitor is not authenticated in the system. **2.** The EcoLink web application is operational.		
Postconditions:	**- Success:** The public homepage layout, promotional banners, feature overviews, and public guides are rendered successfully. **- Failure:** The system data remains unchanged, and an informative fallback notice is displayed if connectivity fails.		
Normal Flow:	**NF-01:** The Guest accesses the website root URL. **NF-02:** The system retrieves dynamic platform highlights, active banners, featured eco-articles, and renders the Homepage layout. **NF-03:** The Guest browses the feature overview cards, public waste sorting guides, or FAQs. **NF-04:** The system displays the corresponding section details and educational media content.		
Alternative Flow:	**AF-01:** At NF-01, the Guest clicks "Sign In" or "Register" in the top navigation bar. The system redirects the Guest to the Login screen or Registration screen.		
Extension Flow:	**ETF-01:** N/A		
Exceptions Flow:	**EF-01:** At NF-01, if network connectivity drops or the application server is temporarily unreachable, the system displays a fallback static notice: "Service temporarily unavailable. Please retry later."		
Business Rule	**BR-AUTH-08:** Access is role-based. Guests access only public information; each role accesses only its own functions and data.		

8.2 Register as User

Primary Actor(s):	Guest	**Secondary Actor(s):**	Google OAuth, Gmail
Description:	Allows an unauthenticated household citizen or individual to create a new personal User account using local email credentials or Google OAuth authentication, in order to participate in waste sorting and recycling handovers.		
Preconditions:	**1.** The Guest is not authenticated. **2.** The Guest is on the User Registration screen.		
Postconditions:	**- Success:** A new User account is created in the system with status Active (Google OAuth) or Pending_Verification (local registration), and confirmation is displayed. **- Failure:** Registration is halted, error notices are displayed, and no account record is created.		

Normal Flow:	**NF-01:** The Guest enters their Full Name, Email Address, Password, Confirm Password, and Phone Number → clicks "Create Account". **NF-02:** The system validates input formats, password strength (minimum 8 characters with letters and numbers), and uniqueness of the email address across all roles. **NF-03:** The system hashes the password using salted hashing, creates the account in Pending_Verification status, dispatches a verification email via Gmail, and displays MSG-AUTH-08. **NF-04:** The Guest opens the verification email and clicks the activation link. **NF-05:** The system validates the token, activates the account (Active status), and displays MSG-AUTH-09.
Alternative Flow:	**AF-01:** At NF-01, the Guest clicks "Continue with Google". The system redirects to the Google OAuth consent screen. The Guest authorizes access. The system validates the Google token and reads the email_verified claim; if email_verified is true, the system creates and activates the User account immediately, generates an active session, and redirects to the Homepage.
Extension Flow:	**ETF-01:** At NF-01, if the Guest clicks "Already have an account? Sign In", the system cancels the registration flow and navigates to the Login screen.
Exceptions Flow:	**EF-01:** At NF-01, if the Guest leaves any required field blank, the system highlights the field and displays MSG-AUTH-02. **EF-02:** At NF-01, if any input exceeds the allowed character limit, the system displays MSG-NTF-03 under the field. **EF-03:** At NF-01, if the entered email is already registered to any User account, the system prevents submission and displays an in-line message: "This email address is already in use. Please sign in or use another email." **EF-04:** At AF-01, if Google returns email_verified = false, the system creates the account in Pending_Verification status and requires the standard email verification flow before activation.
Business Rule	**BR-AUTH-01:** One email address can be linked to only one account, across both roles (User/Recycler) and both sign-up methods (local/Google). **BR-AUTH-02:** During Google sign-in/sign-up, the system reads the email_verified claim from Google's ID token/UserInfo response. If email_verified = true, the email is marked verified immediately with no separate verification step. If email_verified = false, the standard email-verification flow is required before the account is activated. **BR-AUTH-07:** Google sign-in requires the OAuth token to be verified with Google on every login. Accounts created via Google are bound to the verified Google email and default to the User role. **BR-AUTH-08:** Access is role-based. Guests access only public information; each role accesses only its own functions and data. **BR-NTF-03:** Notifications are sent only to the account's verified email; changing the email requires re-verification.

8.3 Apply for Recycler Upgrade

Primary Actor(s):	User	**Secondary Actor(s):**	Cloud Object Storage, Gmail

Description:	Allows an authenticated citizen (User) to submit facility information, GPS location coordinates, accepted scrap categories, reference pricing, and legal business licenses/identity documents from within their Profile to apply for an upgrade to a verified Recycler partner role.
Preconditions:	**1.** The User is authenticated. **2.** The User has navigated to Profile / Settings and opened the 'Register as Recycler Partner' screen.
Postconditions:	**- Success:** An upgrade application is submitted with status PENDING_VERIFY; the User remains an active citizen User, and the facility location remains hidden from public search until verified. **- Failure:** Registration is rejected, uploaded documents are discarded, and error notices are displayed.
Normal Flow:	**NF-01:** The system auto-populates the registered Full Name, Email, and Phone number from the active User profile. The User inputs facility name, operating address, accepted scrap categories, reference price list, and operating hours. **NF-02:** The system verifies the format of the facility details and location coordinates. **NF-03:** The User uploads the business registration license / identity document photo and clicks "Submit Application". **NF-04:** The system validates file formats and sizes, uploads the documents to Cloud Object Storage, and saves the partner upgrade application in PENDING_VERIFY status linked to the User ID. **NF-05:** The system dispatches an acknowledgment email to the applicant through Gmail and displays MSG-MTH-03.
Alternative Flow:	**AF-01:** At NF-01, the User clicks "Detect Current Location". The system requests browser GPS permissions, captures coordinates, and auto-populates the latitude and longitude fields.
Extension Flow:	**ETF-01:** At NF-01, if the User clicks "Cancel", the system discards all entered form data and navigates back to the Homepage.
Exceptions Flow:	**EF-01:** At NF-01, if the User already holds an active application in PENDING_VERIFY status, the system prevents duplicate submission and informs the User that their upgrade application is currently under review. **EF-02:** At NF-01, if any required field is left empty, the system displays MSG-AUTH-02 under the field. **EF-03:** At NF-03, if an uploaded file is not a supported image format (JPG or PNG), the system displays MSG-CLS-01. **EF-04:** At NF-03, if an uploaded file exceeds the maximum allowed size, the system displays MSG-CLS-02. **EF-05:** At NF-03, if the applicant submits without selecting at least one accepted waste category or providing a valid location, the system displays MSG-MTH-06.

Business Rule	**BR-AUTH-01:** One email address can be linked to only one account, across both roles (User/Recycler) and both sign-up methods (local/Google). **BR-MTH-01:** A Recycler must be approved by Staff before appearing in search results or receiving handover requests. **BR-MTH-03:** A Recycler profile can be published only with at least one accepted waste category, collection conditions, and a valid location. **BR-NTF-03:** Notifications are sent only to the account's verified email; changing the email requires re-verification.

8.4 Login System

Primary Actor(s):	System User	**Secondary Actor(s):**	Google OAuth
Description:	Authenticates a registered user (User, Recycler, Staff, Admin) using email and password credentials or Google Single Sign-On, establishes an active authenticated session, and redirects the user to their role-specific workspace.		
Preconditions:	**1.** The user has an existing account in the system. **2.** The System User is on the Login screen.		
Postconditions:	**- Success:** The user identity is verified, an authenticated session and JWT tokens are created, and the user is redirected to their role dashboard. **- Failure:** Authentication is denied, no session is established, and a generic security error message is displayed.		
Normal Flow:	**NF-01:** The System User navigates to the Login screen. **NF-02:** The System User enters their registered email and password , clicks "Sign In". **NF-03:** The system verifies credentials, checks account status (must be Active), generates JWT access/refresh tokens, and establishes the user session. **NF-04:** The system redirects the authenticated user to their role-specific landing page (Homepage for User, Facility Portal for Recycler, Operations Workspace for Staff, Admin Dashboard for Admin).		
Alternative Flow:	**AF-01:** At NF-02, the System User clicks "Sign in with Google". The system authenticates the OAuth token directly with Google, confirms email_verified = true, verifies account status, creates the session, and redirects to the appropriate dashboard.		
Extension Flow:	**ETF-01:** At NF-02, the System User clicks "Forgot Password?" → the system navigates to the Reset Password screen. **ETF-02:** At NF-02, the System User clicks "Register" → the system navigates to the User Registration screen.		

Exceptions Flow:	**EF-01:** At NF-02, if the email or password field is left empty, the system displays MSG-AUTH-02 under the field. **EF-02:** At NF-02, if the credentials are invalid or the account does not exist, the system displays a generic in-line message MSG-AUTH-01 (without disclosing whether the email or password was wrong). **EF-03:** At NF-02, if the account status is Deactivated or Suspended, the system rejects the login attempt, invalidates any existing sessions immediately, and displays MSG-AUTH-04. **EF-04:** At NF-02, if {N} consecutive failed login attempts occur, the system temporarily locks the account for {M} minutes and displays MSG-AUTH-03.
Business Rule	**BR-AUTH-03:** When a sign-in fails, the message must not reveal whether the username exists or the password was wrong. **BR-AUTH-04:** After {N} consecutive failed sign-ins, the account is temporarily locked for {M} minutes. **BR-AUTH-05:** A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately. **BR-AUTH-07:** Google sign-in requires the OAuth token to be verified with Google on every login. Accounts created via Google are bound to the verified Google email and default to the User role. **BR-AUTH-08:** Access is role-based. Guests access only public information; each role accesses only its own functions and data.

8.5 Forgot Password

Primary Actor(s):	System User	**Secondary Actor(s):**	Gmail
Description:	Allows a User or Recycler who has forgotten their password to request a secure one-time reset link by email and set a new password, in order to regain access to their account.		
Preconditions:	**1.** The User or Recycler has an existing account with a registered email address and is not signed in. **2.** The email delivery service (Gmail) is operational. **3.** The actor is on the Login / Forgot Password screen.		
Postconditions:	**- Success:** The new password is saved in hashed form, the reset link is invalidated, all existing sessions are invalidated, and the user is redirected to the Login screen. **- Failure:** If any operation fails, the system data remains unchanged, the existing password is not changed, and an informative error message is displayed.		

Normal Flow:	**NF-01:** The actor clicks "Forgot Password?" on the Login screen. **NF-02:** The system displays the email input form. **NF-03:** The actor enters their registered email → clicks "Send Request". **NF-04:** The system validates rate limits, generates a single-use token expiring in {X} minutes, dispatches a reset link by email via Gmail, and displays toast MSG-AUTH-05 (identical response whether or not the email exists). **NF-05:** The actor opens the email → clicks the reset link → the system validates the token and displays the New Password form. **NF-06:** The actor enters and confirms the new password → clicks "Confirm". **NF-07:** The system validates password complexity, updates the hashed password in the database, invalidates the token, revokes all current sessions, and redirects to the Login screen.
Alternative Flow:	**AF-01:** At NF-04, if the email is not received, the actor clicks "Resend" to request another link, subject to the rate limit.
Extension Flow:	**ETF-01:** At NF-02 or NF-06, if the actor clicks "Back to Login", the system terminates the reset process and returns to the Login screen without changing the password.
Exceptions Flow:	**EF-01:** At NF-03, if the email field is blank, the system displays MSG-AUTH-02 and prevents submission. **EF-02:** At NF-03, if requests exceed the rate limit ({N} per {X} minutes per email, or {M} per hour per IP), the system silently ignores excessive requests and displays MSG-AUTH-07. **EF-03:** At NF-05, if the link is invalid, expired, or already used, the system displays MSG-AUTH-06. **EF-04:** At NF-06, if the new password field is left empty, the system displays MSG-AUTH-02. **EF-05:** At NF-06, if the password does not meet complexity rules or confirmation does not match, the system displays an in-line validation error. **EF-06:** At NF-04, if the email cannot be delivered, the system logs the failure and retries without blocking the request.
Business Rule	**BR-AUTH-06:** A password-reset link is single-use and expires after {X} minutes. The response is identical whether or not the email exists. Resetting invalidates all existing sessions. **BR-AUTH-09:** A password-reset request for the same email is limited to {N} requests per {X} minutes, and requests from the same IP are limited to {M} per hour. Requests beyond the limit are silently ignored (the generic response is still shown). **BR-NTF-01:** Security and transactional emails (verification, password reset, dispute) cannot be disabled by the user. **BR-NTF-03:** Notifications are sent only to the account's verified email; changing the email requires re-verification.

	BR-NTF-05: Notifications must not contain sensitive data (full contact details, tokens), except one-time verification/reset links. **BR-NTF-06:** A failed email delivery is logged and retried up to {N} times, and never blocks or rolls back the business operation.

8.6 View & Update Profile

Primary Actor(s):	System User	**Secondary Actor(s):**	Cloud Object Storage
Description:	Allows any authenticated system user (User, Recycler, Staff, Admin) to view their profile details, update editable contact information (Full Name, Phone Number, Address), and upload an updated avatar image.		
Preconditions:	**1.** The System User is authenticated with an active session. **2.** The System User is on the Profile Settings screen.		
Postconditions:	**- Success:** Updated profile information and avatar are validated and persisted to the database. **- Failure:** Profile updates are rejected, unsaved changes are discarded, and an informative error message is displayed.		
Normal Flow:	**NF-01:** The System User navigates to the "Profile Settings" screen. **NF-02:** The system retrieves and renders current profile information (Full Name, Phone Number, Registered Email [Read-only], Physical Address, Avatar preview). **NF-03:** The System User modifies editable fields (Name, Phone, Address) or selects a new avatar image. **NF-04:** The system displays an immediate local preview of the avatar and validates input formats. **NF-05:** The System User clicks "Save Changes". **NF-06:** The system uploads the avatar to Cloud Object Storage (if modified), saves the profile data to the database, and displays a success confirmation toast.		
Alternative Flow:	**AF-01:** At NF-03, if the System User clicks "Cancel" or navigates away without saving, the system discards unsaved entries and preserves the original profile data.		
Extension Flow:	**ETF-01:** N/A		
Exceptions Flow:	**EF-01:** At NF-03, if an uploaded avatar file is not in JPG or PNG format, the system displays MSG-CLS-01. **EF-02:** At NF-03, if the avatar file exceeds the maximum allowed upload size, the system displays MSG-CLS-02. **EF-03:** At NF-03, if a mandatory profile field is cleared, the system displays MSG-AUTH-02 under the field. **EF-04:** At NF-03, if an input exceeds the maximum length limit, the system displays MSG-NTF-03. **EF-05:** At NF-05, if the user session has expired, the system invalidates the session, displays MSG-NTF-01, and redirects to the Login screen.		
Business Rule	**BR-AUTH-08:** Access is role-based. Guests access only public information; each role accesses only its own functions and data. **BR-NTF-03:** Notifications are sent only to the account's verified email; changing the email requires re-verification.		

8.7 Change Password

Primary Actor(s):	System User	**Secondary Actor(s):**	Gmail
Description:	Allows an authenticated user possessing local credentials to change their account password by verifying their current password, saving a new salted hash, and receiving a security confirmation email.		
Preconditions:	**1.** The System User is logged in with an active session. **2.** The account has a local password credential (non-OAuth-only). **3.** The System User is on the "Change Password" tab.		
Postconditions:	**- Success:** The new password hash is saved to the database and a security alert email is dispatched. **- Failure:** The password update is rejected, the existing password remains unchanged, and an error message is displayed.		
Normal Flow:	**NF-01:** The System User enters Current Password, New Password, and Confirm New Password → clicks "Update Password". **NF-02:** The system verifies the Current Password against the stored cryptographic hash and validates the New Password against complexity policies. **NF-03:** The system updates the password hash in the database, dispatches a security alert email to the registered address via Gmail, and displays a success confirmation toast.		
Alternative Flow:	**AF-01:** At NF-01, if the account registered exclusively via Google OAuth and has no local password, the system displays an informative notice: "Your account is authenticated via Google. Password management is handled through your Google Account."		
Extension Flow:	**ETF-01:** N/A		
Exceptions Flow:	**EF-01:** At NF-01, if any required password field is left empty, the system displays MSG-AUTH-02 under the field. **EF-02:** At NF-01, if the entered Current Password does not match the stored credentials, the system rejects the update and displays MSG-AUTH-01. **EF-03:** At NF-01, if the New Password matches the Current Password, the system displays an in-line error: "New password must be different from current password." **EF-04:** At NF-01, if Confirm New Password does not match the New Password, the system displays an in-line error: "Password confirmation does not match." **EF-05:** At any point, if the session has expired, the system displays MSG-NTF-01 and redirects to the Login screen.		
Business Rule	**BR-AUTH-04:** After {N} consecutive failed sign-ins, the account is temporarily locked for {M} minutes. **BR-AUTH-05:** A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately. **BR-NTF-01:** Security and transactional emails (verification, password reset, dispute) cannot be disabled by the user.		

8.8 View Notifications

Primary Actor(s):	System User	**Secondary Actor(s):**	N/A

Description:	Allows an authenticated user to view incoming system notifications, monitor unread counts, mark items as Read, and navigate directly to the related transaction, dispute, or appointment details.
Preconditions:	**1.** The System User is authenticated with an active session. **2.** The System User is on any authorized system screen.
Postconditions:	**- Success:** Notifications are retrieved and displayed; clicked notifications are marked as Read in the database. **- Failure:** An informative error notification is displayed if notification retrieval fails.
Normal Flow:	**NF-01:** The System User clicks the Notification Bell icon on the top navigation bar. **NF-02:** The system opens a notification dropdown displaying the latest alerts (handover request updates, collection time slots, dispute resolutions, point awards) and an unread count badge. **NF-03:** The System User clicks a specific notification item. **NF-04:** The system marks the notification as Read, decrements the unread badge counter, and navigates directly to the associated entity detail page.
Alternative Flow:	**AF-01:** At NF-02, the System User clicks "Mark all as read". The system updates all user notifications to Read status and resets the badge counter to zero. **AF-02:** At NF-02, the System User clicks "View All Notifications". The system navigates to the dedicated Notification Center page.
Extension Flow:	**ETF-01:** N/A
Exceptions Flow:	**EF-01:** At NF-01, if the session has expired, the system displays MSG-NTF-01 and redirects to the Login page. **EF-02:** At NF-03, if the referenced business entity has been cancelled, deleted, or is inaccessible, the system displays an in-line notice: "The referenced item is no longer available."
Business Rule	**BR-NTF-02:** Every status change of a handover or dispute notifies all affected parties. **BR-NTF-05:** Notifications must not contain sensitive data (full contact details, tokens), except one-time verification/reset links.

8.9 Logout System

Primary Actor(s):	System User	**Secondary Actor(s):**	N/A
Description:	Allows an authenticated user to terminate their active session, invalidate refresh tokens, clear client authentication state, and return to unauthenticated public access.		
Preconditions:	**1.** The System User is logged in with an active session token. **2.** The System User is on any authenticated screen.		
Postconditions:	**- Success:** The active session is revoked, refresh tokens are invalidated, local storage is cleared, and the user is redirected to the public Homepage. **- Failure:** If a network failure occurs during the logout call, client-side session tokens are forcibly purged and the user is redirected to the Homepage.		

Normal Flow:	**NF-01:** The System User clicks "Sign Out" or "Logout" from the user account menu. **NF-02:** The system displays a confirmation prompt: "Are you sure you want to log out?" **NF-03:** The System User confirms logout. **NF-04:** The system calls the logout endpoint, revokes the refresh token in the database/cache, clears client-side tokens/cookies, and redirects to the public Homepage.
Alternative Flow:	**AF-01:** At NF-02, if the System User clicks "Cancel", the system closes the prompt and retains the active session on the current screen.
Extension Flow:	**ETF-01:** N/A
Exceptions Flow:	**EF-01:** At NF-03, if an API timeout or network drop occurs, the client application forcibly destroys local tokens and redirects to the Homepage.
Business Rule	**BR-AUTH-05:** A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately. **BR-AUTH-08:** Access is role-based. Guests access only public information; each role accesses only its own functions and data.

8.10 Manage User & Recycler Accounts

Primary Actor(s):	Staff	**Secondary Actor(s):**	
Description:	Allows an authenticated Staff member to search User and Recycler accounts, view their history, and lock or unlock accounts that violate the policies with a recorded reason, in order to keep the platform safe and trustworthy.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** The Staff has been granted the account management permission. **3.** The Staff is on the User Management screen.		
Postconditions:	**- View:** The system displays the account profile, history, and complaints. **- Lock:** The account is suspended, its active sessions are invalidated, and the reason is recorded. **- Unlock:** The account access is restored and the reason is recorded. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Staff.		

Normal Flow:	**View Account** **NF-01:** The Staff selects "Account Management". **NF-02:** The system displays the account list with filters. **NF-03:** The Staff searches by phone number or email → clicks "View Details". **NF-04:** The system displays the profile, transaction history, and complaints. **Lock Account** **NF-05:** The Staff clicks "Lock Account". **NF-06:** The system displays a form for the duration and reason. **NF-07:** The Staff enters the reason → clicks "Confirm". **NF-08:** The system locks the account → invalidates its active sessions → writes an audit log entry → displays MSG-ADM-01. **Unlock Account** **NF-05 (from NF-04):** The Staff clicks "Unlock Account". **NF-06:** The system displays a form for the reason. **NF-07:** The Staff enters the reason → clicks "Confirm". **NF-08:** The system restores the account access → writes an audit log entry → displays MSG-ADM-01.
Alternative Flow:	**AF-01:** At NF-02, the Staff filters the accounts by role, status, or complaint history.
Extension Flow:	**ETF-01:** At NF-06 (Lock/Unlock), if the Staff clicks "Cancel", the system closes the form and keeps the account status unchanged.
Exceptions Flow:	**EF-01:** At NF-07, if the reason is missing, the system displays MSG-AUTH-02 and prevents the action. **EF-02:** At NF-05, if the Staff attempts to change another Staff or an Admin account, the system denies the action and displays MSG-ADM-04. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-AUTH-05:** A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately. **BR-ADM-03:** Staff may change User/Recycler account status only with a recorded reason, and cannot change Admin or other Staff accounts. **BR-ADM-04:** Accounts with transaction history can only be deactivated, never deleted. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

8.11 Manage Staff Accounts

Primary Actor(s):	Admin	**Secondary Actor(s):**	Gmail
Description:	Allows an authorized Admin to create, view, update, and deactivate Staff accounts (Staff cannot self-register), in order to control who has access to the operational functions of the platform.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has Staff account management privileges. **3.** The Admin is on the Staff Account Management screen.		

Postconditions:	**- View:** The system displays the latest list of Staff accounts. **- Create:** A new Staff account is created and the initial credentials are sent to the Staff by email. **- Update:** The Staff profile is updated. **- Deactivate:** The Staff account is set to Inactive and all its active sessions are terminated. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Admin.
Normal Flow:	**View Staff Accounts** **NF-01:** The Admin navigates to the "Staff Accounts" section. **NF-02:** The system displays the Staff list (Full Name, Corporate Email, Department, Status, Actions). **Create Staff Account** **NF-03:** The Admin clicks "Add New Staff". **NF-04:** The system displays the Staff creation form. **NF-05:** The Admin enters the Full Name, Corporate Email, Phone Number, and department → clicks "Save". **NF-06:** The system validates the email uniqueness and the creation policy → creates the account with temporary credentials → sends an onboarding email through Gmail → writes an audit log entry → displays MSG-ADM-03 → refreshes the list. **Update Staff Account** **NF-03 (from NF-02):** The Admin clicks "Edit" on a Staff row. **NF-04:** The system displays the form pre-populated with the current data. **NF-05:** The Admin modifies the fields → clicks "Save Changes". **NF-06:** The system validates the input → updates the profile → writes an audit log entry → displays a success confirmation. **Deactivate Staff Account** **NF-03 (from NF-02):** The Admin clicks "Deactivate" on a Staff row. **NF-04:** The system displays a confirmation dialog. **NF-05:** The Admin confirms the action. **NF-06:** The system sets the status to Inactive → terminates all active sessions of the account → writes an audit log entry → displays a success notification.
Alternative Flow:	**AF-01:** At NF-02, the Admin searches by name or email, or filters by department or status, and the system updates the displayed Staff list.
Extension Flow:	**ETF-01:** At NF-05 (Create/Update/Deactivate), if the Admin clicks "Cancel", the system closes the dialog and discards all unsaved entries.

Exceptions Flow:	**EF-01:** At NF-06 (Create), if the email already exists on any role of the platform, the system displays an error message and prevents saving. **EF-02:** At NF-06 (Create/Update), if a required field is empty, the system displays MSG-AUTH-02 and prevents saving. **EF-03:** At NF-03 (Deactivate), if the Admin attempts to permanently delete an account with historical operational activity, the system blocks the deletion and only allows deactivation. **EF-04:** At NF-06 (Create), if the onboarding email cannot be delivered, the system displays MSG-NTF-02 and keeps the account created. **EF-05:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-ADM-01:** Staff and Admin accounts are created only by Admin, with no self-registration; Staff cannot create or modify Admin accounts. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-04:** Accounts with transaction history can only be deactivated, never deleted. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log. **BR-AUTH-01:** One email address can be linked to only one account, across both roles (User/Recycler) and both sign-up methods (local/Google). **BR-AUTH-05:** A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately.

8.12 Manage Staff Permissions

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to grant or revoke granular operational permissions of Staff accounts in accordance with the separation of duties, in order to ensure each Staff member only accesses the functions required for the assigned role.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The target Staff account is in Active status. **3.** The Admin is on the Staff Permissions Configuration screen.		
Postconditions:	**- Success:** The updated permission matrix is saved and takes effect immediately. The action is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, the permission settings are not changed, and an informative error message is displayed to the Admin.		

Normal Flow:	**NF-01:** The Admin navigates to the "Staff Permissions Configuration" screen. **NF-02:** The system displays the active Staff members with their current permissions. **NF-03:** The Admin selects a Staff member. **NF-04:** The system displays the permission matrix (Dispute Resolution, AI Dataset Labeling, Recycler Verification, Reward Inventory, Survey Configuration). **NF-05:** The Admin toggles the permissions in accordance with the role separation → clicks "Save Permissions". **NF-06:** The system validates that no administrative governance right is granted to a Staff account → saves the permissions → writes an audit log entry → displays a success confirmation → refreshes the permission view.
Alternative Flow:	**AF-01:** At NF-05, the Admin selects a preset role template (e.g., "Dispute Officer", "AI Data Specialist") and the system fills in the corresponding permissions.
Extension Flow:	**ETF-01:** At NF-05, if the Admin clicks "Cancel", the system resets the permissions to the currently saved values without saving any change.
Exceptions Flow:	**EF-01:** At NF-06, if the selected permissions would grant an administrative governance right to a Staff account, the system rejects the change and displays MSG-ADM-04. **EF-02:** At NF-05, if a non-Admin account attempts to assign permissions, the system denies the request and displays MSG-ADM-04. **EF-03:** At NF-06, if the target Staff account has been deactivated in the meantime, the system rejects the change and refreshes the list. **EF-04:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-ADM-01:** Staff and Admin accounts are created only by Admin, with no self-registration; Staff cannot create or modify Admin accounts. **BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.

9. Blog Content Management

9.1 Create Blog Post Draft

Primary Actor(s):	Staff	**Secondary Actor(s):**	Admin
Description:	Allows an authenticated Staff member to write articles about green living and waste sorting and either save them as drafts or submit them for Admin approval, in order to publish quality educational content for Users.		
Preconditions:	**1.** The Staff is authenticated and has an active session with role Staff. **2.** The Staff is on the Blog/News Management screen.		

Postconditions:	**- Draft:** The post is saved internally in Draft status. **- Submit:** The post status is set to Pending_Approval, the Admin is notified, and MSG-ADM-06 is displayed. **- Failure:** If any operation fails, the system data remains unchanged, any uploaded temporary files are discarded, and an informative error message is displayed to the Staff.
Normal Flow:	**NF-01:** The Staff selects "Blog Posts" → clicks "New Post". **NF-02:** The system displays the rich text editor. **NF-03:** The Staff enters the title, uploads a cover image, and writes the content. **NF-04:** The Staff clicks "Preview" to check the layout. **NF-05:** The Staff clicks "Submit for Approval". **NF-06:** The system validates the input → sets the post status to Pending_Approval → notifies the Admin → displays MSG-ADM-06.
Alternative Flow:	**AF-01:** At NF-05, the Staff clicks "Save Draft" and the system saves the post in Draft status without notifying the Admin.
Extension Flow:	**ETF-01:** At NF-03, if the Staff clicks "Cancel", the system discards the unsaved content and returns to the Blog/News Management screen.
Exceptions Flow:	**EF-01:** At NF-06, if the title or cover image is missing, the system displays MSG-AUTH-02 and prevents submission. **EF-02:** At NF-03, if the cover image is in an unsupported format or exceeds the size limit, the system displays MSG-CLS-01 or MSG-CLS-02 and rejects the image. **EF-03:** At any point, if the Staff's session has expired, the system displays MSG-NTF-01 and redirects the Staff to the Login page.
Business Rule	**BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-08:** A blog post authored by Staff must be in Draft or Pending_Approval status. It is published publicly in the app only after Admin approves it (Approved status).

9.2 Approve / Reject Blog Post

Primary Actor(s):	Admin	**Secondary Actor(s):**	Staff
Description:	Allows an authorized Admin to review blog posts written by Staff and either approve them for public publication or reject them back to draft with feedback, in order to ensure the content quality before it reaches Users.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** At least one blog post is in Pending_Approval status. **3.** The Admin is on the Blog Moderation Queue screen.		
Postconditions:	**- Approve:** The post status is set to Approved and the post is published to Users. The decision is recorded in the audit log and the Staff is notified. **- Reject:** The post status is set to Draft with the feedback notes and the Staff is notified.		

	- Failure: If any operation fails, the system data remains unchanged, the post remains in Pending_Approval status, and an informative error message is displayed to the Admin.
Normal Flow:	**Approve Post** **NF-01:** The Admin navigates to the "Blog Moderation Queue". **NF-02:** The system displays the posts awaiting review (title, author, submission date, thumbnail). **NF-03:** The Admin opens a post to preview the full text and media. **NF-04:** The Admin clicks "Approve". **NF-05:** The system sets the post status to Approved and publishes it → writes an audit log entry → notifies the Staff → displays MSG-ADM-05. **Reject Post** **NF-04 (from NF-03):** The Admin clicks "Reject". **NF-05:** The system displays a feedback form. **NF-06:** The Admin enters the revision feedback → clicks "Send". **NF-07:** The system sets the post status to Draft → writes an audit log entry → notifies the Staff with the feedback.
Alternative Flow:	**AF-01:** At NF-02, the Admin searches the queue by title or author, or sorts it by submission date.
Extension Flow:	**ETF-01:** At NF-04 or NF-06, if the Admin closes the preview or clicks "Cancel", the system keeps the post in Pending_Approval status without any change.
Exceptions Flow:	**EF-01:** At NF-06 (Reject), if no feedback is entered, the system displays MSG-AUTH-02 and prevents submission. **EF-02:** At NF-05, if the post was recalled to Draft by the Staff before the decision, the system notifies the Admin that the post is no longer pending and refreshes the queue. **EF-03:** At NF-04, if a non-Admin account attempts to approve a post, the system denies the request and displays MSG-ADM-04. **EF-04:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-ADM-02:** Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey configuration, and account status. **BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log. **BR-ADM-08:** A blog post authored by Staff must be in Draft or Pending_Approval status. It is published publicly in the app only after Admin approves it (Approved status).

10. System Analytics & Reporting

10.1 View Dashboard & Analytics

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to monitor platform analytics (waste collection volumes, user growth, handover completion, dispute rate, and AI classification performance) in read-only mode, in order to evaluate the operation of the system and support decisions.		

Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The system contains completed and confirmed operational records. **3.** The Admin is on the administration home screen.
Postconditions:	**- Success:** The system displays the KPI cards and charts calculated from completed and confirmed data. No data is modified. **- Failure:** If any operation fails, the system data remains unchanged and an informative error message is displayed to the Admin.
Normal Flow:	**NF-01:** The Admin selects the "Dashboard & Analytics" module on the administration sidebar. **NF-02:** The system queries the completed and confirmed operational records. **NF-03:** The system displays the KPI cards (Total Recycled Waste Weight, Completed Handover Count, New User Registrations, Dispute Rate). **NF-04:** The system displays the analytical charts (Waste Distribution by Material, AI Classification Accuracy Rate, Partner Geographic Density). **NF-05:** The Admin selects a metric card to view its detailed breakdown.
Alternative Flow:	**AF-01:** At NF-03, the Admin changes the filters (time period: week / month / quarter / year, waste category, or region) and the system recalculates and updates the charts.
Extension Flow:	**ETF-01:** At NF-05, if the Admin closes the detailed view or navigates to another administration screen, the system returns to the dashboard without any change.
Exceptions Flow:	**EF-01:** At NF-02, if the database connection fails or the analytics service is unreachable, the system displays an error message. **EF-02:** At NF-02, if no data exists for the selected filters, the system displays an empty-state message. **EF-03:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-ADM-07:** Dashboard analytics are calculated only from completed and confirmed data, and are read-only.

10.2 Export Data Reports

Primary Actor(s):	Admin	**Secondary Actor(s):**	
Description:	Allows an authorized Admin to export analytical summaries, completed transaction records, waste statistics, and partner metrics as downloadable CSV or Excel reports, in order to support reporting and analysis outside the system.		
Preconditions:	**1.** The Admin is authenticated and has an active session with role Admin. **2.** The Admin has reporting and export privileges. **3.** The Admin is on the Report Export screen.		
Postconditions:	**- Success:** The report file (CSV or Excel) is generated and downloaded, and the export is recorded in the audit log. **- Failure:** If any operation fails, the system data remains unchanged, no file is generated, and an informative error message is displayed to the Admin.		

Normal Flow:	**NF-01:** The Admin navigates to the "Export Data Reports" screen. **NF-02:** The Admin selects the report type (Completed Trade Records, Waste Stream Statistics, Recycler Performance Overview). **NF-03:** The Admin sets the filters (Start Date, End Date, Region, Transaction Status). **NF-04:** The Admin selects the file format (CSV or Excel). **NF-05:** The Admin clicks "Export". **NF-06:** The system aggregates the records → excludes or anonymizes personal contact and banking details → writes an audit log entry → starts the file download → displays MSG-ADM-02.
Alternative Flow:	**AF-01:** At NF-03, if the Admin does not specify a date range, the system applies the default range of the last 30 days.
Extension Flow:	**ETF-01:** At NF-05, if the Admin clicks "Cancel", the system aborts the file generation and keeps the selected filters.
Exceptions Flow:	**EF-01:** At NF-06, if no record matches the filters, the system displays a message that no data exists and cancels the download. **EF-02:** At NF-05, if a non-authorized account attempts to export reports, the system denies the request and displays MSG-ADM-04. **EF-03:** At NF-06, if the report generation fails because of a system error, the system displays an error message and no file is downloaded. **EF-04:** At any point, if the Admin's session has expired, the system displays MSG-NTF-01 and redirects the Admin to the Login page.
Business Rule	**BR-ADM-05:** Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log. **BR-ADM-06:** Exported reports contain aggregated or anonymized data by default, exclude personal contact details, and every export is logged.

III. Functional Requirements

1. User Authentication & Profile Management

1.1 Guest Onboarding & Authentication

1.1.1 Landing Homepage & System Introduction (SCR-GST-01)



- **Mapped Use Case:** G-01: View Homepage
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup / Wireframe: SCR-GST-01 — Hero banner with environmental mission, AI waste classification showcase, value proposition for households & recyclers, and portal navigation bar]

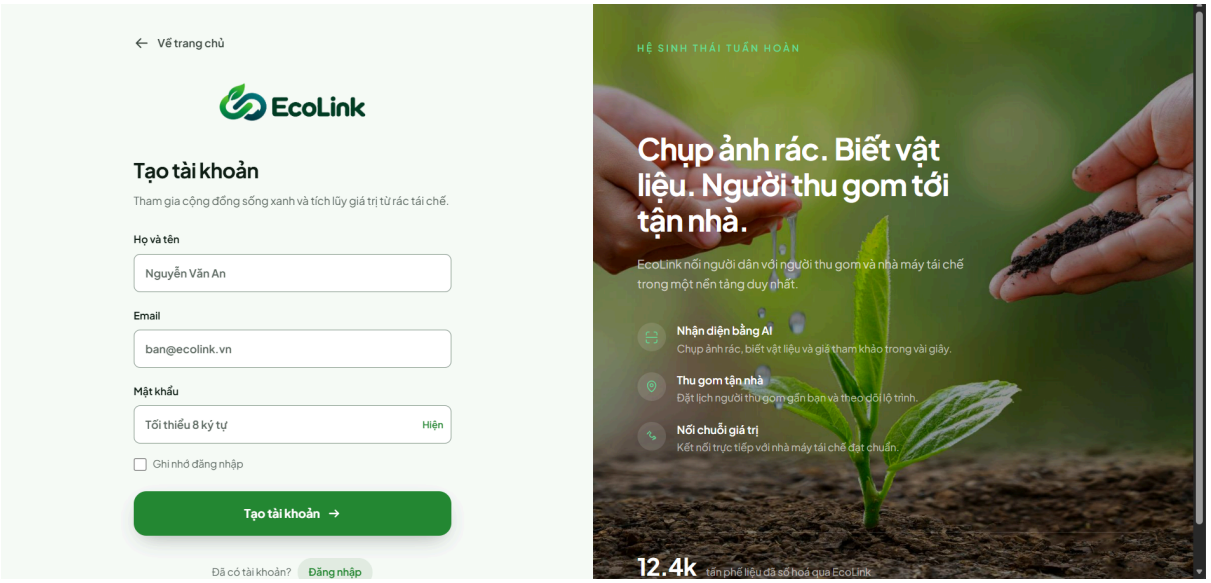
This screen allows a Guest to:

- Read EcoLink's environmental mission and browse the platform overview.
- Review featured presentations on AI waste classification technology and citizen reward incentives.
- Learn about household waste recycling workflows, partner networks, and featured environmental blogs.
- Click the Call-To-Action (CTA) "Start Scanning Waste Now" button — because AI photo classification is strictly restricted to authenticated User accounts, the system prompts the Guest to authenticate and automatically redirects them to the Sign In screen (SCR-SYS-04) or Registration screen (SCR-GST-02).
- Navigate to the User Registration screen, Recycler Registration screen, or Sign In screen.

Field Description:

Field Name	Description
Hero Banner Text	Headline statement highlighting EcoLink's mission; read-only text.
CTA "Start Scanning" Button	Call-to-action button to initiate waste classification; upon click, prompts unauthenticated guests to sign in and redirects them to the Sign In screen (SCR-GST-04).
Navigation Register User	Action button navigating to the User Registration screen (SCR-GST-02).
Navigation Register Recycler	Action button navigating to the Recycler Partner Registration screen (SCR-GST-03).
Navigation Sign In	Action button navigating to the System Authentication screen (SCR-GST-04).

1.1.2 User Account Registration (SCR-GST-02)



- **Mapped Use Case:** G-02: Register as User

  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-GST-02 — Standard registration card with Full Name, Phone, Email, Password, Confirm Password, and Terms checkbox]

This screen allows a Guest to:

- Input personal credentials to establish a standard User account.
- Receive real-time validation feedback on phone number format, email syntax, and password strength.
- Read and accept the platform Terms of Service and Privacy Policy.
- Tap "Create Account" to submit registration and trigger an account verification email.

Field Description:

Field Name	Description
Full Name Input	User's full legal name; alphabetic string, length between 2 and 50 characters; mandatory.
Email Address Input	Electronic mail address; standard valid email format (e.g., name@domain.com); maximum 100 characters; mandatory; globally unique.
Phone Number Input	Mobile phone number in Vietnam format; 10 digits starting with 0; mandatory; globally unique.
Password Input	Account secret password; minimum 8 characters, containing at least 1 uppercase letter, 1 lowercase letter, and 1 numeric digit; mandatory.
Confirm Password Input	Password re-entry; must match Password Input exactly.
Terms Agreement Checkbox	Consent checkbox acknowledging terms of service; boolean; must be checked (True).

1.1.3 Recycler Partner Upgrade Application (SCR-USR-17)

- Mapped Use Case: U-17: Apply for Recycler Upgrade

[Content #1: UI layout (Wireframe or Mockup for the screen)]
[Placeholder for UI Mockup: SCR-USR-17 — In-App Partner upgrade application form accessible from User Profile, featuring Facility Name, Business License uploader, GPS Location picker, and Accepted Material checklist]

This in-app screen allows an authenticated User (Citizen) to:

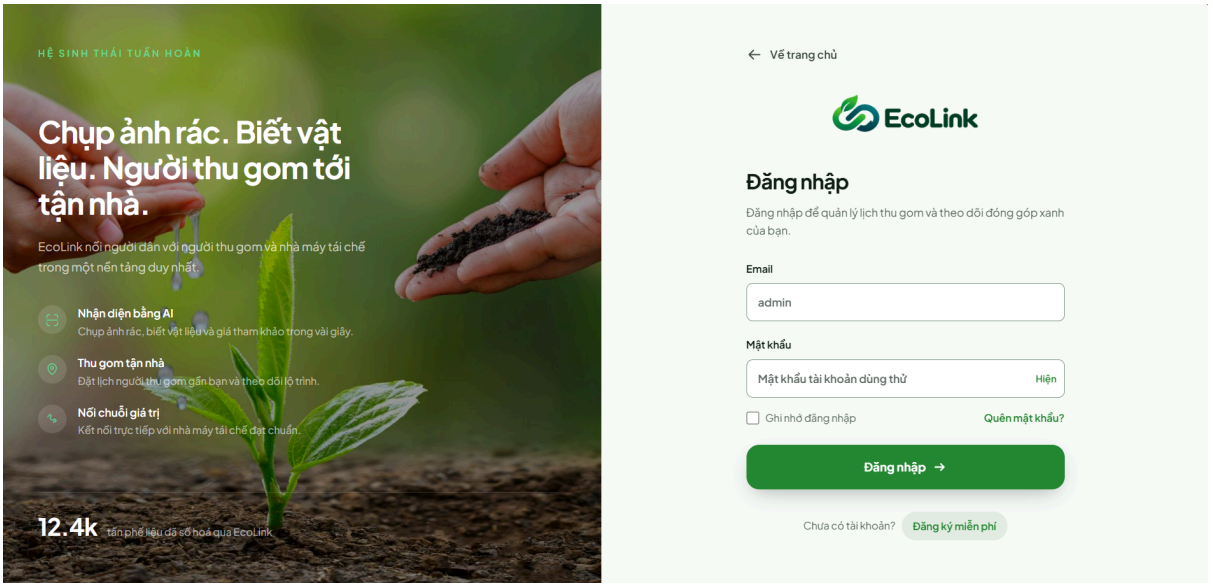
- Input business entity credentials including facility name, legal representative name, and registered business telephone.
- Upload scanned copies of Business Licenses or National ID documents to Cloudflare R2 storage for Staff verification.
- Pin the precise facility location on an interactive map to capture GPS coordinates for spatial discovery.
- Select initial waste categories that the facility is equipped to purchase.
- Submit the application for Staff appraisal pursuant to BR-MTH-01.

Field Description:

Field Name	Description
Facility Name Input	Registered trade name of the collection facility or scrap depot; length 5 to 100 characters; mandatory.
Representative Name Input	Full name of the authorized legal representative or proprietor; length 2 to 50 characters; mandatory.
Business Phone Input	Primary corporate contact number; 10 digits; mandatory.
Address Street Input	Exact physical street address (house number, street name, ward, district); maximum 150 characters; mandatory.
Map Coordinate Picker	Spatial geographic coordinates comprising Longitude and Latitude; automatically captured upon pinning location on the map.
License Document Uploader	Upload component for business registration licenses or identity proof; maximum file size 10 MB; stored securely in Cloudflare R2.
Initial Accepted Categories	Checklist of supported waste categories; requires selecting at least one valid category pursuant to BR-MTH-03.

1.1.4 System Authentication & Sign In (SCR-SYS-04)

- **Mapped Use Case:** SU-01: Login



This screen allows Any System Actor to:

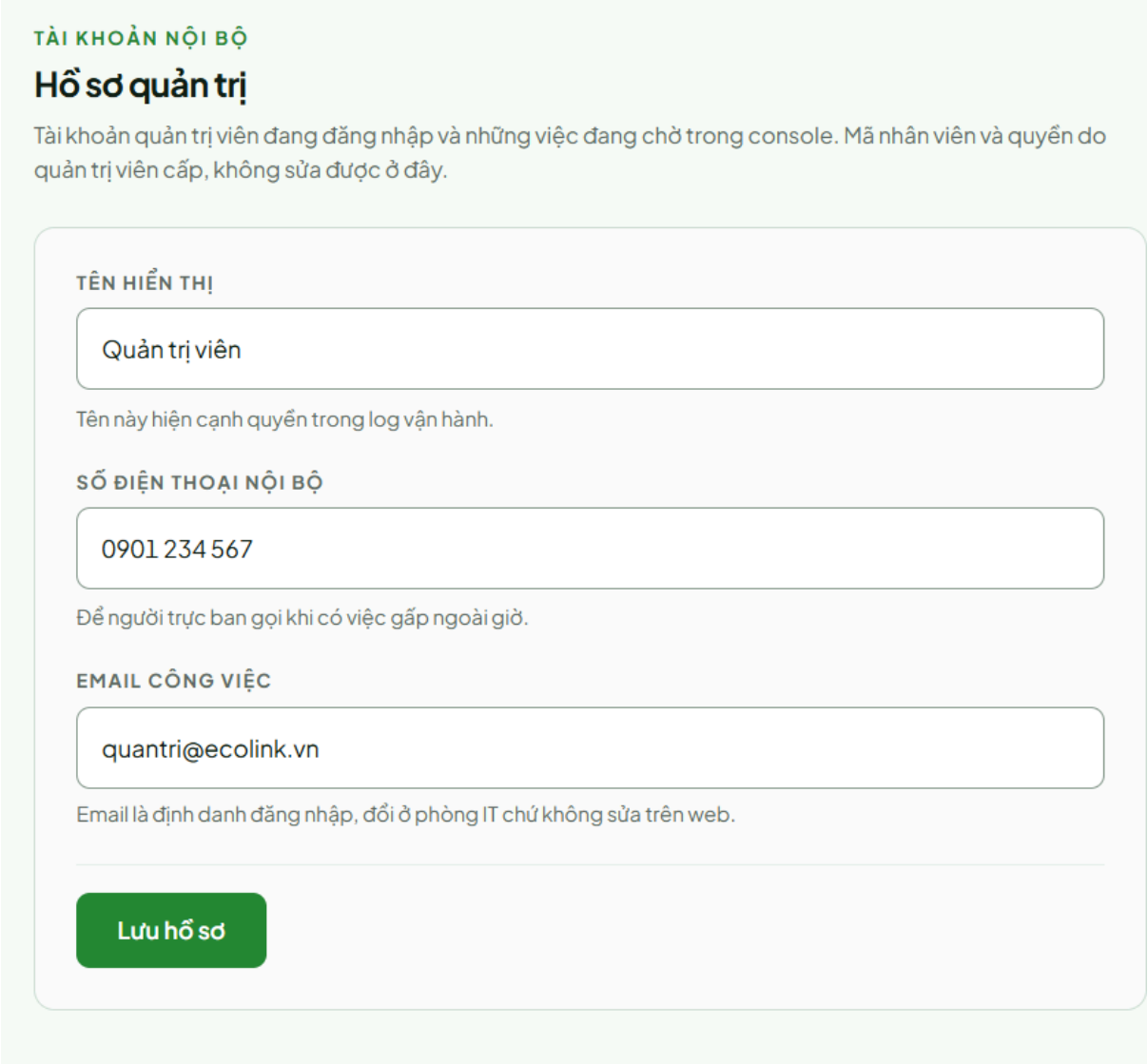
- Authenticate using registered Email (or Mobile Phone) and secret Password.
- Authenticate using Single Sign-On via Google Identity Platform (OAuth 2.0 PKCE).
- Tap "Forgot Password?" to launch the credential recovery workflow (SU-06).
- Upon successful authentication, the system dynamically routes the actor to their designated dashboard based on the assigned role (User, Recycler, Staff, Admin).

Field Description:

Field Name	Description
Account Identifier Input	Registered Email Address or Mobile Phone Number; maximum 100 characters; mandatory.
Login Password Input	Account password string; masked character input; mandatory.
Remember Me Checkbox	Session persistence checkbox; extending JWT token validity from 24 hours to 30 days when checked.
Google SSO Button	Action button initiating Google OAuth 2.0 authentication flow over secure transport.
Forgot Password Link	Hyperlink navigating to the One-Time Password password recovery workflow.

1.2 System User Subsystem (Common Inherited Features)

1.2.1 Profile Overview & Update (SCR-SYS-01)



- **Mapped Use Case:** SU-03: View & Update Profile
  [Content #1: UI layout (Wireframe or Mockup for the screen)]

[Placeholder for UI Mockup: SCR-SYS-01 — Profile overview with avatar photo uploader, basic info inputs, default address, and save button]

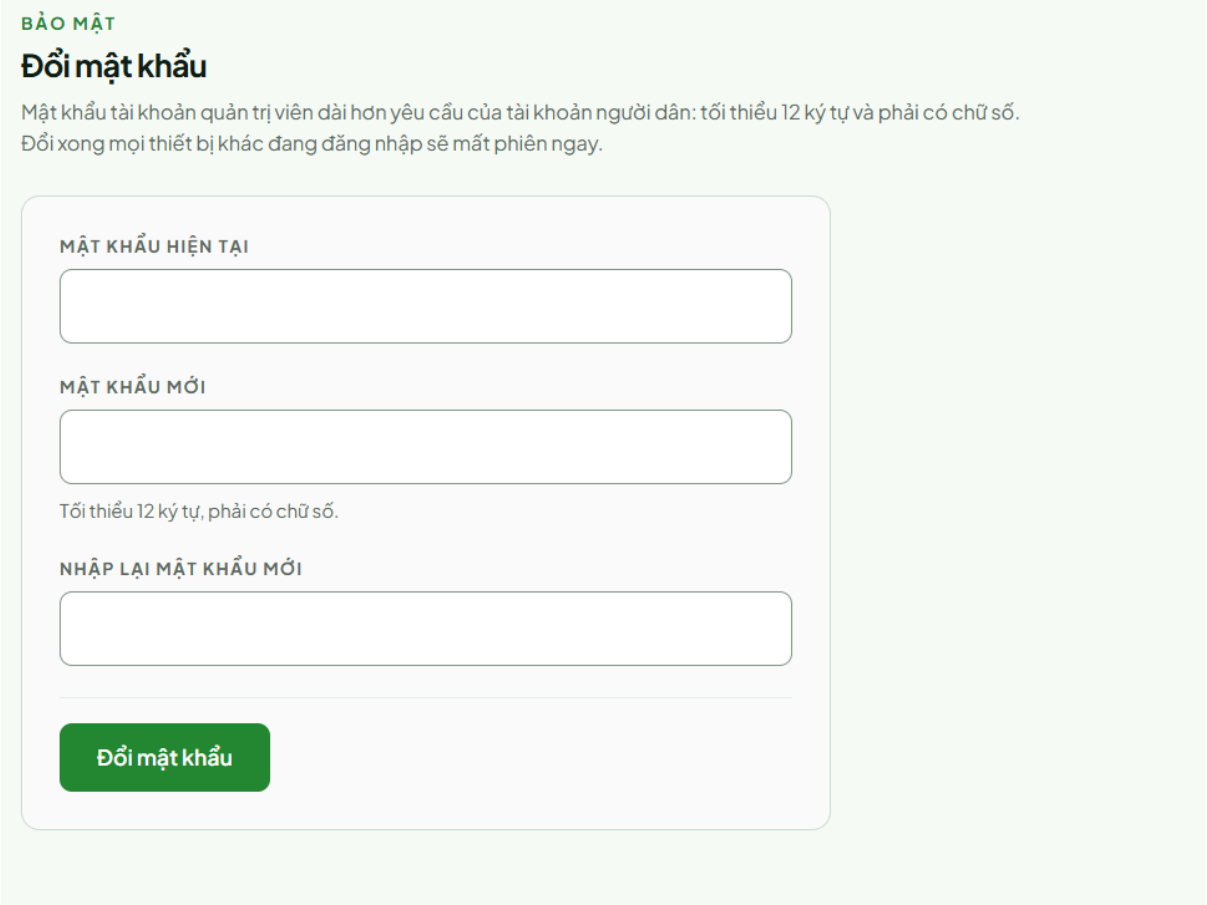
This screen allows an Authenticated User to:

- Inspect current profile information, avatar portrait, assigned role, and membership registration date.
- Upload and crop a new profile avatar image hosted in Cloudflare R2 storage.
- Modify display name, date of birth, gender, and default collection address.
- Tap "Save Changes" to commit modifications to the database.

Field Description:

Field Name	Description
Avatar Image Uploader	Profile portrait image file; JPG/PNG format, maximum 5 MB; stored in Cloudflare R2.
Display Name Input	Public screen name; string between 2 and 50 characters; mandatory.
Email Text (Read-only)	Verified account email address; immutable display field on this screen.
Phone Text (Read-only)	Verified mobile telephone number; immutable display field.
Default Address Textbox	Primary residential collection address pre-filling future trade requests; maximum 150 characters.

1.2.2 Password Change (SCR-SYS-02)



- **Mapped Use Case:** SU-02: Change Password

[Content #1: UI layout (Wireframe or Mockup for the screen)]
[Placeholder for UI Mockup: SCR-SYS-02 — Change password card with Current Password, New Password, Confirm New Password, and Update button]

This screen allows an Authenticated User to:

- Validate ownership by supplying their existing password.
- Specify a new password complying with security complexity standards.
- Re-type the new password for equivalence checking.
- Submit the change, prompting the system to re-hash the new credential with BCrypt and invalidate prior tokens.

Field Description:

Field Name	Description
Current Password Input	Existing account password; verified against the stored BCrypt hash in the database.
New Password Input	Desired replacement password; minimum 8 characters with at least 1 uppercase, 1 lowercase, and 1 numeric digit; must differ from existing password.
Confirm New Password Input	Confirmation entry; must match New Password Input identically.

1.2.3 System Notification Center (SCR-SYS-03)

- **Mapped Use Case:** SU-04: View Notifications
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-SYS-03 — Notification feed with unread filter, category tabs (Transactions, System, Rewards), and mark-all-read action]

This screen allows an Authenticated User to:

- Review an inbox of real-time alerts (handover request status changes, incoming messages, appointment reminders, point credits).
- Filter alerts by unread status or functional classification.
- Tap "Mark All as Read" to clear pending badges.
- Click any alert entry to navigate directly to the referenced transaction record.

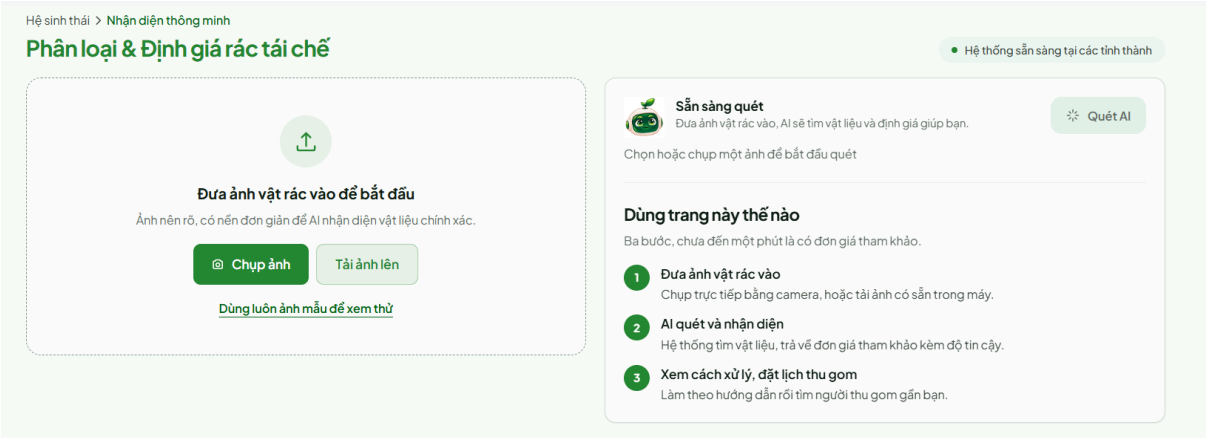
Field Description:

Field Name	Description
Notification Item Row	Feed entry displaying event icon, brief summary title, descriptive snippet, relative timestamp (e.g., "5 minutes ago"), and read indicator.
Mark All As Read Button	Batch action updating read flags across all notifications in the user's inbox.
Filter Tabs	Classification chips: All, Handover Trades, Eco-Rewards, Security Notices.

2. USER Subsystem (Household Waste Sorters)

2.1 AI Waste Classification & Guidance

2.1.1 AI Waste Scanner Viewfinder (SCR-USR-01)



- **Mapped Use Case:** U-01: Identify Waste via Photo (UC10)
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-01 — Desktop web waste classification portal on PC featuring dual input modalities: live PC webcam viewfinder with instant capture button, and drag-and-drop file upload zone with interactive image preview and analyze CTA]

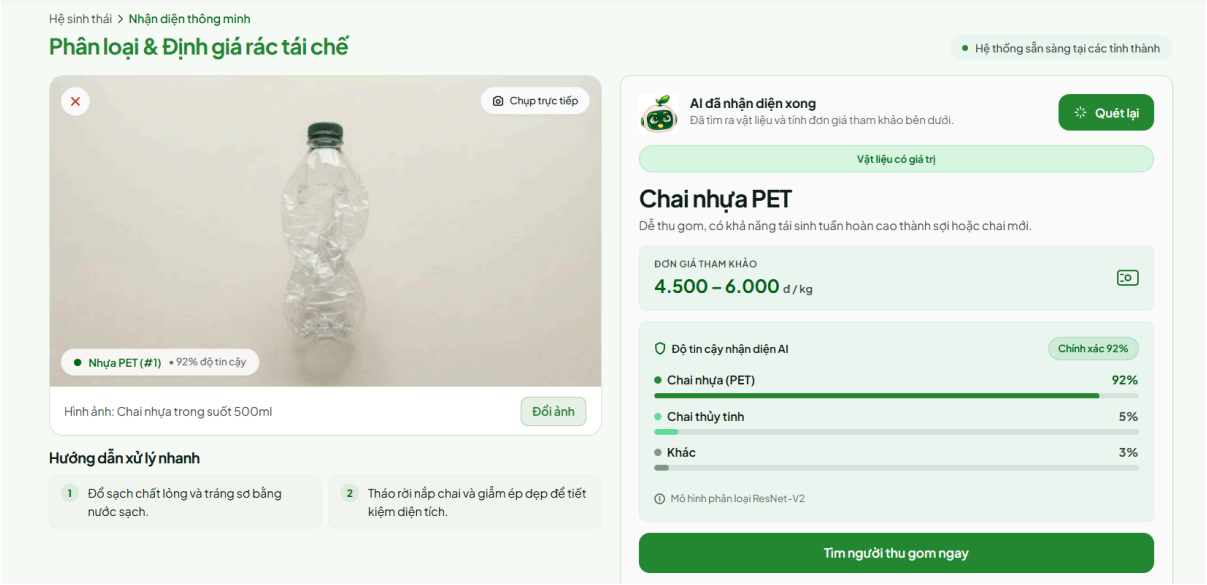
This screen allows the Citizen User on the desktop web application (PC / Laptop) to:

- Activate the PC/laptop webcam to capture waste images directly via a real-time video feed with an alignment framing guide.
- Upload existing waste photos from local PC storage via a responsive drag-and-drop zone or system file browser dialog (File Explorer).
- Review an instant image preview to verify image sharpness and waste clarity prior to submission.
- Click "Retake / Replace" to clear the preview buffer and capture or select a different photo.
- Click "Analyze Waste with AI" to upload the image to Cloudflare R2 storage and trigger model inference.

Field Description:

Field Name	Description
Webcam Viewfinder Feed	Live video stream container rendering camera input from PC/laptop webcam; 16:9 or 4:3 aspect ratio with alignment guide overlay.
Webcam Capture Button	Shutter action button capturing the active webcam frame and freezing it into the preview buffer.
Drag-and-Drop Upload Area	Interactive drop zone accepting image file uploads from PC local storage; supports image/jpeg, image/png, image/webp; maximum 10 MB.
Image Preview Container	Visual container displaying the captured or uploaded photo before initiating analysis.
Retake / Replace Button	Action button resetting the image preview, re-enabling webcam feed or allowing a new file selection.
Analyze Waste Button	Primary call-to-action button; enabled once a valid image is staged; persists photo to Cloudflare R2 and navigates to the result dialog (SCR-USR-02).

2.1.2 Scan Result & Feedback Dialog (SCR-USR-02)



- **Mapped Use Case:** U-01: Identify Waste via Photo (UC10), U-04: Report Incorrect AI Classification (UC13)
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-02 — Modal/Card displaying scanned image preview, AI top prediction, confidence score badge, quick action buttons, and incorrect reporting dropdown]

This screen allows the User to:

- Review the predicted waste classification, percentage confidence score, and recyclability badge.
- Inspect summary pre-treatment instructions and scrap market price ranges.
- Tap "View Detailed Guide" to transition to the sorting handbook (U-02).
- Tap "Find Recyclers" to locate nearby collectors accepting this item (U-05).
- Tap "Report Incorrect Classification" to submit correction feedback with the true category (U-04).

Field Description:

Field Name	Description
Scanned Image Preview	Thumbnail image preview sourced from Cloudflare R2 storage; read-only.
Predicted Category Name	Text label of the detected waste category (e.g., "PET Plastic Bottle", "Aluminum Can"); read-only.
Confidence Score Badge	Numerical percentage confidence (0% – 100%); rendered green if >= 70%, orange if between 50% and 69%.
Recyclable Status Badge	Classification badge: "Recyclable" vs "Non-Recyclable".
Scrap Price Estimate	Market price range per kilogram (e.g., "3,000 – 5,000 VND/kg"); currency: VND.
Reported Category Dropdown	Dropdown selector displaying all active waste categories for reporting misclassifications; mandatory upon reporting.
Feedback Note Input	Optional narrative explanation regarding the misclassification; maximum 250 characters.

2.1.3 Waste Catalog & Keyword Search (SCR-USR-03)

- **Mapped Use Case:** U-03: Search & View Waste Details (UC12)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-03 — Search bar with autocomplete, material filter chips (Plastic, Paper, Metal, Glass), and grid of waste category cards]

This screen allows the User to:

- Search waste types manually via keywords with real-time autocomplete suggestions.
- Filter the catalog by material family (Plastic, Paper, Metal, Glass, E-Waste).
- Browse category cards with visual icons and recyclability tags.
- Tap any card to navigate directly to its detailed handling guide (U-02).

Field Description:

Field Name	Description
Search Keyword Input	Search string input; triggers autocomplete suggestions after 2 characters; maximum 50 characters.
Material Filter Chips	Filter selectors: All, Plastic, Paper, Metal, Glass, E-Waste; default: "All".
Category Card Item	Grid item displaying: Category Icon (Cloudflare R2), Category Name, Material Family, Recyclable Badge.

2.1.4 Detailed Sorting Guidance (SCR-USR-04)

- **Mapped Use Case:** U-02: Get Sorting Guidance (UC11)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-04 — Step-by-step visual handling guide (Clean, Dry, Flatten), disposal bin recommendation, and recycler CTA button]

This screen allows the User to:

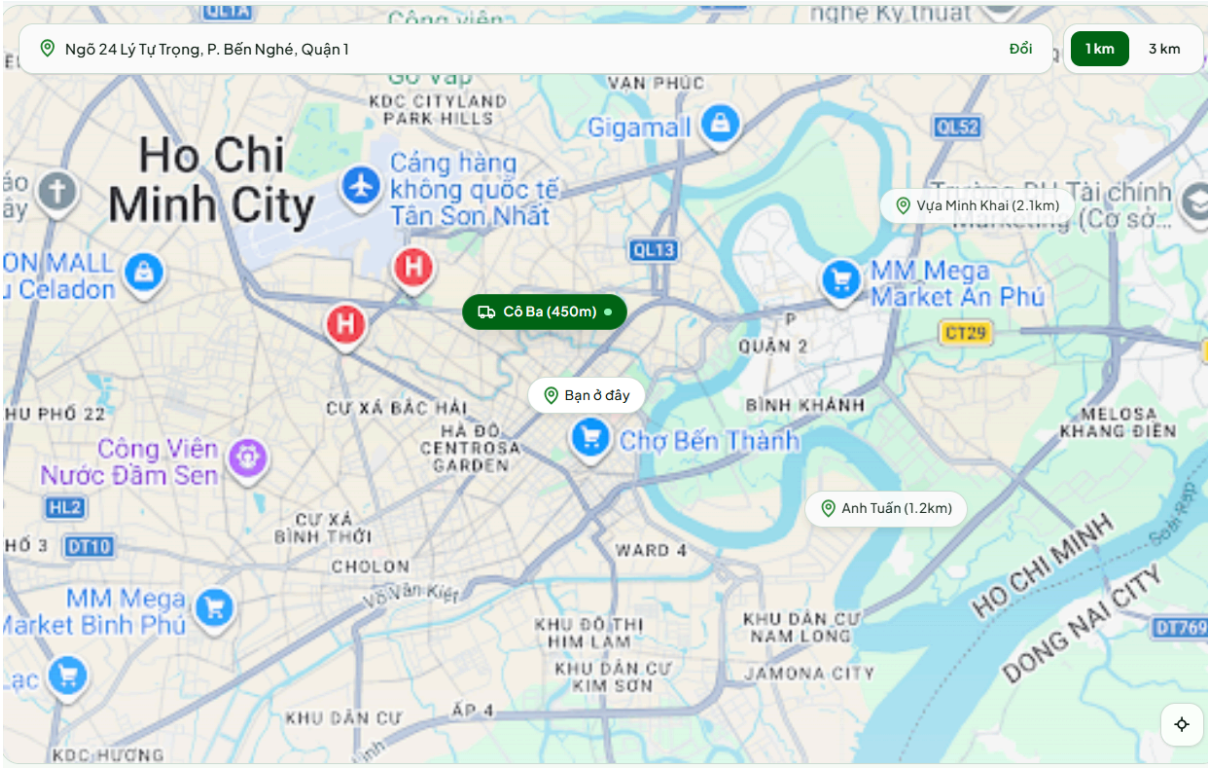
- Read structured sequential pre-treatment instructions (Empty residue, Rinse with water, Dry, Flatten/Bundle).
- Identify the designated municipal bin color/type (Yellow recyclable bin or organic bin).
- Read environmental impact insights and common sorting mistakes.
- Tap "Find Recyclers for this Item" to transition to spatial discovery (U-05).

Field Description:

Field Name	Description
Category Header Title	Display name and icon of the selected waste category; prominent header.
Step-by-Step Guide List	Numbered procedural handling steps (1 to N); each containing an action headline, detailed explanation, and illustration.
Disposal Bin Type Badge	Designated municipal container badge (e.g., "Yellow Bin - Household Recyclables"); read-only.
Environmental Impact Tip	Ecological impact narrative highlighting resources conserved by recycling this item.

2.2 Scrap Trade Matching & Handover Requests

2.2.1 Nearby Recyclers Map & Directory (GPS) (SCR-USR-05)



- **Mapped Use Case:** U-05: Find Nearby Recyclers (GPS) (UC14)
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-05 — Interactive map view with recycler pins, radius slider, material filter bar, and sliding bottom profile sheet]

This screen allows the User to:

- View their current GPS location (blue pulse) alongside verified scrap collection facilities (green markers).
- Adjust the search radius slider between 2 km and 20 km (default: 5 km).
- Filter facilities by accepted material categories.
- Tap a facility marker to open a bottom summary sheet showing business hours, price sheets, and rating.
- Tap "Book Handover" on the facility sheet to initiate a trade request (U-06).

Field Description:

Field Name	Description
Interactive Map View	OpenStreetMap / Leaflet map viewport supporting pinch-to-zoom and pan gestures.
Radius Slider	Slider input modifying search radius; bounds: 2 km to 20 km; step: 1 km; default: 5 km.
Waste Type Multi-Filter	Multi-select chip filter displaying only facilities accepting all selected waste categories.
Recycler Pin Marker	Geographic marker representing an active facility; highlights upon user selection.
Recycler Name	Legal trade name of the facility; string up to 100 characters.

Operating Hours Badge	Daily operating schedule badge; green if open, gray if closed.
Price Table Preview	Condensed summary table of per-kg scrap prices; displays stale price warning if unedited over 30 days.
Rating Star Average	Cumulative score between 1.0 and 5.0 stars with total review count.

2.2.2 Create Handover Request & Time Slot Selection (SCR-USR-06)

- **Mapped Use Case:** U-06: Create Handover Request (UC15)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-USR-06 — Multi-item waste entry form, photo uploader preview, calendar & time slot picker, and address confirmation]

This screen allows the User to:

- Add multiple recyclable items to sell, specifying estimated weight (kg) for each category.
- Attach 1 to 5 photos of the actual waste items uploaded to Cloudflare R2.
- Pick a preferred collection date and **desired collection time slot** within the facility's working schedule.
- Verify pickup address and tap "Submit Handover Request".

Field Description:

Field Name	Description
Recycler Summary Banner	Displays target facility name and estimated distance; read-only.
Waste Item Row	Dynamic entry row: Category Selector (Dropdown) and Estimated Weight (Positive decimal > 0, max 2 decimals, e.g., 15.50 kg).
Photo Upload Area	Image file uploader; min 1 photo, max 5 photos; JPG/PNG; stored in Cloudflare R2.
Collection Date Picker	Calendar date selector; allows selecting today or future dates up to 7 days ahead.
Preferred Time Slot	Dropdown selector of collection windows (e.g., "08:00 – 10:00", "14:00 – 16:00"); must be at least 2 hours in advance.
Pickup Address Text	Detailed street pickup location; defaults to user profile address; editable.
Additional Notes Input	Optional directional instructions or special handling notes; maximum 200 characters.

2.2.3 Handover Request Details & Order Tracking (SCR-USR-07)

- **Mapped Use Case:** U-07: Update / Cancel Handover Request (UC16)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-USR-07 — Status progression bar (Pending, Accepted, Scheduled, Completed), order summary card, edit/cancel action buttons]

This screen allows the User to:

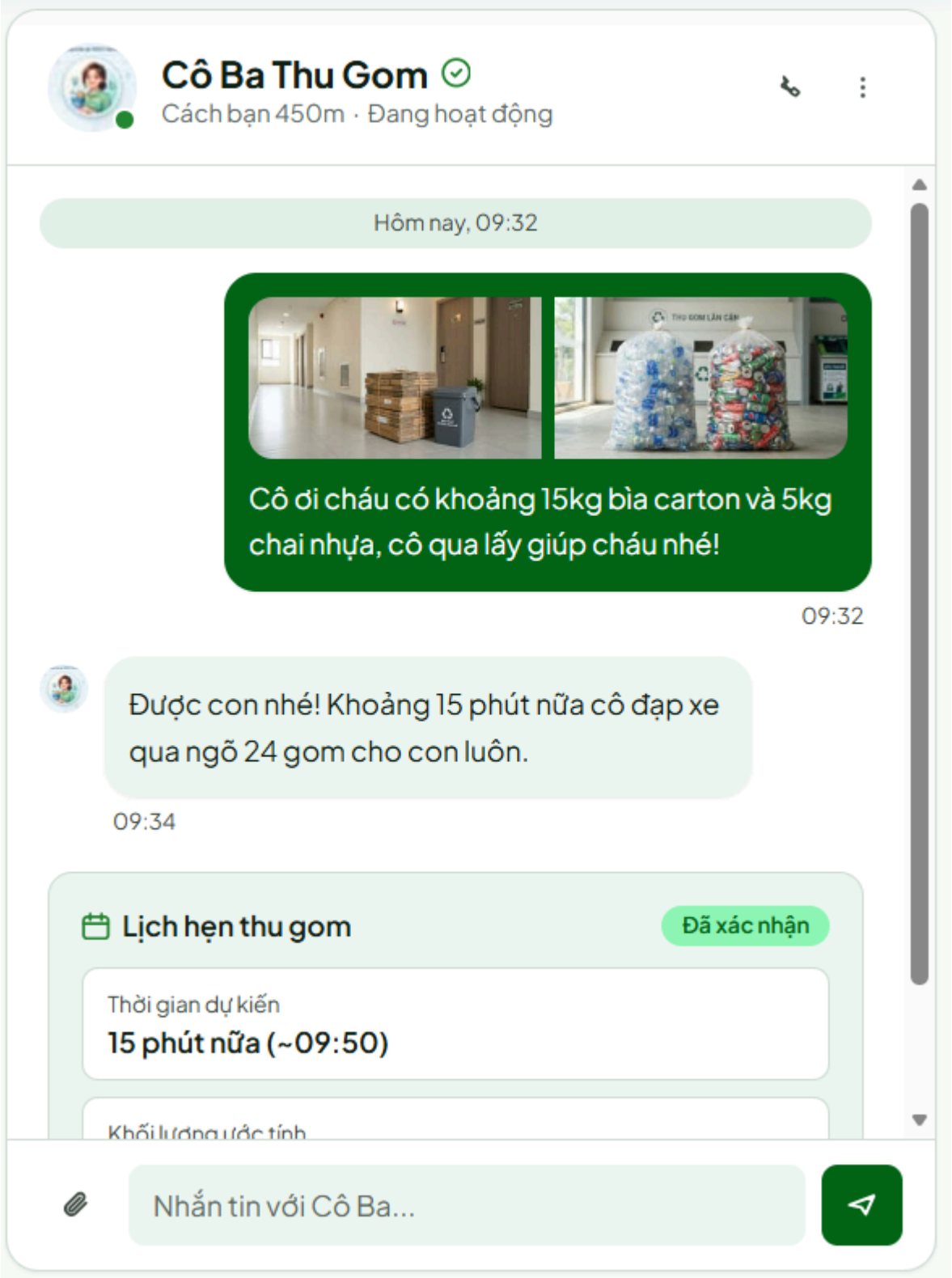
- Track real-time order lifecycle status (Pending, Accepted, In-Progress, Completed, or Cancelled).
- Inspect itemized quantities, photo attachments, and agreed time slots.

- Edit request parameters if the order remains in Pending status.
- Cancel the request (providing a mandatory reason) prior to completion.
- Tap "Message Recycler" to launch the negotiation chat thread (U-08).

Field Description:

Field Name	Description
Request ID Label	Unique order tracking identifier (e.g., #HND-20261005-0012); read-only.
Status Progression Bar	Linear visual tracker: Pending, Accepted, Completed; dynamic color states.
Confirmed Time Slot	Locked collection window agreed upon by both parties.
Recycler Contact Card	Facility telephone and contact details; **unmasked only after request is Accepted** pursuant to BR-HND-08.
Cancellation Reason Input	Form input requiring a cancellation reason; mandatory upon cancellation.

2.2.4 In-App Real-Time Negotiation Chatbox (SCR-USR-08)



- **Mapped Use Case:** U-08: Negotiate via Chatbox (UC17)
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-08 — Real-time chat stream with buyer, price counter-proposal bubble, image attachment button, and message input box]

This screen allows the User to:

- Engage in two-way real-time messaging with the assigned collector scoped to the specific order.
- Transmit photos of scrap items or alleyway landmarks uploaded to Cloudflare R2.
- Review and respond to formal price or schedule counter-proposals sent by the collector.
- Verify message transmission status and sent timestamps.

Field Description:

Field Name	Description
Chat Message Stream	Chronological message thread; user messages right-aligned, collector messages left-aligned.
Message Input Textbox	Text entry field; maximum 500 characters per message.
Attachment Button	File picker for uploading photos; maximum 5 MB per image; hosted on Cloudflare R2.
Counter-Proposal Bubble	Structured proposal card displaying adjusted unit price or rescheduled time window with "Accept" and "Decline" buttons.

2.2.5 Handover Receipt Confirmation & Net Payout (SCR-USR-09)

- **Mapped Use Case:** U-09: Confirm Handover Receipt (UC18)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-USR-09 — Scale verification photo preview, actual weighed weight breakdown table, total payout amount, and Confirm/Dispute action buttons]

This screen allows the User to:

- Audit the digital receipt generated on-site by the collector: actual scale weights, agreed unit prices, and sub-totals.
- Inspect high-resolution scale photos uploaded by the collector to verify weighing accuracy.
- Receive scrap payment directly from the collector (via physical cash or peer-to-peer direct bank transfer outside the platform).
- Tap "Confirm Receipt & Complete" upon receiving funds to close the order and claim Green Sprout points.
- Tap "Dispute Receipt" if weight or payout discrepancies cannot be resolved on-site (U-11).

Field Description:

Field Name	Description
Receipt ID	Unique digital receipt identifier; read-only.
Scale Photo Proof	Photo of the scale readout uploaded to Cloudflare R2; tap to view full screen.
Weighed Items Table	Itemized table: Category Name, Measured Weight (kg), Unit Rate (VND/kg), Total (VND).
Total Net Payout	Total cash payout figure handed directly to the user; formatted in VND.
Deviation Warning Banner	Visual alert banner triggered if measured weight deviates by more than {X}% from estimates (MSG-HND-04).

Eco-Points Preview Badge	Calculated Green Sprout point reward to be credited immediately upon confirmation.

2.2.6 Recycler Review & Star Rating Modal (SCR-USR-10)

- **Mapped Use Case:** U-10: Rate Recycler (UC19)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-USR-10 — 5-star interactive rating component, quick compliment tags, optional comment textarea, and submit button]

This screen allows the User to:

- Assign an overall star rating (1 to 5 stars) to the collector following a Completed transaction.
- Select quick compliment tags (e.g., "Punctual", "Accurate Scale", "Courteous", "Fair Price").
- Provide an optional written qualitative review.
- Tap "Submit Review" or "Skip".

[Content #3: Provide further descriptions for the screen's components/fields using table format below]

Field Description:

Field Name	Description
Star Rating Component	Interactive rating component; integer value between 1 and 5; mandatory.
Quick Compliment Tags	Multi-select chips: "Accurate Weighing", "Polite", "On-Time", "Fast Payment".
Written Review Input	Text input for qualitative feedback; maximum 300 characters.

2.2.7 Transaction Dispute & Complaint Submission (SCR-USR-11)

- **Mapped Use Case:** U-11: Submit Complaint (UC20)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-USR-11 — Dispute reason dropdown, detailed incident description, photo evidence upload, and submit ticket button]

This screen allows the User to:

- Select a formal complaint reason (Scale Discrepancy, Payment Withheld, Unprofessional Conduct, Other).
- Enter a detailed narrative description of the dispute incident.
- Upload supporting photo evidence stored in Cloudflare R2.
- Submit the dispute to the Staff arbitration queue (S-05) pursuant to BR-DSP.

Field Description:

Field Name	Description
Complaint Reason Category	Dropdown selector (mandatory): "Weight Inaccuracy", "Payment Discrepancy", "Misconduct", "Other".
Incident Description Textbox	Narrative explanation field; min 20 characters, max 1,000 characters; mandatory.
Evidence Photo Upload Area	Upload zone for supporting photos; max 3 photos; stored in Cloudflare R2; optional.

2.3 Gamification, Rewards & In-System Purchases

2.3.1 EcoLink Service Packages & Dynamic VietQR Checkout (SCR-USR-12)

- **Mapped Use Case:** U-16: Purchase In-System Services via VietQR (UC21)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-USR-12 — Pricing tier cards (EcoPlus, Ad-Free), checkout modal with dynamic VietQR code, countdown timer, and banking transfer details]

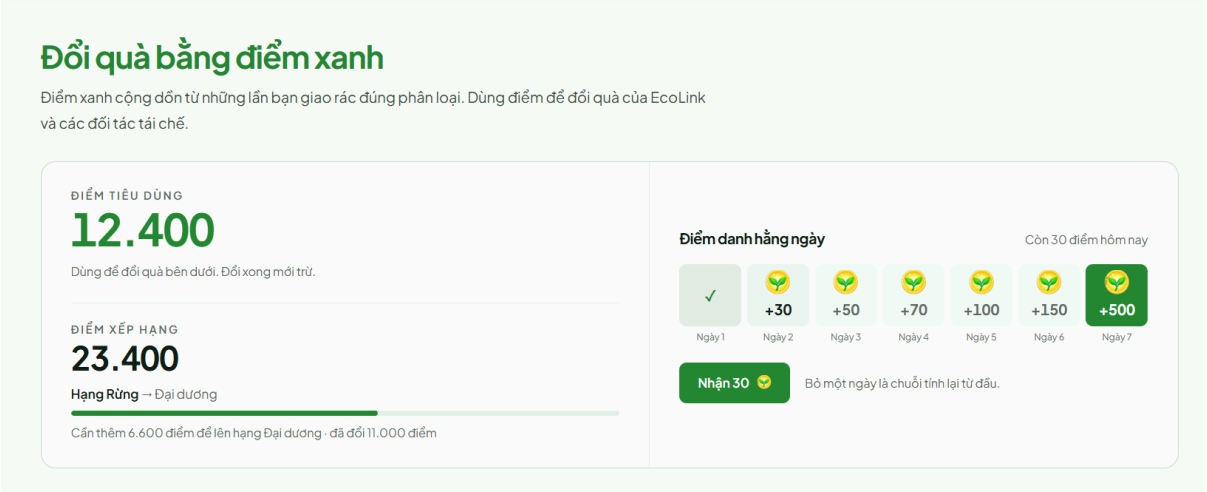
This screen allows the User to:

- Compare subscription tiers for in-system privileges (Ad-Free Experience, EcoPlus Green Contributor Badge, Priority Matching).
- Select a billing cycle (1 Month, 6 Months, 12 Months).
- Tap "Upgrade Now" to generate an order with a dynamic NAPAS 24/7 VietQR code.
  - Scan the QR code using any banking mobile app or copy account details to execute payment.
- Enjoy automatic privilege activation immediately upon webhook verification (BR-PAY-01, BR-PAY-02).

Field Description:

Field Name	Description
Package Selection Cards	Display cards highlighting features, entitlements, and prices configured via A-10.
Order Reference Code	Unique transaction reference string (e.g., SRV-20261005-9921); read-only.
Dynamic VietQR Image	Dynamically generated NAPAS 24/7 QR code embedding EcoLink's corporate account, exact price, and order reference.
Countdown Timer	15:00-minute countdown timer invalidating the checkout code upon reaching 00:00 (MSG-PAY-03).
Bank Account Details Block	Formatted copyable fields: Beneficiary Bank, Account Number, Account Name, Exact Amount, Memo.

2.3.2 Daily Check-in & Streak Reward Tracker (SCR-USR-13)



- **Mapped Use Case:** U-13: Daily Check-in (UC22)

[Content #1: UI layout (Wireframe or Mockup for the screen)]
[Placeholder for UI Mockup: SCR-USR-13 — 7-day streak calendar tracker, animated check-in button, current wallet balance, and milestone reward progress]

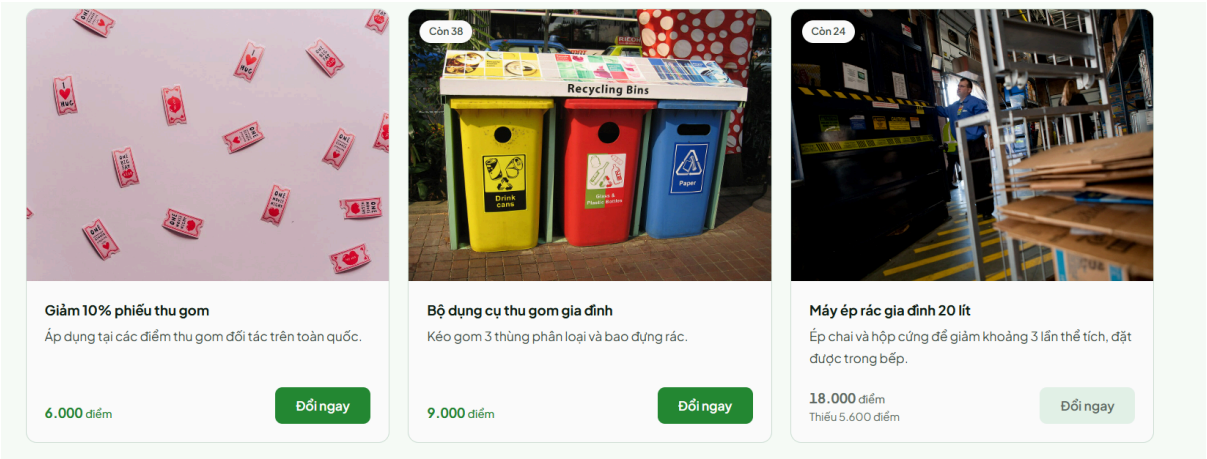
This screen allows the User to:

- Monitor their consecutive check-in streak count and weekly 7-day progress calendar.
- Preview today's base points and cumulative streak milestone bonus.
- Tap "Check-in Now" to claim daily Green Sprout points.
- View their total wallet point balance in real time.

Field Description:

Field Name	Description
Streak Day Counter	Integer count of consecutive active check-in days; resets to 1 upon an unexcused lapse.
7-Day Calendar Tracker	Graphical week tracker showing check marks for completed days and milestone rewards on Day 7.
Check-in Action Button	Action button enabled when eligible; transitions to disabled state with "Checked-in Today" label post-execution.
Wallet Point Balance	Total accumulated Green Sprout points in the user's account.

2.3.3 Eco-Rewards Gift Store & Voucher Redemption (SCR-USR-14)



- **Mapped Use Case:** U-14: Redeem Points for Gifts (UC23)
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-14 — Gift catalog grid with point badges, category filter (Voucher vs Physical), and confirm redemption modal]

This screen allows the User to:

- Browse reward catalog offerings including e-vouchers, sustainable merchandise, seeds, and potted plants.
- Filter items by reward type or sort by point cost.
- Verify live inventory availability before redeeming.
- Tap "Redeem Reward" to initiate an atomic point deduction and receive digital voucher codes or submit delivery details.

Field Description:

Field Name	Description
User Point Balance Display	Prominent point balance banner for quick purchasing power evaluation.
Gift Item Card	Catalog element displaying: Product Image (Cloudflare R2), Title, Point Cost, Stock Count, Category Badge.
Delivery Address Input	Shipping address field for physical merchandise redemptions; defaults to user profile.
Voucher Code Display	Revealed digital voucher code block with a one-tap copy button upon redemption.

2.3.4 My Virtual Tree & Eco Facts Garden (SCR-USR-15)

- **Mapped Use Case:** U-15: Upgrade Virtual Tree (UC24)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-15 — Interactive virtual garden, tree growth animation, level progress bar, upgrade CTA button, and unlocked Eco Fact cards]

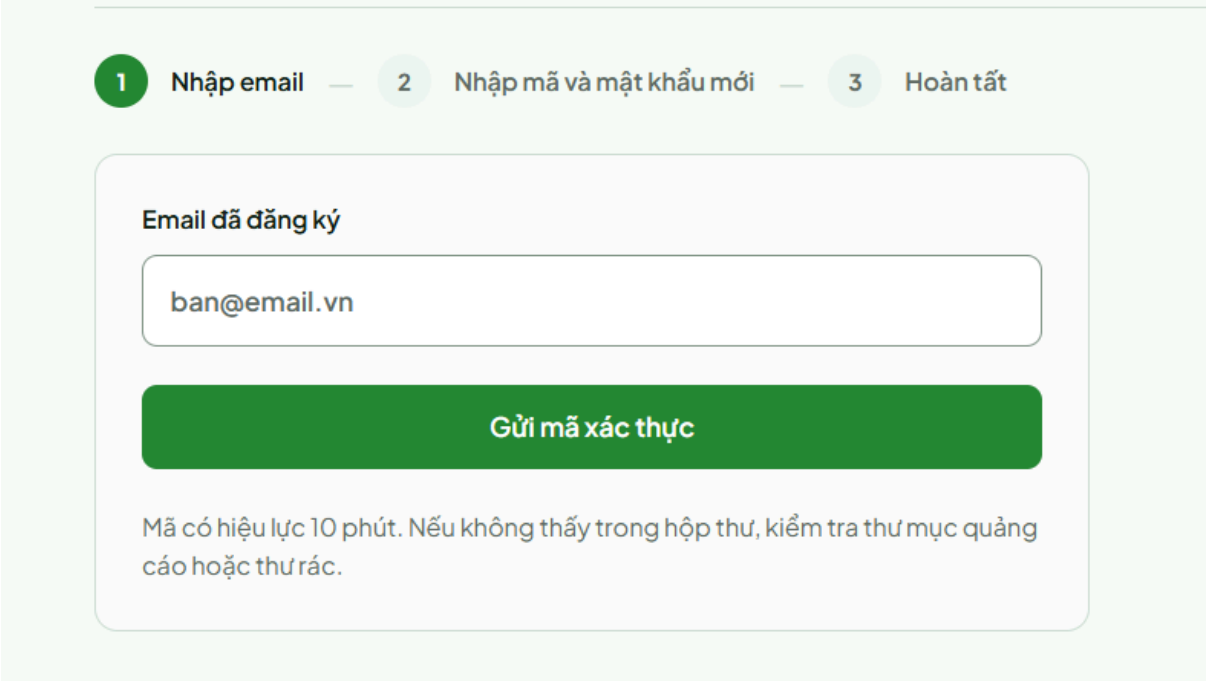
This screen allows the User to:

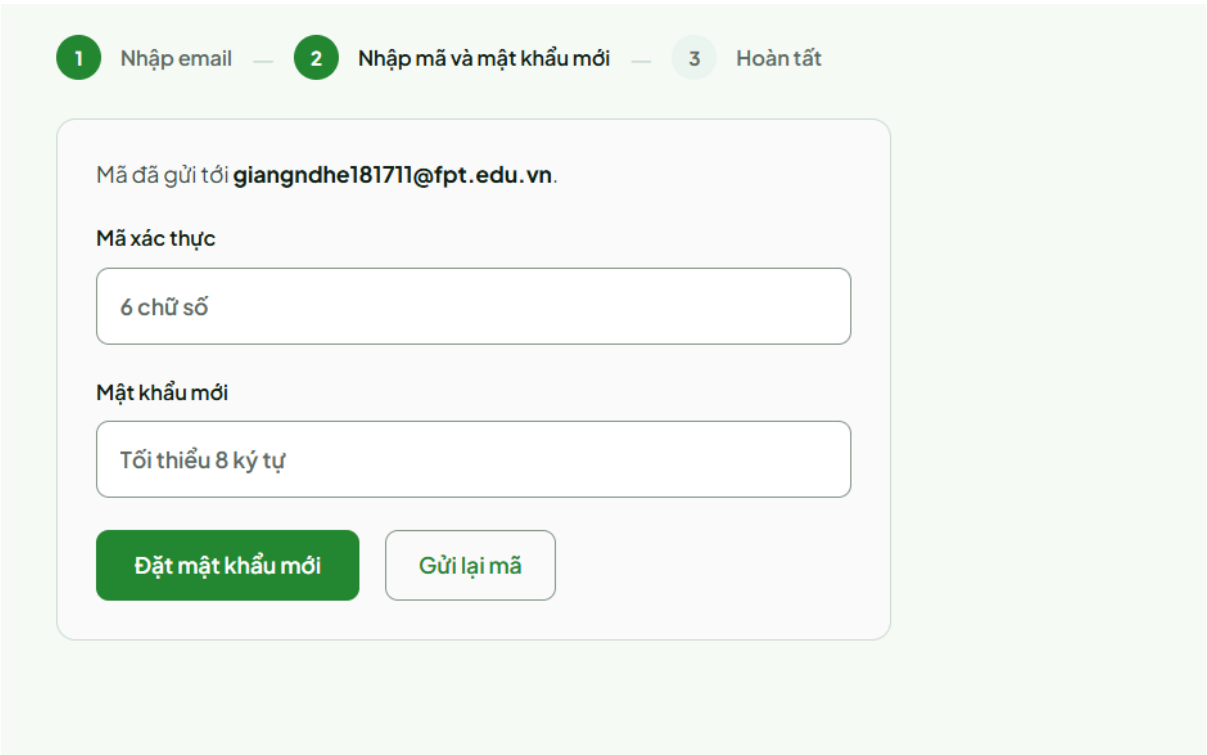
- Interact with their visual Virtual Tree across developmental stages (Seedling, Sprout, Young Tree, Ancient Canopy).
- Track growth percentage progress and point costs required for the next growth threshold.
- Tap "Water & Upgrade Tree" to invest Green Sprout points, evolving the tree graphic and triggering celebration animations.
- Access the "Eco Facts Handbook" to review all environmental facts unlocked throughout their journey.

Field Description:

Field Name	Description
Virtual Tree Graphic Area	Animated interactive canvas rendering the tree corresponding to its current level (Level 1 to Max Level).
Current Tree Level Label	Developmental stage title (e.g., "Level 3: Flourishing Young Tree").
Level Progress Bar	Visual percentage indicator measuring accumulated points toward the next level.
Upgrade Point Cost Badge	Points required for next evolution step.
Eco Fact Card List	Collection cards of unlocked environmental insights; each featuring an illustration and educational fact.

2.3.5 Forgot Password & Email OTP Verification (SCR-USR-16)





- **Mapped Use Case: SU-06**: Forgot Password
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-USR-16 — Email entry form, 6-digit OTP input boxes with countdown timer, and secure new password form]

This screen allows the User to:

- Input their registered email address to request a secure password recovery code.
- Enter the 6-digit One-Time Password (OTP) received via SendGrid email within a 10-minute validity window.
- Request a code re-send if not received after 60 seconds.
- Configure and confirm a new secure password to restore account access.

Field Description:

Field Name	Description
Registered Email Input	Account email address; standard valid email format (e.g., name@domain.com); maximum 100 characters; mandatory.
6-Digit OTP Input	Six individual numerical digit input cells; numeric characters only (0-9); auto-advancing cursor.
Resend OTP Countdown	60-second cooldown timer; unlocks resend trigger upon reaching zero.
New Password Input	Password field requiring at least 8 characters with uppercase, lowercase, and numbers.
Confirm Password Input	Password verification field; must match New Password Input identically.

3. RECYCLER Subsystem (Commercial Scrap Collectors)

3.1 Scrap Category & Pricing Administration

3.1.1 Accepted Categories & Scrap Pricing Sheet (SCR-RCY-01)

- **Mapped Use Case:** R-01: Manage Accepted Waste & Pricing

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-RCY-01 — List of master categories with toggle switch, min/max scrap price input per kg, and bulk update action]

This screen allows the Recycler to:

- Browse all system master waste categories and toggle purchase acceptance for each category.
- Configure purchasing prices (VND/kg) for each active category.
- Specify minimum weight thresholds and special collection criteria.
- Save the price sheet to publish rates to the public map and refresh pricing freshness timestamps.

Field Description:

Field Name	Description
Category Acceptance Toggle	Boolean toggle switch; active (True) indicates the facility purchases this material.
Purchase Price Input	Unit purchase price per kilogram; positive integer; currency: VND/kg (e.g., 4,500 VND/kg).
Minimum Quantity Input	Minimum threshold weight required for collection; decimal > 0; unit: kg (default: 1.0 kg).
Collection Conditions Text	Special acceptance conditions notes; maximum 200 characters.
Last Updated Timestamp	Freshness indicator; triggers stale pricing flags if unedited over 30 days (JOB-04).

3.1.2 Operating Status & GPS Online/Offline Switch (SCR-RCY-02)

- **Mapped Use Case:** R-02: Toggle Active Status

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

[Placeholder for UI Mockup: SCR-RCY-02 — Prominent online/offline switch banner, daily operating schedule picker, and immediate pause button]

This screen allows the Recycler to:

- Toggle their real-time operating availability (Online vs Offline) with a single tap.
- When Online: The facility displays an active green marker on user maps (U-05) and can receive incoming orders.
- When Offline: The facility is temporarily concealed from search results, while existing active orders remain operational (BR-MTH-06).

Field Description:

Field Name	Description
Online Availability Switch	Primary operational toggle; states: Online (Accepting Orders) vs Offline (Resting); takes effect immediately.
Operating Hours Schedule	Fixed weekly operating hours schedule (e.g., Monday – Saturday: 08:00 – 17:30).
Temporary Busy Reason Text	Optional status note explaining temporary closures (e.g., "Warehouse Capacity Reached"); max 100 characters.

3.2 Collection Fulfillment & Trade Settlement

3.2.1 Incoming Requests & Schedule Management (SCR-RCY-03)

- **Mapped Use Case:** R-03: Accept/Reject Handover Request, R-04: Confirm/Cancel Collection

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-RCY-03 — Kanban board or list of requests (Pending, Accepted, In-Progress), request cards with proposed time slots, Accept/Decline action buttons]

This screen allows the Recycler to:

- Monitor incoming trade requests dispatched by nearby users in real time.
- Review requested waste materials, estimated quantities, photo evidence, and **user-proposed time slots**.
- Tap "Accept" to lock in the appointment, officially **confirming the proposed collection time window** (BR-NEG-05).
- Tap "Reject" (supplying a reason) if capacity is constrained.
- Cancel a scheduled collection prior to execution if unforeseen emergencies arise (R-04).

Field Description:

Field Name	Description
Request Item Summary	Summary card showing Order ID, Seller Name, Distance (km), Material List, and Requested Time Slot.
Accept Action Button	Confirmation action advancing order to Accepted status, confirming the schedule, and unmasking seller contact info.
Reject Action Button	Rejection trigger prompting a mandatory rejection rationale modal.
Cancel Scheduled Button	Order cancellation button for confirmed appointments; requires mandatory cancellation reason (BR-HND-06).

3.2.2 Recycler Negotiation Chatbox (SCR-RCY-04)

- **Mapped Use Case:** R-05: Negotiate via Chatbox

  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-RCY-04 — Recycler chat view with quick proposal generator button for adjusting prices or proposing alternate time windows]

This screen allows the Recycler to:

- Chat with sellers in real time to coordinate logistical directions or resolve scheduling conflicts.
- Send structured price counter-proposals or proposed alternate time slots directly into the chat thread.
- Inspect seller scrap photos hosted on Cloudflare R2 before arriving on site.

Field Description:

Field Name	Description
Chat Conversation Feed	Persistent two-way message log between collector and seller; immutable audit trail (BR-NEG-01).
Proposal Generator Button	Modal trigger to generate formal price or schedule adjustments directly inside the conversation.
Photo Viewer Modal	High-resolution image viewer for examining waste quality on Cloudflare R2.

3.2.3 Digital Handover Receipt & Scale Proof Upload (SCR-RCY-05)

- **Mapped Use Case:** R-06: Create Handover Receipt

  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-RCY-05 — On-site digital receipt creation with actual weight input rows, scale photo capture button, price computation, and submit button]

This screen allows the Recycler (executed on-site during weighing):

- Enter the actual measured weight (kg) for each scrap item.
- Capture photos of the digital scale readout and upload them to Cloudflare R2.
- Compute total payout amounts automatically based on agreed unit rates.
- Tap "Issue Digital Receipt" to push confirmation requests to the seller's smartphone (U-09).
- Settle payment directly with the seller in cash or direct bank transfer.

Field Description:

Field Name	Description
Actual Measured Weight Input	Weight recorded on-site; positive decimal > 0; max 2 decimal places (e.g., 12.35 kg); mandatory.
Scale Photo Proof Capture	Direct camera capture of digital scale readout; uploaded to Cloudflare R2; mandatory minimum 1 photo (BR-HND-04).
Agreed Unit Price Display	Locked rate per kg agreed upon in the order; read-only.

Calculated Total Payout	Auto-computed payout sum (Weight multiplied by Unit Rate); currency: VND.
Submit Receipt Button	Action button dispatching the receipt to the seller and advancing order to Awaiting User Confirmation.

3.2.4 Partner Tier Upgrades via VietQR (SCR-RCY-06)

- **Mapped Use Case:** R-07: Purchase In-System Services via VietQR

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-RCY-06 — Recycler partner upgrade tiers (Priority Map Marker, Verified Blue Badge), dynamic VietQR checkout screen]

This screen allows the Recycler to:

- Browse commercial partner packages: Verified Partner Badge, Featured Map Marker, and Expanded Search Radius.
- Select subscription terms (3 Months, 6 Months, 12 Months).
- Scan dynamic NAPAS 24/7 VietQR codes to settle platform service fees.
- Receive automated badge activation upon webhook payment confirmation (BR-PAY-01, BR-PAY-02).

Field Description:

Field Name	Description
Recycler Package Card	Tier card outlining partner perks and prices set by Admin via A-10.
Partner VietQR Code	Dynamic QR image embedding partner reference codes and exact subscription prices; 15-minute validity.
Activation Status Badge	Visual verified partner badge displayed on the collector's public profile upon activation.

3.2.5 Recycler Password Reset via Email OTP (SCR-RCY-07)

- **Mapped** SU-06: Forgot Password

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-RCY-07 — Recycler password reset card with registered business email, 6-digit OTP boxes, and secure new password form]

This screen allows the Recycler to:

- Request credential resets by entering their registered corporate email address.
- Enter the 6-digit OTP code dispatched via SendGrid email within 10 minutes.
- Set a compliant new password to resume business operations.

Field Description:

Field Name	Description
Business Email Input	Registered business entity email address; standard valid email format (e.g., name@domain.com); mandatory.
6-Digit OTP Input	Numerical verification cells; 10-minute validity (BR-AUTH-08); protected by rate limits (BR-AUTH-09).
New Secure Password	High-strength password satisfying complexity standards.

4. Platform Governance, Moderation & Analytics

4.1 Operational Moderation & Quality Assurance (Staff Workbench)

4.1.1 AI Misclassification Review & Labeling Console (SCR-STF-01)

- **Mapped Use Case:** S-01: Review Misclassified Images, S-02: Label Image Data

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-01 — Two-pane review console: original image with AI prediction vs user suggestion on the left, master category labeling tool on the right]

This screen allows Support Staff to:

- Audit misclassification incident reports submitted by users from U-04.
- Compare the original photo against model predictions and user-suggested classes.
- Approve reports (confirming AI error) or Reject reports (if AI was correct or photo is invalid).
- Assign the verified true class label (S-02) to ready the image for retraining pipelines.

Field Description:

Field Name	Description
Review Queue Table	Tabular queue: Scan ID, Image Thumbnail, AI Class (% confidence), User Class, Date, Review Status.
High-Resolution Image Viewer	Zoomable image canvas displaying the original photo hosted on Cloudflare R2.
Verified Label Dropdown	Master catalog dropdown selector assigning the official ground-truth class.
Review Action Buttons	Action controls: "Approve & Label" (routes to dataset), "Reject" (dismisses report), "Escalate".
Review Notes Textarea	Auditor remarks explaining labeling decisions; maximum 200 characters.

4.1.2 Verified Training Dataset Management (SCR-STF-02)

- **Mapped Use Case:** S-03: Update Verified Dataset

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-02 — Dataset catalog with class distribution charts, train/validation split toggles, and dataset export button]

This screen allows Support Staff to:

- Monitor sample distributions across waste categories (plastic, metal, paper...).
- Allocate verified images into Train and Validation sets, ensuring strict zero-leakage separation (BR-CLS-07).
- Tag dataset milestone versions ready for YOLOv8 model retraining cycles.

Field Description:

Field Name	Description
Class Distribution Chart	Histogram chart displaying image sample counts per class with deficiency warnings (< 100 samples).
Split Allocation Ratio	Partition ratio selector: Default 80% Train / 20% Validation.

Export Dataset Trigger	Export action packing images and annotations for Background Job JOB-06.

4.1.3 Recycler Facility Verification Queue (SCR-STF-03)

- **Mapped Use Case:** S-04: Verify Recycler Application

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-03 — Partner verification queue, split-screen viewing applicant details, uploaded business license, and Approve/Reject modal]

This screen allows Support Staff to:

- Review pending partner registration filings submitted via G-03.
- Examine uploaded Business Registration Certificates and National ID scans hosted on Cloudflare R2.
- Cross-check declared street addresses against pinned satellite map coordinates.
- Approve applications to activate partner discovery and dispatch congratulations emails (MSG-MTH-04).
- Reject applications with mandatory descriptive feedback for remediation (BR-MTH-02, MSG-MTH-05).

Field Description:

Field Name	Description
Applicant Detail Cards	Overview cards showing Facility Name, Representative, Phone, and Pinned Address.
Document Preview Pane	Embedded viewer rendering license images and PDF documents hosted on Cloudflare R2.
Map Pin Verification	Satellite map view displaying facility coordinates for physical plausibility auditing.
Rejection Reason Textbox	Mandatory feedback text dispatched to the applicant upon rejection.

4.1.4 Dispute Arbitration & Complaint Resolution (SCR-STF-04)

- **Mapped Use Case:** S-05: Handle Complaints

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-04 — Ticket resolution interface displaying transaction history, chat log transcripts, scale photos, and arbitration ruling editor]

This screen allows Support Staff to:

- Adjudicate transaction disputes lodged by sellers or collectors (U-11).
- Examine immutable case files: Scale proof photos, itemized receipts, and entire chat transcripts (BR-DSP-06).
- Formulate binding arbitration rulings (BR-DSP-04): Dismiss dispute, Recommend reimbursement, or Issue conduct warnings.
- Publish formal outcomes, notifying both parties via email and push alerts (MSG-DSP-02).

Field Description:

Field Name	Description

Dispute Ticket Header	Ticket ID, Order Reference, Complainant Name, Reason Category, Priority Indicator.
Audit Trail Evidence Viewer	Unified dossier showing scale photos, estimated vs weighed weights, and full chat transcripts.
Ruling Outcome Dropdown	Adjudication selector: "Dismiss Complaint", "Uphold - Require Payout Adjustment", "Official Partner Warning".
Settlement Amount Input	Recommended reimbursement amount (VND).
Formal Resolution Summary	Mandatory written arbitration decision dispatched to both parties.

4.1.5 Account Management & Moderation Console (SCR-STF-05)

- **Mapped Use Case:** S-06: View Users/Buyers Accounts, S-07: Lock/Delete Accounts

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-05 — Paginated user directory with role filters, transaction summary, warning count, and account status toggle with reason prompt]

This screen allows Support Staff to:

- Search and filter all registered User and Recycler accounts.
- Review transaction logs, dispute histories, and platform reliability scores.
- Suspend or deactivate offending accounts with mandatory recorded justifications (BR-ADM-03).
- Enforce soft-deletion/deactivation policies, preventing deletion of accounts with historical trades (BR-ADM-04).

Field Description:

Field Name	Description
Account Directory Table	Data table displaying: Name, Email, Role, Joined Date, Completed Trades, Disputes, Status.
Account Detail Drawer	Sliding view rendering transaction logs, feedback ratings, and moderation audit trails.
Lock Account Modal	Confirmation dialog requiring a mandatory "Reason for Suspension" input.

4.1.6 Environmental Blog Authoring & Markdown Editor (SCR-STF-06)



- **Mapped Use Case:** S-08: Create Blog Post
  [Content #1: UI layout (Wireframe or Mockup for the screen)]
  [Placeholder for UI Mockup: SCR-STF-06 — WYSIWYG / Markdown blog editor with cover image uploader, tag selector, preview tab, and Submit for Approval button]

This screen allows Support Staff to:

- Compose environmental articles, recycling guides, and platform updates.
- Upload banner images hosted on Cloudflare R2 storage.
- Apply category classification tags.
- Preview mobile-responsive layouts.
- Submit drafts for Administrator publication approval (BR-ADM-08, MSG-ADM-06).

Field Description:

Field Name	Description

Blog Title Input	Article headline; length 10 to 150 characters; mandatory.
Cover Image Uploader	Banner image uploader; 16:9 aspect ratio, max 5 MB; hosted on Cloudflare R2.
Category Tags Select	Multi-select input for article topics.
Markdown / Rich-Text Editor	Formatting editor supporting headings, lists, quotes, and embedded images.
Submit for Approval Button	Transition action setting status to Pending_Approval for Admin review.

4.1.7 Eco Fact Knowledge Base Editor (SCR-STF-07)

- **Mapped Use Case:** S-09: Manage Eco Fact Content

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-07 — Catalog of Eco Facts organized by tree level milestones, fact editor form with graphic upload]

This screen allows Support Staff to:

- Create, curate, and order environmental education cards (Eco Facts).
- Map specific facts to Virtual Tree developmental milestones (U-15).
- Upload vector graphics and illustrations stored in Cloudflare R2.

Field Description:

Field Name	Description
Eco Fact Title Input	Short educational headline; max 80 characters.
Fact Content Textarea	Engaging environmental fact text; max 300 characters.
Tree Level Milestone Dropdown	Target tree level unlocking this card (e.g., Level 1, Level 2, Level 5).
Illustration Image Uploader	Graphic illustration file; hosted in Cloudflare R2.

4.1.8 Reward & Voucher Inventory Management (SCR-STF-08)

- **Mapped Use Case:** S-10: Manage Gift Inventory

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-08 — Inventory dashboard with stock alerts, new gift modal, voucher code batch importer, and physical dispatch queue]

This screen allows Support Staff to:

- Maintain the reward catalog in the Gift Store (U-14).
- Restock physical merchandise quotas or batch-import digital voucher codes.
- Set Green Sprout point redemption pricing.
- Fulfill physical merchandise shipments by inspecting shipping addresses and updating dispatch statuses.

Field Description:

Field Name	Description
Gift Name Input	Title of the merchandise or e-voucher; max 100 characters; mandatory.
Reward Type Radio	Classification: "Digital E-Voucher" vs "Physical Merchandise".

Required Points Input	Point redemption cost; positive integer > 0.
Current Stock Quantity	Live inventory count decremented atomically upon user redemption (BR-GAM-03).
Batch Voucher Import Textarea	Multi-line text field for importing batches of unique promo codes.
Dispatch Status Dropdown	Shipping lifecycle status: "Pending Packing", "In Transit", "Delivered".

4.1.9 Staff Work Shift Attendance Check-in (SCR-STF-09)

- **Mapped Use Case:** S-11: Staff Daily Check-in

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-STF-09 — Staff workspace dashboard with daily shift check-in button, attendance history log, and assigned queue metrics]

This screen allows Support Staff to:

- Clock in for their operational shift.
- View assigned review queues (pending partner audits, dispute tickets, misclassification reports).
- Inspect monthly attendance logs and shift records.

Field Description:

Field Name	Description
Shift Check-in Button	Clock-in execution trigger logging shift start timestamps into the audit ledger.
Attendance History Table	Shift records displaying: Work Date, Start Time, End Time, Tasks Completed.

4.2 Executive Analytics & System Administration (Admin Portal)

4.2.1 Executive Analytics Dashboard & KPIs (SCR-ADM-01)

- **Mapped Use Case:** A-01: View Dashboard & Analytics

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-01 — Executive dashboard with KPI metric cards, waste volume charts by material family, trade value graphs, and user growth analytics]

This screen allows the Administrator to:

- Inspect high-level Key Performance Indicators (KPIs) in real time: Total Diverted Waste (kg), Gross Trade Volume, New Accounts, and AI Top-1 Accuracy.
- Filter analytics across dynamic timeframes (Today, Past 7 Days, Month-to-Date, Year-to-Date).
- Analyze waste composition charts by material family.
- All metrics are read-only and derived strictly from Completed transactions (BR-ADM-07).

Field Description:

Field Name	Description
Date Range Filter Picker	Analytics timeframe filter supporting preset cycles and custom date pickers.
KPI Metric Cards	Summary cards displaying Total Scrap Weight (kg), Platform Revenue (VND), Active Users, Active Collectors.

Waste Material Breakdown Chart	Doughnut chart showing scrap weight percentage distribution across material families.
Transaction Volume Trend Line	Longitudinal chart tracking completed trade request volume over time.

4.2.2 Data Export Center & Immutable Audit Log Viewer (SCR-ADM-02)

- **Mapped Use Case:** A-02: Export Data Reports

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-02 — Export configuration form with format selectors (CSV, XLSX), report type dropdown, and immutable audit trail viewer]

This screen allows the Administrator to:

- Generate and download operational data reports for environmental reporting and financial audits.
- Select output formats: CSV or Microsoft Excel (XLSX).
- Automatically anonymize personal identifiable information pursuant to BR-ADM-06.
- Audit the immutable system activity log recording all Staff and Admin actions (BR-ADM-05).

Field Description:

Field Name	Description
Report Type Dropdown	Report classification: "Handover Trades", "Scrap Volumes", "Service Revenue", "Audit Logs".
Export Format Radio	Target format: "Microsoft Excel (.xlsx)" or "Comma Separated Values (.csv)".
Date Range Constraint	Timeframe boundaries; capped at 365 days per single bulk extraction.
Export Execution Button	Generation trigger recording export actor IDs and initiating download (MSG-ADM-02).
Audit Log Table	Tabular audit ledger: Timestamp, Actor ID, Client IP, Operation, Impacted Entity.

4.2.3 Staff Account Management & RBAC Permissions (SCR-ADM-03)

- **Mapped Use Case:** A-03: Manage Staff Accounts, A-04: Manage Staff Permissions

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-03 — Staff account directory, Create Staff modal, and Role-Based Access Control permission matrix checkboxes]

This screen allows the Administrator to:

- Maintain the roster of authorized Support Staff members.
- Provision new Staff accounts (Staff accounts are created exclusively by Admin pursuant to BR-ADM-01).
- Suspend or restore Staff access privileges.
- Fine-tune Role-Based Access Control (RBAC) permissions across operational domains (A-04).

Field Description:

Field Name	Description

Staff Name Input	Staff legal name; string between 2 and 50 characters; mandatory.
Staff Corporate Email Input	Assigned internal corporate email; mandatory; unique.
Temporary Password Input	Initial system-generated password; mandatory change required upon first sign-in.
Permission Matrix Checkboxes	Domain authorization matrix: AI Curation, Dispute Adjudication, Facility Audits, Inventory, Blog.
Account Status Toggle	Operational status switch: Active vs Deactivated.

4.2.4 Master Waste Categories Catalog Management (SCR-ADM-04)

- **Mapped Use Case:** A-05: Manage Master Waste Categories

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-04 — Master category tree list, Add/Edit category modal with icon uploader, recyclability flags, and deactivation toggle]

This screen allows the Administrator to:

- Manage the system master waste category hierarchy (PET, HDPE, Cardboard, Aluminum...).
- Add new categories, edit naming nomenclature, material family classifications, and vector icons (Cloudflare R2).
- Deactivate obsolete categories, enforcing foreign key integrity so referenced historical categories are never purged (BR-SRT-04).

Field Description:

Field Name	Description
Category Code Identifier	Machine identifier string (e.g., PLASTIC_PET_01); uppercase alphanumeric; unique.
Category Name Input	Official category label; length 3 to 80 characters; mandatory.
Material Family Dropdown	Family group: Plastic, Paper, Metal, Glass, E-Waste.
Recyclable Boolean Flag	Recyclability designation: Recyclable (True) vs Non-Recyclable (False).
Category Icon Uploader	Vector SVG or transparent PNG icon file; hosted on Cloudflare R2.
Category Active Status	Operational state: Active vs Inactive.

4.2.5 Rule Engine & Sorting Policy Configuration (SCR-ADM-05)

- **Mapped Use Case:** A-06: Configure Sorting Policy (Rule Engine)

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-05 — Rule Engine policy editor with sequential handling step builder, bin assignment dropdown, and publish action]

This screen allows the Administrator to:

- Formulate and maintain procedural handling policies in the Rule Engine for each waste category.
- Construct step-by-step cleaning procedures and map designated municipal bin receptacles.

- Publish new policy revisions; updates take effect on subsequent queries without modifying historical scan records (BR-SRT-01, BR-SRT-02).

Field Description:

Field Name	Description
Target Waste Category	Category bound to this rule; selected from Master Waste Categories.
Handling Step Item	Procedural item: Step Order (1, 2, 3...), Action Headline, Step Narrative.
Target Disposal Bin Type	Designated container type: Organic, Yellow Recyclables, General Waste, Hazardous Depot.
Average Market Price Range	Market scrap reference pricing range (Floor – Ceiling per kg).
Policy Status & Version	Policy revision tag (e.g., v2.1); exactly one active version per category (BR-SRT-01).

4.2.6 Gamification Settings & Tree Growth Config (SCR-ADM-06)

- **Mapped Use Case:** A-07: Manage Gamification Config

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-06 — Gamification configuration console with point conversion rates, daily caps, tree level thresholds, and streak multiplier tables]

This screen allows the Administrator to:

- Calibrate point accrual rates per kilogram of recycled material (e.g., 1 kg plastic = 10 points).
- Impose daily non-transaction point caps to prevent artificial point farming (BR-GAM-02).
- Configure daily check-in base rewards and consecutive streak multipliers.
- Establish Virtual Tree developmental stages and evolutionary point thresholds (U-15).

Field Description:

Field Name	Description
Point Conversion Rate Matrix	Point award formula per kilogram of verified Completed scrap.
Non-Transaction Daily Point Cap	Maximum point accumulation cap per calendar day for non-trade actions (e.g., 50 points/day).
Daily Check-in Base Points	Base reward credited per daily check-in; default: 5 points.
Streak Bonus Increment	Incremental point bonus per consecutive streak day; capped at 20 points/day.
Tree Level Threshold Table	Evolution matrix: Stage Level (1, 2, 3...), Title, Required Point Threshold.

4.2.7 Leaderboard Cycles & Reset Configuration (SCR-ADM-07)

- **Mapped Use Case:** A-08: Configure Leaderboard Cycles

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

[Placeholder for UI Mockup: SCR-ADM-07 — Leaderboard cycle settings with Weekly/Monthly/Quarterly radio, snapshot schedule picker, and top reward allocations]

This screen allows the Administrator to:

- Configure ecological leaderboard competitive cycles: Weekly, Monthly, or Quarterly.
- Schedule automated snapshot archival and point reset runs executed by Background Job JOB-05.
- Allocate bonus point honors for Top 1, Top 2, and Top 3 eco-champions.

Field Description:

Field Name	Description
Leaderboard Cycle Mode	Competition frequency: Weekly (Monday 00:00) / Monthly (1st of month) / Quarterly.
Snapshot Archive Retention	Archival retention duration for historical rankings; default: 12 months.
Top Rank Reward Points	Bonus point allocations granted to Top 1, 2, and 3 finishers upon cycle conclusion.

4.2.8 Blog Editorial Review & Publication Approval (SCR-ADM-08)

- **Mapped Use Case:** A-09: Approve / Reject Blog Post

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

  [Placeholder for UI Mockup: SCR-ADM-08 — Blog editorial queue, full-page preview with mobile simulation, Approve and Reject with Feedback buttons]

This screen allows the Administrator to:

- Review draft articles submitted by Support Staff from S-08.
- Preview formatted content and responsive graphics on an interactive mobile simulator.
- Tap "Approve & Publish" to release the post to the public live application (BR-ADM-08, MSG-ADM-05).
- Tap "Reject" to return the draft to the author with editorial feedback.

Field Description:

Field Name	Description
Editorial Review Queue	Submission table: Title, Author (Staff), Category, Submission Date, Status (Pending_Approval).
Article Live Preview	Full simulation view rendering Cloudflare R2 banner graphics, typography, and embedded tags.
Editorial Decision Actions	Decision triggers: "Approve & Publish" vs "Reject with Notes".
Editorial Feedback Textarea	Editorial critique remarks dispatched directly to the authoring Staff member upon rejection.

4.2.9 Service Package Fees & Financial Reconciliation (SCR-ADM-09)

- **Mapped Use Case:** A-10: Configure Transaction Fee, A-11: Manage Financial Reconciliation

  [Content #1: UI layout (Wireframe or Mockup for the screen)]

[Placeholder for UI Mockup: SCR-ADM-09 — In-system service package fee configuration tables, daily VietQR transaction ledger, reconciliation anomaly flags, and bank payout audit]

This screen allows the Administrator to:

- Set subscription prices for EcoLink in-system services for Users and Recyclers (Ad-Free, Verified Badges).
- Inspect the real-time financial ledger of all incoming NAPAS 24/7 VietQR payments.
- Flag and reconcile payment anomalies (e.g., customer transferred an inaccurate amount).
- Export financial reconciliation reports for accounting compliance.

Field Description:

Field Name	Description
Service Package Fee Table	Fee configuration: Package Name, Target Role (User/Recycler), Duration (1/6/12 Months), Official Price (VND).
VietQR Transaction Ledger	Audit ledger: Order Reference, Bank Transaction Code, Payer Account, Amount Received, Timestamp, Status.
Discrepancy Anomaly Flag	Red visual tag marking amount mismatches with an action button for manual reconciliation.
Corporate Bank Account Setup	Secure configuration block for EcoLink's official corporate beneficiary bank account.

5. Background Job Inventory

5.0 Overview Table

Job ID	Background Job Name	Group	Trigger / Execution Frequency	Primary Business Purpose	Linked Business Rules
JOB-01	Auto-Complete Handover Receipt	Scheduled	Every 60 minutes	Automatically completes pending handover receipts and credits Green Seed points if the User takes no action or submits no dispute within 48 hours.	**BR-HND-04**, **BR-GAM-01**, **BR-HND-07**
JOB-02	Reset Daily Check-in & Validate Consecutive Streak	Scheduled	Daily at 00:00 (ICT / GMT+7)	Refreshes daily check-in availability; evaluates user activity from the prior day and resets consecutive check-in streak to Day 1 if a day was missed.	**BR-GAM-09**
JOB-03	Expire Pending Handover Requests	Scheduled	Every 30 minutes	Automatically transitions pending scrap handover requests to Expired if no recycling	**BR-HND-02**, **BR-NTF-02**

				facility accepts or rejects them within the configured SLA window.	
JOB-04	Flag Outdated Scrap Price Lists	Scheduled	Daily at 01:00 (ICT / GMT+7)	Attaches a visual "Outdated Pricing" warning badge to recycler facility profiles if their scrap buying price list has not been updated for over 30 days.	**BR-MTH-03**
JOB-05	Close Leaderboard Cycle & Reset Top Rankings	Scheduled	Periodic (Weekly / Monthly / Quarterly configured by Admin)	Archives previous leaderboard cycle rankings into historical audit logs and resets active cycle points to zero to initiate a clean competition.	**BR-GAM-07**, **BR-ADM-05**
JOB-06	Export Curated Scrap Images for AI Retraining	Scheduled	Weekly on Sunday at 02:00 (ICT)	Automatically extracts staff-verified and re-annotated misclassification images to augment the dataset pipeline for model retraining.	**BR-CLS-06**, **BR-CLS-07**
JOB-07	Purge Stale Unverified Accounts	Scheduled	Daily at 03:00 (ICT / GMT+7)	Deactivates or purges unverified draft registrations exceeding the verification timeframe to release email and phone number identities.	**BR-AUTH-01**, **BR-ADM-04**
JOB-08	Enforce Dispute Window Expiration	Scheduled	Every 60 minutes	Closes dispute submission eligibility on completed handover transactions once the statutory dispute period (over {N} days) has elapsed.	**BR-DSP-01**
JOB-09	Send Handover Appointment Reminders	Automated Event Trigger	Triggered {X} hours prior to scheduled appointment	Dispatches automated push notifications reminding both the User and the Recycler of their upcoming scrap collection handover time and location.	**BR-NTF-04**, **BR-NEG-05**, **BR-NTF-05**

JOB-10	Cancel Expired In-System VietQR Payment Orders	Scheduled	Every 5 minutes	Scans pending in-system service orders (Verified Badges, Ad-Free packages) and transitions them to Expired if uncompleted past the 15-minute countdown.	**BR-PAY-01**, **BR-PAY-02**
JOB-11	Auto-Unlock Temporarily Locked Accounts	Scheduled	Every 5 minutes	Restores normal active status for accounts temporarily locked due to consecutive failed sign-in attempts once the {M}-minute lockout cooldown elapses.	**BR-AUTH-04**
JOB-12	Expire Unused Reward Vouchers	Scheduled	Daily at 00:05 (ICT / GMT+7)	Scans redeemed merchant voucher codes and marks expired ones as Invalid to prevent fraudulent redemption at partner outlets.	**BR-GAM-05**

5.1 Auto-Complete Handover Receipt (JOB-01)

Trigger: Scheduled — every 60 minutes (at the top of each hour).
Purpose: Protects recycling facilities from stalled settlements by automatically closing transactions and crediting user reward points when the user takes no action within 48 hours after receipt issuance.
Processing steps:

1. The system queries all handover receipts residing in Pending_User_Confirmation status created more than 48 hours prior to the current timestamp.
2. For each identified receipt, the system automatically transitions the transaction status to Completed.
3. Calculates the earned Green Seed points based on verified scrap weight and configured conversion matrices, then deposits the points into the user's reward balance (BR-GAM-01).
4. Seals the transaction record into immutable historical transaction ledgers where further editing or deletion is strictly prohibited (BR-HND-07).
5. Dispatches an automated push notification informing the user that their handover transaction has been auto-completed and reward points have been credited.

Failure behavior:

Scenario	System behavior
Database unreachable	Job pauses execution, records a warning event in system operational logs, and retries on the next scheduled execution cycle. No partial data mutations occur.
Open dispute exists on receipt	The system skips this receipt, preserving its current state pending customer support staff adjudication; auto-completion is strictly inhibited.

5.2 Reset Daily Check-in & Validate Consecutive Streak (JOB-02)

Trigger: Scheduled — daily at exactly 00:00 (ICT / GMT+7).

Purpose: Refreshes daily check-in eligibility for the new calendar day and resets consecutive check-in streaks if a user missed checking in on the previous day.
Processing steps:

1. Scans all active User and Staff accounts registered across the platform.
2. Resets the daily check-in state flag for all accounts to Unchecked.
3. Evaluates the most recent check-in timestamp for each user. If the user failed to perform a check-in during the preceding calendar day, the system resets their consecutive streak counter back to Day 1 pursuant to BR-GAM-09.
4. Records the count of initialized accounts into system maintenance logs.

Failure behavior:

Scenario	System behavior
Midnight scheduled execution outage	Upon system restoration, the job immediately verifies current calendar date checkpoints and executes compensation sweeps prior to allowing any user check-in operations.

5.3 Expire Pending Handover Requests (JOB-03)

Trigger: Scheduled — every 30 minutes.
Purpose: Prevents scrap collection requests from remaining indefinitely unresolved when recycling facilities do not respond within configured response windows.
Processing steps:

1. Scans all collection handover requests remaining in Pending status whose submission timestamp exceeds the administrative SLA timeout threshold (e.g., 24 hours).
2. Transitions the status of qualifying requests to Expired pursuant to BR-HND-02.
3. Dispatches informational push notifications to initiating users, advising them that their request has expired and suggesting alternative nearby recycling facilities (BR-NTF-02).

Failure behavior:

Scenario	System behavior
Recycler accepts request during sweep	The user-driven acceptance action takes precedence; transaction transitions to Accepted, and the background job skips the record.

5.4 Flag Outdated Scrap Price Lists (JOB-04)

Trigger: Scheduled — daily at 01:00 AM (ICT / GMT+7).
Purpose: Enforces pricing transparency by alerting users when a recycling facility has not maintained scrap purchasing rates within the past 30 days.
Processing steps:

1. Evaluates all active recycling facility profile records within the platform.
2. Compares the last updated timestamp of each facility's scrap price list against the current system date. If the delta exceeds 30 calendar days, the system attaches an Outdated Pricing Warning badge to the public facility profile (BR-MTH-03).
3. Generates an automated portal reminder advising the facility owner to review and republish current scrap prices.

Failure behavior:

Scenario	System behavior
Push notification service outage	Logs the error; the application of the visual warning badge on the database and public profile proceeds unaffected.

5.5 Close Leaderboard Cycle & Reset Top Rankings (JOB-05)

Trigger: Scheduled — periodic execution at cycle conclusion configured by Administrator (Weekly, Monthly, or Quarterly).

Purpose: Archives completed competition cycle rankings into permanent historical records and initializes a zero-point competition period for active participants.

Processing steps:

1. Compiles and snapshots final ranking standings and cycle points into historical archive tables (BR-ADM-05).
2. Resets cycle competition points to zero for all participants (while preserving lifetime reward wallet balances).
3. Enforces calculation criteria: points are derived exclusively from completed, non-disputed handover transactions (BR-GAM-07).
4. Dispatches congratulatory award notifications and honors to top leaderboard finishers.

Failure behavior:

Scenario	System behavior
Snapshot archival fails	The job immediately aborts the point reset routine, preserves existing leaderboard standings intact, and issues a critical alert to prevent historical data loss.

5.6 Export Curated Scrap Images for AI Retraining (JOB-06)

Trigger: Scheduled — weekly on Sunday at 02:00 AM (ICT / GMT+7).

Purpose: Extracts high-quality verified waste images re-annotated by Support Staff to feed downstream model fine-tuning workflows.

Processing steps:

1. Identifies classification dispute records where Support Staff verified, resolved, and corrected item bounding boxes and category labels during the week (BR-CLS-06).
2. Cross-references image hashes to guarantee selected samples do not exist in test benchmark splits, preventing AI data leakage (BR-CLS-07).
3. Bundles verified image assets alongside standardized JSON annotations and synchronizes them to the training dataset storage partition.
4. Marks processed records as exported to prevent redundant extraction in subsequent cycles.

Failure behavior:

Scenario	System behavior
No newly curated images available	Job completes immediately with zero operations; logs a normal informational status.

5.7 Purge Stale Unverified Accounts (JOB-07)

Trigger: Scheduled — daily at 03:00 AM (ICT / GMT+7).

Purpose: Releases orphaned usernames, email addresses, and phone numbers held by abandoned registrations that failed email verification.

Processing steps:

1. Scans user account entities in Unverified status whose registration creation timestamp exceeds the statutory cutoff (e.g., 7 days).
2. Deactivates or purges draft records to liberate identity identifiers for prospective legitimate users (BR-AUTH-01).
3. Enforces historical protection rules: accounts associated with past transactions are strictly immune from deletion (BR-ADM-04).

Failure behavior:

Scenario	System behavior
Database write timeout	Rolls back current batch, logs warning event, and schedules retry for the subsequent night.

5.8 Enforce Dispute Window Expiration (JOB-08)

Trigger: Scheduled — every 60 minutes.
Purpose: Closes dispute submission eligibility on completed handover transactions once the statutory dispute period has elapsed, preserving financial ledger stability.
Processing steps:

1. Queries handover transactions in Completed status whose completion timestamp exceeds the maximum dispute eligibility window ({N} days).
2. Transitions dispute eligibility status to Dispute_Window_Closed pursuant to BR-DSP-01.
3. Deactivates the dispute initiation control within the user interfaces of both participating parties.

Failure behavior:

Scenario	System behavior
Database connectivity interruption	Retries automatically on the subsequent execution interval with zero collateral data impact.

5.9 Send Handover Appointment Reminders (JOB-09)

Trigger: Automated Event Trigger — executes {X} hours prior to scheduled appointment timestamp.
Purpose: Minimizes appointment no-shows and scheduling lapses by proactively notifying both user and collector before the agreed collection slot.
Processing steps:

1. Identifies collection handover agreements finalized by mutual mutual consent scheduled for fulfillment within the reminder window (BR-NEG-05).
2. Dispatches real-time push notifications to mobile and web clients of both User and Recycler (BR-NTF-04).
3. Formulates reminder notifications containing appointment time, address location, and waste preparation reminders, strictly omitting sensitive payment data (BR-NTF-05).

Failure behavior:

Scenario	System behavior
User client offline / unreachable	Push payload is retained in Firebase Cloud Messaging message queue and delivered upon client reconnect.

5.10 Cancel Expired In-System VietQR Payment Orders (JOB-10)

Trigger: Scheduled — every 5 minutes.
Purpose: Automatically cancels dynamic VietQR payment orders that exceed the 15-minute countdown window without payment confirmation, ensuring inventory and order integrity.
Processing steps:

1. Scans pending in-system service orders (e.g., Verified Green Badges, Ad-Free subscriptions) remaining in Pending_Payment status older than 15 minutes.
2. Transitions order status to Expired and deactivates the associated dynamic VietQR code pursuant to BR-PAY-01.
3. Records session termination in the payment ledger to prevent accidental post-expiry service provisioning (BR-PAY-02).

Failure behavior:

Scenario	System behavior
Bank webhook arrives during cancellation	System verifies bank transfer timestamp: if payment was confirmed before the 15-minute threshold, payment takes precedence and service is activated.

5.11 Auto-Unlock Temporarily Locked Accounts (JOB-11)

Trigger: Scheduled — every 5 minutes.
Purpose: Restores sign-in access automatically for accounts temporarily locked due to consecutive invalid credentials once the lockout period has expired.
Processing steps:

1. Queries all account records residing in Temporarily_Locked status resulting from exceeding failed password attempt thresholds.
2. Evaluates the lockout timestamp against current system time; if the elapsed duration meets or exceeds {M} minutes (pursuant to BR-AUTH-04), resets failed attempt count to zero.
3. Restores account status to Active, enabling the user to resume normal authentication attempts.

Failure behavior:

Scenario	System behavior
No locked accounts identified	Job finishes cleanly with zero database modifications.

5.12 Expire Unused Reward Vouchers (JOB-12)

Trigger: Scheduled — daily at 00:05 AM (ICT / GMT+7).
Purpose: Invalidates redeemed merchant vouchers that have passed their designated expiration date, preventing invalid redemption disputes with partner vendors.
Processing steps:

1. Evaluates all redeemed voucher records held in user reward wallets that remain in Unredeemed status.
2. Compares voucher expiration validity dates against the current date; transitions expired codes to Invalid_Expired pursuant to BR-GAM-05.
3. Updates the user wallet view, displaying expired vouchers with a muted visual state and an "Expired" indicator.

Failure behavior:

Scenario	System behavior
Network timeout during status batch update	Job captures warning log and schedules a catch-up execution batch.

6. External API Inventory

6.0 Overview Table

Service ID	External System / Interface Name	Provider	Data Exchange Direction	Primary Business Capability	Linked Business Rules

EXT-01	AI Waste Classification Service (FastAPI YOLOv8)	Internal AI Subsystem	EcoLink Backend ↔ AI Service	Ingests waste photograph uploads and returns predicted category classifications alongside detection confidence metrics.	**BR-CLS-01**, **BR-CLS-02**, **BR-CLS-03**
EXT-02	Cloud Storage Service (Cloudflare R2 Storage)	Cloudflare	EcoLink Backend ↔ Cloudflare R2	Securely uploads, stores, and serves waste photos, handover receipts, profile avatars, and chat attachments via S3 API with zero egress fees.	**BR-CLS-01**, **BR-CLS-08**
EXT-03	Google Identity Service (Google OAuth 2.0)	Google Identity Platform	EcoLink Client/Backend ↔ Google	Supports rapid, secure user authentication utilizing verified Google identity credentials.	**BR-AUTH-01**, **BR-AUTH-02**, **BR-AUTH-07**
EXT-04	Dynamic VietQR Payment Gateway	VietQR.io	EcoLink Backend ↔ VietQR Gateway	Generates dynamic NAPAS 24/7 QR codes and processes IPN webhooks for in-system platform service packages (Verified Badges, Ad-Free).	**BR-PAY-01**, **BR-PAY-02**
EXT-05	Map & Geocoding Service (OpenStreetMap / Leaflet)	OpenStreetMap Community	EcoLink Client/Backend ↔ Map Server	Visualizes interactive digital maps, geocodes physical addresses, and displays nearby recycling facilities.	**BR-MTH-04**, **BR-MTH-05**

EXT-06	Web Push Notification Service (Firebase Cloud Messaging)	Google Firebase	EcoLink Backend → User Web Browser	Dispatches instant Web Push Notifications upon receipt acceptance, chat interactions, and appointment milestones.	**BR-NTF-01**, **BR-NTF-02**, **BR-NTF-05**
EXT-07	Transactional Email Service (Twilio SendGrid / SMTP)	Twilio SendGrid	EcoLink Backend → User Mailbox	Transmits security OTPs, single-use password reset links, and formal account status notices.	**BR-AUTH-06**, **BR-AUTH-09**, **BR-NTF-01**, **BR-NTF-03**, **BR-NTF-06**

6.1 AI Waste Classification Interface (EXT-01)

URL	POST: /api/v1/ai/waste/classify (Internal Microservice Endpoint)
Purpose	Receives image payloads of scrap waste and returns predicted material categories with bounding box coordinates and confidence ratings.
Caller	EcoLink Backend Core Subsystem
Auth	Internal Service-to-Service API Key / Bearer Token Header

6.2 Cloudflare R2 Cloud Object Storage Interface (EXT-02)

URL	PUT / GET: https://.r2.cloudflarestorage.com// (S3 Compatible API)
Purpose	Persists waste inspection photos, handover receipts, user avatars, and in-chat attachments with enterprise high availability and zero egress fees.
Caller	EcoLink Backend File Storage Manager
Auth	AWS S3 Signature v4 (Access Key ID & Secret Access Key)

6.3 Google OAuth 2.0 Identity Interface (EXT-03)

URL	GET / POST: https://accounts.google.com/o/oauth2/v2/auth & https://oauth2.googleapis.com/token
Purpose	Facilitates single-sign-on (SSO) authentication for Users and Recyclers using verified Google accounts.
Caller	EcoLink Web Frontend & Backend Identity Manager
Auth	OAuth 2.0 Authorization Code Flow (Client ID & Client Secret)

6.4 VietQR Payment Gateway Interface (EXT-04)

URL	Dynamic QR Generation: POST: https://api.vietqr.io/v2/generate Inbound Instant Payment Webhook (IPN): POST: /api/v1/payments/vietqr/webhook
Purpose	Generates dynamic banking QR codes and receives instant payment webhooks for in-system EcoLink service packages (Verified Badges, Ad-Free subscriptions).
Caller	EcoLink Backend (Generation) & VietQR / Partner Bank Gateway (Inbound Webhook)
Auth	API Key / Client ID (Outbound) & HMAC-SHA256 Cryptographic Signature Verification (Inbound Webhook)

6.5 OpenStreetMap & Nominatim Geocoding Interface (EXT-05)

URL	GET: https://nominatim.openstreetmap.org/search & https://tile.openstreetmap.org/{z}/{x}/{y}.png
Purpose	Converts human-readable street addresses to geographic coordinate pairs (Latitude/Longitude) and serves interactive raster map tiles.
Caller	EcoLink Web Frontend Map Component & Backend Address Resolver
Auth	None required; compliant with OSM Foundation HTTP User-Agent Identification Policy

6.6 Firebase Cloud Messaging (FCM) Push Service (EXT-06)

URL	POST: https://fcm.googleapis.com/v1/projects/{project_id}/messages:send
Purpose	Dispatches real-time browser push notifications to desktop and mobile browsers for immediate operational alerts.
Caller	EcoLink Backend Notification Dispatcher
Auth	Google Service Account OAuth 2.0 Bearer Token

6.7 Twilio SendGrid Transactional Email Interface (EXT-07)

URL	POST: https://api.sendgrid.com/v3/mail/send
Purpose	Transmits critical security emails including single-use password reset tokens, account activation notices, and dispute rulings.
Caller	EcoLink Backend Email Dispatch Service
Auth	Twilio SendGrid Bearer API Secret Key

IV. Non-Functional Requirements

1. External Interfaces

The EcoLink system communicates properly with end users, client devices, internal services, and designated external third-party service providers through the following standardized interfaces:

ID	Interface	Description	Protocol / Format
EI-01	User Interface (Web)	Responsive React web application serving Guest, User, Recycler, Staff, and Administrator actors. Access to client hardware features (device camera for AI waste capture and Geolocation API for GPS recycler discovery) requires explicit runtime user consent.	HTTPS, HTML5, CSS3, JavaScript
EI-02	AI Classification Service	Internal inference microservice hosting the self-trained computer vision waste classification model. The backend transmits image payloads over a secure private network and receives top-N predicted categories, material groups, and confidence scores.	REST / JSON (Internal Virtual Network)

EI-03	Map & Geolocation Service	Integrates with map service provider (OpenStreetMap / Google Maps) and PostGIS database extension for address geocoding, interactive map rendering, distance computation, and spatial radius query execution.	REST (Map Provider), SQL (PostGIS)
EI-04	Object Storage (Cloudflare R2)	Cloud object storage storing waste photos, handover receipt evidence, dispute proof photos, facility business licenses, user avatars, and blog media. Relational database stores unique object keys only.	S3-Compatible API (Private Bucket)
EI-05	Google OAuth 2.0	External authentication provider for Single Sign-On (SSO) and quick account registration. The system verifies identity tokens directly with Google and validates the email_verified claim on every login (BR-AUTH-02, BR-AUTH-07).	OAuth 2.0 / OpenID Connect (HTTPS)
EI-06	Gmail / SMTP Mail Service	Automated email delivery service used to dispatch account verification links, password reset links/OTPs, handover status alerts, dispute resolution conclusions, and periodic survey invitations.	SMTP over TLS (Port 587)
EI-07	Real-time Messaging Channel	Full-duplex real-time communication channel powering negotiation chat (NEG) between User and Recycler and delivering instantaneous in-app system notifications (NTF).	WebSocket (WSS) with polling fallback
EI-08	Database Interface	High-performance relational database storage with PostGIS extension for structured business entities, relational constraints, transaction records, and spatial queries.	TCP / TLS (Private Virtual Network)
EI-09	Data Export Interface	Generates downloadable business analytics, transaction ledgers, and environmental statistics reports for administrative audit and external reporting.	.xlsx / .csv (UTF-8 Encoded)

2. Quality Attributes

The required software quality attributes and behavioral characteristics of the EcoLink platform are specified below across distinct quality dimensions:

2.1 Usability

ID	Requirement	Measure
NFR-USA-01	A first-time User must be able to complete the core flow ('scan waste -> view sorting guidance') without formal training.	<= 3 operational steps; >= 90% of test users complete unaided on first attempt
NFR-USA-02	All primary actions (AI scan, upload photo, view sorting guide, find nearby recycler, create handover request, daily check-in) must be directly accessible from the main dashboard.	<= 2 clicks or taps from the home landing view
NFR-USA-03	System error, confirmation, and validation messages must be presented in clear, non-technical language and conform strictly to the System Messages catalog (MSG-*).	100% of user-facing dialogs; system stack traces and database errors are strictly suppressed
NFR-USA-04	The user interface must provide comprehensive bilingual language support for Vietnamese (default) and English.	Instantaneous toggle across all screens without loss of entered form data or page reload
NFR-USA-05	The web application interface must provide a consistent, responsive layout supporting desktop, tablet, and mobile browsers.	Seamless viewport scaling across widths ranging from 360 px to 1920 px
NFR-USA-06	The application interface must adhere to international accessibility and readability guidelines.	Conforms to WCAG 2.1 AA standards; all interactive touch targets are at least 44x44 px
NFR-USA-07	Long-running asynchronous operations (AI image classification, media upload, report generation) must display active visual progress indicators.	Visual progress state displayed within <= 200 ms of user trigger; includes cancel/retry options
NFR-USA-08	A Recycler must be able to review incoming handover requests, negotiate terms, create digital receipts, and confirm completion in a concise flow.	<= 5 steps; <= 2 minutes total operational time for a trained Recycler
NFR-USA-09	Operations Staff must be able to compose rich-text educational blog articles and submit them for administrative review without technical formatting knowledge.	<= 5 steps from clicking 'New Post' to final submission for Admin review

2.2 Performance

ID	Requirement	Target	Condition
NFR-PER-01	AI classification response latency	Result returned within <= 5 seconds (95th percentile); hard timeout at 15 seconds	Image size <= 10 MB, up to 100 concurrent classification requests
NFR-PER-02	Web page initial load time	All non-media web views render fully within <= 3 seconds (95th percentile)	Up to 100 concurrent active users on standard 4G mobile connection

NFR-PER-03	Nearby-recycler spatial search latency	Spatial search results and map marker coordinates returned within <= 2 seconds	Search radius up to 20 km, up to 100 concurrent spatial search queries
NFR-PER-04	Standard transactional API response latency	System acknowledges submission (login, create request, confirm receipt, check-in, redeem) within <= 2 seconds (P95)	Up to 100 concurrent users active simultaneously
NFR-PER-05	Real-time negotiation chat message delivery	End-to-end message transmission latency <= 2 seconds	Standard network connectivity between User and Recycler
NFR-PER-06	Dashboard analytics rendering & report export	Dashboard metrics render within <= 5 seconds; report generation completes within <= 30 seconds	Aggregated datasets containing up to 100,000 transaction records
NFR-PER-07	Concurrent active user capacity	System remains fully operational with zero request loss or stability degradation	100 concurrent active users on single-server deployment baseline
NFR-PER-08	Sorting catalog & leaderboard lookup	Active sorting policies and leaderboard rankings retrieved within <= 2 seconds	High-concurrency cached read paths under normal system load

2.3 Security

ID	Requirement	Description	Related Rules
NFR-SEC-01	Mandatory authentication gate	All application screens and API endpoints require authenticated sessions, excluding public landing pages, waste catalog search, approved blog posts, login, registration, and password recovery.	BR-AUTH-08
NFR-SEC-02	Cryptographic password protection	Passwords must never be stored in plaintext or reversible encryption; stored using salted adaptive cryptographic hashing (bcrypt or Argon2). Minimum length 8 characters including uppercase letters and digits.	BR-AUTH-04
NFR-SEC-03	Account lockout protection	Accounts are temporarily locked for 15 minutes after 5 consecutive failed sign-in attempts to prevent brute-force credential stuffing; automatically unlocked by background job JOB-01.	BR-AUTH-04, JOB-01

NFR-SEC-04	Role-based access control (RBAC)	Enforces strict server-side authorization and separation of duties across Guest, User, Recycler, Staff, and Administrator roles in accordance with the Permission Matrix.	BR-AUTH-08, BR-ADM-02
NFR-SEC-05	Transport layer encryption	All client-to-server and inter-service communications must be encrypted in transit using HTTPS/TLS 1.2+ with HTTP Strict Transport Security (HSTS) headers enforced.	Industry Standard
NFR-SEC-06	Secure session lifecycle	JWT access token lifetime is limited to <= 60 minutes [TBD]; all active user sessions are invalidated immediately upon password change, account suspension, or deactivation.	BR-AUTH-05, BR-AUTH-06
NFR-SEC-07	Anti-enumeration defenses	Authentication failure and password recovery responses provide identical generic messages regardless of whether the target username or email exists in the database.	BR-AUTH-03, BR-AUTH-06
NFR-SEC-08	API rate limiting & abuse mitigation	Sensitive public endpoints enforce strict rate limiting: AI scan and recycler search <= 30 req/min/user; password reset restricted per email/IP; login restricted per IP.	BR-AUTH-09
NFR-SEC-09	File upload sanitization & inspection	All uploaded media files are inspected for valid magic numbers, restricted to JPG/PNG, capped at <= 10 MB, and stripped of sensitive EXIF geolocation metadata upon ingestion.	BR-CLS-01
NFR-SEC-10	Private object storage authorization	Stored media files (waste photos, receipts, dispute proof) remain in private buckets, served strictly via short-lived signed URLs generated after server-side access verification.	BR-CLS-08, BR-DSP-06
NFR-SEC-11	Location data privacy protection	Precise user GPS coordinates are utilized solely on the server side to compute distances and are never exposed to Recyclers. Exact pickup address is shared only after request acceptance.	BR-MTH-05, BR-HND-08
NFR-SEC-12	Application vulnerability defenses	Application is hardened against OWASP Top 10 vulnerabilities: parameterized queries prevent SQL injection, strict output encoding prevents XSS, and anti-forgery tokens prevent CSRF attacks.	OWASP Top 10

NFR-SEC-13	Secrets & credentials management	Database credentials, cloud storage keys, and OAuth client secrets must reside in isolated environment variables or secret vaults; hardcoding in source code is strictly prohibited.	Industry Standard
NFR-SEC-14	Immutable audit logging	All administrative operations, role modifications, recycler approvals/rejections, dispute decisions, blog approvals, and data exports are logged to an append-only audit trail; retained for 90 days.	BR-ADM-05, JOB-02
NFR-SEC-15	Transaction record immutability	Negotiation chat transcripts, accepted offer snapshots, confirmed receipts, and dispute arbitration evidence cannot be edited or deleted by any user or administrator.	BR-NEG-01, BR-HND-07
NFR-SEC-16	OAuth token server-side verification	Google OAuth ID tokens must be cryptographically validated with Google public keys on every login; accounts registered via Google default strictly to the User role.	BR-AUTH-07
NFR-SEC-17	QR code redirect safety	Scanned QR codes must validate against internal system-issued identifiers; external URL redirections are strictly blocked to prevent open redirect vulnerabilities.	BR-SRT-05
NFR-SEC-18	Notification data protection	Automated email and push notifications must never contain sensitive authorization tokens or full contact details, except one-time account verification or password reset links.	BR-NTF-05
NFR-SEC-19	Personal data regulatory compliance	Data collection complies with Vietnam Personal Data Protection Decree (Decree 13/2023/ND-CP); user accounts with historical transactions are deactivated rather than hard-deleted.	BR-ADM-04, BR-ADM-06
NFR-SEC-20	Content moderation integrity gate	Blog articles drafted by Staff remain completely inaccessible to public users until an Administrator formally reviews and approves the submission.	BR-ADM-08

2.4 Reliability & Availability

ID	Requirement	Target	Condition

NFR-REL-01	Overall system availability	>= 99.0% service uptime measured monthly	Excludes announced scheduled maintenance windows (<= 4 hours/month)
NFR-REL-02	AI classification fault tolerance	Displays user-friendly notification (MSG-CLS-05); auto-retries once; prompts user to retry or browse waste catalog manually	AI inference microservice temporarily unreachable or times out (> 15 s)
NFR-REL-03	Handover state transition integrity	Strict one-way lifecycle progression (Pending -> Accepted -> Scheduled -> Completed); states are never lost, skipped, or duplicated	Concurrent User and Recycler updates handled via database transactions and optimistic locking
NFR-REL-04	Atomic points & reward transactions	Deducting Eco-points, updating voucher inventory stock, and issuing unique voucher codes execute as a single atomic transaction; point balance cannot be negative	High-concurrency voucher redemptions and daily check-in events
NFR-REL-05	Graceful third-party degradation	Transient failures of external services (Google Maps, Gmail SMTP, WebSockets) must not crash core web applications or block scan/handover flows	External third-party API outage or network disruption
NFR-REL-06	Reliable email notification delivery	Failed email transmissions are logged and retried automatically up to {N}=3 times; delivery failures never roll back core business transactions	Temporary SMTP gateway disruption or mail server timeout (BR-NTF-06)
NFR-REL-07	Background worker resilience	Scheduled background workers (account unlock JOB-01, log cleanup JOB-02, reminder dispatch, request expiry) are idempotent with automatic retry and error alerts	Worker execution failures or temporary database timeouts

2.5 Scalability

- NFR-SCA-01: Microservice AI Architecture The AI Classification Service is deployed and scaled independently in an isolated container/microservice, ensuring heavy computer-vision inference loads do not degrade web and transaction responsiveness.
- NFR-SCA-02: Spatial Index Optimization PostGIS GiST spatial indexing on facility coordinates ensures nearest-neighbor and radius queries execute within <= 2 seconds even as the directory scales to over 10,000 recycling partners.
- NFR-SCA-03: Stateless Backend Services The .NET API backend is designed to be fully stateless, enabling seamless horizontal scale-out across multiple container instances behind a reverse proxy/load balancer.
- NFR-SCA-04: Database Capacity Sizing The PostgreSQL database schema and indexing strategy are optimized to handle up to 100,000 registered users and 1,000,000 waste scan records without structural redesign.
- NFR-SCA-05: Server-Side Pagination All high-volume data listings (scan history, accounts, transaction logs, reviews, audit logs) enforce server-side pagination with a default page size of 20 items.

2.6 Maintainability

- NFR-MNT-01: Layered Architectural Pattern Backend application is structured following clean layered architecture (Controller â€
- NFR-MNT-02: Automated Test Coverage Core business rules (waste classification mapping, spatial filtering, handover status machine, gamification point calculation) maintain automated unit test coverage of at least 70%.
- NFR-MNT-03: Centralized Dynamic Configuration Waste categories, sorting policies, gamification parameters (daily point cap, streak bonuses, tree upgrade thresholds), AI confidence threshold, lockout rules, and deviation limits are configurable via Admin UI without code redeployment.
- NFR-MNT-04: Parameterization Governance All configurable threshold placeholders {N}, {M}, {X} defined across business rules are maintained in a centralized configuration store.
- NFR-MNT-05: Structured Diagnostic Logging Implements structured JSON logging with unique correlation IDs across all requests, background jobs, and error events to accelerate troubleshooting.
- NFR-MNT-06: Continuous Integration & Version Control Codebase is maintained in Git repositories with mandatory peer code reviews and an automated CI pipeline executing build verification and unit test suites.
- NFR-MNT-07: API Documentation & Versioning All REST APIs are documented according to OpenAPI/Swagger standards and versioned explicitly (e.g., /api/v1) to ensure backward compatibility; database schema changes are managed via versioned migration scripts.

2.7 Compatibility & Portability

- NFR-CMP-01: Cross-Browser Compatibility Web application renders consistently and functions fully on the latest two major releases of Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.
- NFR-CMP-02: Mobile Browser & Device Capabilities Device camera and GPS features operate reliably on iOS Safari and Android Chrome; manual photo selection and manual address input provided as seamless fallbacks.
- NFR-CMP-03: Infrastructure Agnostic Portability Application backend is packaged as Docker containers, allowing deployment to any standard Linux cloud environment supporting .NET and PostgreSQL without vendor lock-in.
- NFR-CMP-04: Object Storage Interoperability Cloud storage interactions utilize standard S3-compatible APIs, allowing cloud object storage providers to be switched without application code modifications.

- NFR-CMP-05: Localization & International Standards System enforces UTF-8 character encoding supporting full Vietnamese diacritics. Timestamps are stored in UTC and presented in UTC+7 (Indochina Time). Currency values are displayed in VND for reference only.

2.8 Data Backup & Recovery

ID	Requirement	Target
NFR-BCK-01	Automated database backup	Full automated database backup executed every 24 hours; backup snapshots retained for >= 7 days in secure isolated storage.
NFR-BCK-02	Object storage redundancy	Cross-region replication or object versioning enabled on Cloudflare R2 for waste photos, receipts, licenses, and blog media.
NFR-BCK-03	Recovery Point Objective (RPO)	RPO <= 24 hours of operational transactional data loss in the event of catastrophic primary datacenter failure.
NFR-BCK-04	Recovery Time Objective (RTO)	Core platform restorable and fully operational within RTO <= 4 hours following critical hardware or cloud disruption.
NFR-BCK-05	Disaster recovery drill validation	Automated database backup restoration test executed at least once per release cycle to verify snapshot viability.
NFR-BCK-06	Permanent business record retention	User accounts, completed handover transactions, receipts, negotiation chats, and disputes are retained permanently; audit logs retained for 90 days.

2.9 AI Model Quality

ID	Requirement	Target	Condition
NFR-AI-01	Classification accuracy benchmark	Top-1 accuracy >= 85%; Top-3 accuracy >= 95%	Evaluated on held-out test dataset covering all supported recyclable waste classes
NFR-AI-02	Low-confidence uncertainty handling	Predictions with confidence below Admin-configured threshold (default 60%) flagged as 'uncertain' (MSG-CLS-04); user must confirm or select manually	Enforced on every AI scan request (BR-CLS-03)

NFR-AI-03	Zero-downtime model update cycle	Model retraining and container redeployment execute without downtime to main web application; instant rollback to previous version supported	Retrained using Staff-verified feedback data queue
NFR-AI-04	Inference traceability & versioning	Each classification record persists model version and confidence score for auditing and longitudinal performance tracking	Stored permanently with user scan history
NFR-AI-05	Dataset partition integrity	Only Staff-reviewed and labeled images enter datasets; strict automated check ensures zero sample overlap between training and validation sets	Enforced by system on dataset update (BR-CLS-06, BR-CLS-07)
NFR-AI-06	Out-of-Distribution (OOD) rejection	Non-waste or unidentifiable items return informative failure (MSG-CLS-05) instead of false confident prediction; unmapped predictions are suppressed	OOD, blurry, or non-waste image inputs (BR-CLS-02)
NFR-AI-07	Operational accuracy monitoring	Real-time classification accuracy, misclassification report rate, and category distribution displayed on Admin dashboard (A-01)	Continuous production monitoring reviewed monthly
NFR-AI-08	Feedback traceability loop	Every misclassification report links source image, AI-predicted label, and user-suggested label into the Staff labeling queue	Enforced on every report submission (BR-CLS-05)

2.10 Auditability & Data Integrity

- NFR-AUD-01: End-to-End Handover Traceability Every recycling trade is auditable from initial request creation, counter-offers, confirmed appointment time, on-site weighing photos, to final receipt confirmation.
- NFR-AUD-02: Non-Custodial Financial Clarification EcoLink strictly records transaction quantities and reference values; the system does not handle, transfer, or escrow funds. Actual settlements occur directly between User and Recycler.
- NFR-AUD-03: Referential Data Protection Categories, accounts, and historical records with transaction ties cannot be deleted from the database; only deactivation is permitted (BR-SRT-04, BR-ADM-04).
- NFR-AUD-04: Dispute Accountability Every dispute outcome issued by Operations Staff must record a formal ruling recommendation and rationale, ensuring explainable moderation (BR-DSP-04).

- NFR-AUD-05: Read-Only Dashboard Metrics High-level analytics cards and environmental tonnage summaries are computed strictly from completed and verified transactions (BR-ADM-07).

2.11 Legal & Compliance

- NFR-LEG-01: Terms of Service & Privacy Policy Users and Recyclers must explicitly review and accept terms of service during registration; explicit runtime consent is requested for camera and GPS access.
- NFR-LEG-02: Personal Data Protection Compliance Adheres to Vietnam Decree 13/2023/ND-CP on Personal Data Protection and the Law on Cybersecurity; all exported data files anonymize personally identifiable information.
- NFR-LEG-03: Community Standards & Content Moderation Public blog posts and community reviews are subject to platform moderation; Staff and Admin possess authorities to remove abusive or fraudulent entries (BR-RVW-03, BR-ADM-08).

V. Requirement Appendix

Code	Module	Main Actors / External Systems	Related Flows
AUTH	Account & Authentication	Guest, User, Recycler, Google	Registration, login, password reset, OAuth, profile, logout
CLS	AI Classification & Model Data	User, Staff	Waste image upload, AI result, misclassification feedback, error-image labeling, validation dataset
SRT	Sorting Guidance & Waste Catalog	User, Admin	Rule Engine lookup, waste search, QR scan, catalog and sorting guide updates
MTH	Recycler Profile & Matching	User, Recycler, Staff	GPS search, accepted categories, Recycler verification
NEG	Negotiation & Collection Time Slot	User, Recycler	Negotiation messages; collection time slot proposed inside the Handover Request
HND	Handover & Transaction	User, Recycler	Handover request (incl. proposed slot), accept/reject, receipt confirmation with photo, settlement (recorded only, no payment processing)
PAY	In-System Service Payment	User, Recycler, Admin, VietQR Service	Purchase of in-system service packages (ad-free, Green Tick badge, map priority) via dynamic VietQR, pricing/fee configuration, revenue reconciliation
DSP	Dispute Resolution	User, Recycler, Staff	Dispute report, Staff recommendation (no automatic refund mechanism)
GAM	Gamification & Rewards	User, Recycler, Staff, Admin	Points (User & Recycler), daily check-in (User/Staff), voucher redemption, Virtual Tree, Eco Facts, leaderboard
RVW	Rating, Review & Survey	User, Recycler, Staff	Partner rating (self-delete or Staff/Admin moderation), periodic service survey

NTF	Notification	All actors, Gmail	Email and in-app notifications
ADM	Administration, Reporting & Blog Approval	Admin, Staff	Staff accounts, account status, dashboard analytics, report export, blog post approval

1. Business Rules

BR-ID	Rule Statement	Applies To	Rationale	Enforced By
BR-AUTH-01	One email address can be linked to only one account, across both roles (User/Recycler) and both sign-up methods (local/Google).	Registration	Prevents duplicate identities and ambiguous ownership of history.	The system automatically on every registration
BR-AUTH-02	During Google sign-in/sign-up, the system reads the `email_verified` claim from Google's ID token/UserInfo response. If `email_verified = true`, the email is marked verified immediately with no separate verification step. If `email_verified = false`, the standard email-verification flow is required before the account is activated.	Registration, activation	Google reliably reports whether it has verified the user's email, so the system can trust this signal.	The system automatically on every Google sign-in/sign-up
BR-AUTH-03	When a sign-in fails, the message must not reveal whether the username exists or the password was wrong.	Sign-in	Prevents account enumeration.	The system automatically on every failed sign-in
BR-AUTH-04	After {N} consecutive failed sign-ins, the account is temporarily locked for {M} minutes.	Sign-in	Mitigates brute-force attacks.	The system automatically on every failed sign-in
BR-AUTH-05	A deactivated or suspended account cannot sign in, regardless of password correctness, and all active sessions are invalidated immediately.	Sign-in, all requests	Ensures banned/departed accounts lose access at once.	The system automatically on every sign-in and request
BR-AUTH-06	A password-reset link is single-use and expires after {X} minutes. The response is identical whether or not the email exists. Resetting	Password reset	Prevents enumeration and token reuse/theft.	The system automatically on every reset request and use

	invalidates all existing sessions.			
BR-AUTH-07	Google sign-in requires the OAuth token to be verified with Google on every login. Accounts created via Google are bound to the verified Google email and default to the User role.	Google OAuth	Prevents forged tokens and role escalation at sign-up.	The system automatically on every OAuth login
BR-AUTH-08	Access is role-based. Guests access only public information; each role accesses only its own functions and data.	All views/requests	Enforces least privilege.	The system automatically on every request
BR-AUTH-09	A password-reset request for the same email is limited to {N} requests per {X} minutes, and requests from the same IP are limited to {M} per hour. Requests beyond the limit are silently ignored (the generic response is still shown).	Password reset	Prevents spam/abuse of the reset-email channel and protects the mail-sending quota.	The system automatically on every reset request
BR-CLS-01	Only images in supported formats (JPG/PNG) and within {X} MB are accepted. Invalid files are rejected before AI processing.	Waste image upload	Protects system resources and model input quality.	The system automatically on every upload
BR-CLS-02	An AI prediction is displayed only if it maps to an active category in the waste catalog.	AI classification result	Prevents wrong or unknown categories from producing incorrect guidance.	The system automatically on every classification
BR-CLS-03	If model confidence is below the threshold configured by Admin, the result is flagged "uncertain" and the user must confirm or choose the category manually.	AI classification result	Avoids presenting low-confidence predictions as fact.	The system automatically on every classification
BR-CLS-04	A user must confirm or correct the suggested category before the classification is recorded as final.	Classification confirmation	Ensures data quality and that the user acts on a verified category.	The system automatically on every scan

BR-CLS-05	Every misclassification report is linked to the original image, the predicted category, and the user-suggested category, and enters the Staff review queue.	Misclassification feedback	Creates a traceable feedback loop for model improvement.	The system automatically on every report
BR-CLS-06	Only Staff-reviewed and labeled images may be added to training/validation datasets.	Dataset management	Prevents noisy or malicious labels from degrading the model.	Staff review; the system blocks unreviewed data
BR-CLS-07	An image must not appear in both the training and validation datasets. Validation dataset changes are made by Staff only.	Validation dataset	Avoids data leakage and unreliable accuracy metrics.	The system automatically on every dataset update
BR-CLS-08	Waste images are accessible only to the owner, the Recycler in an active related handover, Staff reviewing feedback, and Admin.	Waste image access	Protects user privacy.	The system automatically on every image request
BR-SRT-01	Sorting guidance is served only from active rules in the Rule Engine, and each waste category has exactly one active guide version at a time.	Sorting guidance lookup	Guarantees consistent, unambiguous instructions.	The system automatically on every lookup
BR-SRT-02	Only Admin can create, edit, or retire catalog entries and sorting policies. Changes apply to new lookups only.	Waste catalog, sorting policy	Preserves history while keeping guidance current.	The system automatically on every catalog/policy change
BR-SRT-03	A sorting guide must contain at least one ordered handling step and a destination before it can be published.	Sorting guide publishing	Prevents incomplete instructions from reaching users.	The system automatically on every publish
BR-SRT-04	A category referenced by classification results, Recycler profiles, or transactions can only be deactivated, never deleted.	Waste category	Preserves referential integrity and historical records.	The system automatically on every delete attempt
BR-SRT-05	A scanned QR code must resolve to a valid EcoLink-issued target, and redirection is limited to internal EcoLink pages.	QR scan	Prevents malicious or external redirects.	The system automatically on every scan

BR-MTH-01	A Recycler must be approved by Staff before appearing in search results or receiving handover requests.	Recycler registration	Ensures partner legitimacy and safety.	The system automatically on every search
BR-MTH-02	A rejected verification must include a reason, and the Recycler may resubmit a corrected profile.	Recycler verification	Gives transparency and a path to correction.	The system automatically on every Staff decision
BR-MTH-03	A Recycler profile can be published only with at least one accepted waste category, collection conditions, and a valid location.	Recycler profile publishing	Prevents unusable listings.	The system automatically on every publish
BR-MTH-04	Search returns only approved, active Recyclers within the user's search radius that accept the identified waste category, sorted by distance by default.	Recycler search (PostGIS)	Keeps results relevant and actionable.	The system automatically on every search
BR-MTH-05	A user's GPS coordinates are used only to compute distance and never shown to Recyclers. The exact address is shared only after a request is accepted.	Location data	Protects user location privacy.	The system automatically on every search/profile view
BR-MTH-06	A suspended Recycler is hidden from search immediately, but existing active transactions remain visible to both parties.	Recycler visibility	Removes risky partners quickly without breaking in-flight deals.	The system automatically on every status change
BR-MTH-07	Removing an accepted category from a Recycler profile does not affect handover requests already accepted.	Recycler profile update	Protects users' agreed deals.	The system automatically on every profile update
BR-NEG-01	Negotiation messages exist only within a specific handover request and are visible only to its User and Recycler (and assigned Staff during a dispute). The full message history is retained and never deleted, so it can serve as evidence if a dispute arises.	Negotiation messages	Keeps deal discussions confidential while preserving a verifiable record for dispute resolution.	The system automatically on every message and retrieval

BR-NEG-02	Making or changing a price/quantity offer is allowed only while the request is Pending or Negotiating, and is locked once it is Accepted, Rejected, Cancelled, or Expired. The chat channel itself stays open for both parties even after the handover is Completed.	Negotiation offers and chat	Prevents changes to agreed terms while still letting users stay in touch after a deal.	The system automatically on every offer/message
BR-NEG-03	Only the latest offer is valid; each counter-offer supersedes the previous one. Upon acceptance, the agreed terms (price, quantity, time) are stored as an immutable snapshot.	Offers/counter-offers	Removes ambiguity about what was agreed.	The system automatically on every offer/acceptance
BR-NEG-04	The User proposes their desired collection time slot directly inside the Handover Request when creating it. If the slot doesn't work for the Recycler, the two parties negotiate a new time through the chat.	Time slot proposal	Aligns with the one-way Handover flow; removes the need for a separate appointment module.	The system automatically when the request is created
BR-NEG-05	The collection time slot becomes officially confirmed the moment the Recycler clicks "Accept" on the User's Handover Request — no separate confirmation step is needed.	Time slot confirmation	Aligns schedule confirmation with the Handover Request's own Accepted status.	The system automatically when the Recycler accepts the request
BR-HND-01	A handover request can be sent only to an approved, active Recycler that accepts the waste category in the request.	Handover request creation	Prevents requests that cannot be fulfilled.	The system automatically on every request
BR-HND-02	Handover status changes are one-way and permanent: Pending → (Negotiating) → Accepted → Scheduled → Completed, or terminal Rejected/Cancelled/Expired/Disputed. No status may revert.	Handover status	Maintains data integrity and audit trail.	The system automatically on every status change

BR-HND-03	A Recycler can view and respond only to requests addressed to them.	Handover request access	Maintains fairness and privacy between Recyclers.	The system automatically on every query
BR-HND-04	A handover is Completed only when the Recycler confirms with a handover photo and the User confirms the receipt.	Handover completion	Provides evidence and mutual agreement.	The system automatically on every completion attempt
BR-HND-05	Final settlement uses the Recycler-confirmed actual quantity and the agreed unit price. If the actual quantity deviates from the agreed one by more than {X}%, the User may accept or raise a dispute before settlement. Payment of the settlement amount happens directly between the Recycler and the User (e.g., cash or bank transfer at the time of handover) — the system only records the figures, it does not process or hold the money.	Settlement	Protects the User from unfair adjustments.	The system automatically at confirmation
BR-HND-06	Either party may cancel until the handover is Completed. Cancellations within {X} hours of a confirmed appointment are recorded as late cancellations.	Cancellation	Discourages no-shows while keeping cancellation possible.	The system automatically on every cancellation
BR-HND-07	Once Completed, a transaction cannot be edited or deleted, only viewed.	Transaction records	Preserves reliable history for reporting and disputes.	The system automatically after completion
BR-HND-08	Full contact details are revealed only after a request is accepted.	Contact visibility	Protects privacy until both parties commit.	The system automatically on every profile/request view

BR-PAY-01	Payments for in-system services (e.g., ad removal, verified badges, feature upgrades) by a User/Recycler are processed through dynamic VietQR codes linked to a unique order reference and are marked successful only after a verified payment confirmation (valid signature and matching amount).	EcoLink service payment	Ensures accurate, automatic and transparent upgrades of in-system packages/features.	The system automatically on every webhook received
BR-PAY-02	In-system payment events are handled idempotently, so that each VietQR payment is credited and activated exactly once.	In-system VietQR payment	Prevents duplicate processing.	The system automatically on every webhook received
BR-DSP-01	Only a User or Recycler who is a party to the transaction can raise a dispute, within {N} days after completion, with at most one open dispute per transaction.	Dispute creation	Prevents abuse and duplicate disputes.	The system automatically on every submission
BR-DSP-02	A dispute must include a reason category and a description; supporting photos are optional.	Dispute creation	Gives Staff enough information to decide.	The system automatically on every submission
BR-DSP-04	Only Staff can resolve a dispute, and every decision must record an outcome (e.g., the User should receive a full/partial refund from the Recycler, or the dispute is dismissed) and a rationale. Since payment happens directly between the two parties outside the system, the outcome is a binding recommendation communicated to both for them to carry out themselves — the system has no mechanism to automatically transfer or refund money.	Dispute resolution	Ensures accountable, explainable decisions even though the system doesn't process payments.	The system automatically records the decision Staff makes
BR-DSP-06	Dispute data (messages, photos, transaction details) is accessible only to the two parties and the assigned Staff.	Dispute data access	Protects confidentiality.	The system automatically on every request

BR-GAM-01	Points are awarded only for verified events, such as a Completed handover (based on confirmed category and quantity) — credited to both the completing User and the completing Recycler — and only once per event per role. No points for cancelled/disputed transactions; points are reversed if a dispute outcome reverses the transaction.	Point awarding	Prevents unearned or duplicated points.	The system automatically on every qualifying event
BR-GAM-02	Daily point-earning from non-transactional actions is capped at {N} per user/recycler.	Point awarding	Prevents point farming.	The system automatically on every award
BR-GAM-03	Point balance cannot be negative. A redemption deducts points and issues the voucher atomically.	Point redemption	Ensures consistency between points and vouchers.	The system automatically on every redemption
BR-GAM-04	A reward can be redeemed only while active and in stock; each redemption reduces stock atomically.	Reward inventory	Prevents over-redemption.	The system automatically on every redemption
BR-GAM-05	A voucher is single-use, bound to the redeeming account, and expires on its expiry date.	Voucher	Prevents resale, reuse, and stale vouchers.	The system automatically on every voucher use
BR-GAM-06	Virtual Tree stage progression is one-way. Care actions are limited to {N} per day; an unlocked Eco Fact stays unlocked.	Virtual Tree, Eco Facts	Keeps progression fair and consistent.	The system automatically on every care/unlock
BR-GAM-07	The leaderboard counts only Completed, non-disputed transactions per the metric Admin configures, and shows only display names.	Leaderboard	Ensures fair ranking and privacy.	The system automatically on every calculation
BR-GAM-08	Earning points and redeeming them for vouchers is available to both the User and Recycler roles. Virtual Tree care, Eco Facts, and the contribution leaderboard remain scoped to the User role only.	Gamification access	Rewards Recyclers who help complete trades, while keeping consumer-facing features for individual Users.	The system automatically runs whenever there's a request

BR-GAM-09	Each User/Staff account may check in once per day. The reward points increase with the length of the consecutive-day streak, and the streak automatically resets to Day 1 if a day is missed.	Daily check-in	Encourages consistent daily engagement (per use cases U-15, S-11). For Staff, the check-in includes check-in/check-out to log working hours and an attendance streak.	The system automatically runs every time attendance is taken
BR-RVW-01	A rating/review can be submitted only by a party of a Completed transaction, once per party per transaction, within {N} days after completion.	Rating submission	Prevents fake, duplicate, or retaliatory reviews.	The system automatically on every submission
BR-RVW-02	A rating is 1–5 stars with an optional comment. Once submitted, it cannot be edited, but the author may delete it entirely (see BR-RVW-03).	Rating content	Keeps posted content trustworthy while giving the author control to retract it.	The system automatically on every edit/delete attempt
BR-RVW-03	The review's author (User or Recycler, whichever submitted it) may delete their own review at any time; this is permanent and cannot be restored. Exception: Staff/Admin may hide or delete a review if it has been reported and found to violate community standards (e.g., offensive language, fabricated information). Aside from the author and Staff/Admin under this exception, no other party — including the one being rated — can delete or hide a review. The displayed average rating recalculates immediately after any deletion/hiding.	Rating deletion/hiding & display	Gives the author control over their own content and lets Staff/Admin moderate harmful content, without letting the rated party suppress unfavorable-but-legitimate feedback.	The system automatically on every deletion/recalculation; Staff/Admin act on a reported review

BR-RVW-04	Surveys are sent on Staff's schedule, only to accounts with at least one Completed transaction in the period, once per recipient per survey.	Periodic survey	Targets relevant respondents, avoids duplicates.	The system automatically each survey cycle
BR-RVW-05	Survey results shown to Staff/Admin are aggregated and never identify individual respondents.	Survey analytics	Encourages honest feedback.	The system automatically on every report
BR-NTF-01	Security and transactional emails (verification, password reset, dispute) cannot be disabled by the user.	Email notification	Ensures critical information always reaches the user.	The system automatically on every send
BR-NTF-02	Every status change of a handover or dispute notifies all affected parties.	Status notifications	Keeps all parties informed.	The system automatically on every status change
BR-NTF-03	Notifications are sent only to the account's verified email; changing the email requires re-verification.	Email delivery	Prevents notifications going to the wrong person.	The system automatically on every send
BR-NTF-04	Appointment reminders are sent {X} hours before the scheduled time.	Appointment reminder	Reduces no-shows.	The system automatically (scheduled job)
BR-NTF-05	Notifications must not contain sensitive data (full contact details, tokens), except one-time verification/reset links.	Notification content	Reduces data exposure through email.	The system automatically on every notification
BR-NTF-06	A failed email delivery is logged and retried up to {N} times, and never blocks or rolls back the business operation.	Email delivery	Keeps core flows reliable.	The system automatically on every failed send
BR-ADM-01	Staff and Admin accounts are created only by Admin, with no self-registration; Staff cannot create or modify Admin accounts.	Staff/Admin accounts	Prevents unauthorized privileged access.	The system automatically on every account creation
BR-ADM-02	Roles are separated: Admin handles policy, catalog, leaderboard configuration, Staff accounts, and reports. Staff handles disputes, labeling, verification, reward inventory, survey	Role permissions	Enforces separation of duties.	The system automatically on every action

	configuration, and account status.			
BR-ADM-03	Staff may change User/Recycler account status only with a recorded reason, and cannot change Admin or other Staff accounts.	Account status	Ensures accountable moderation.	The system automatically on every status change
BR-ADM-04	Accounts with transaction history can only be deactivated, never deleted.	Account deletion	Preserves financial and audit history.	The system automatically on every delete attempt
BR-ADM-05	Every Admin/Staff action on configuration, account status, verification, and dispute decisions is recorded in an immutable audit log.	Audit trail	Provides accountability and traceability.	The system automatically on every action
BR-ADM-06	Exported reports contain aggregated or anonymized data by default, exclude personal contact details, and every export is logged.	Data report export	Protects personal data.	The system automatically on every export
BR-ADM-07	Dashboard analytics are calculated only from completed and confirmed data, and are read-only.	Dashboard Analytics	Ensures reliable metrics.	The system automatically on every analytics query
BR-ADM-08	A blog post authored by Staff must be in Draft or Pending_Approval status. It is published publicly in the app only after Admin approves it (Approved status). If the Admin rejects it, the post returns to Draft with feedback notes for the Staff to revise; both decisions are audit-logged and the Staff is notified.	Blog content publishing	Ensures content is reviewed before reaching users (per use cases S-08, A-09).	The system automatically enforces the status gate; Admin makes the approval decision

2. System Messages

#	Message code	Message Type	Context	Content
1	**MSG-AUTH-01**	In line	Wrong username/password	Incorrect username or password. Please check again.
2	**MSG-AUTH-02**	Red, under field	Required field empty	The * field is required.
3	**MSG-AUTH-03**	In line	Account temporarily locked	Too many failed attempts. Please try again in {M} minutes.

4	**MSG-AUTH-04**	In line	Account deactivated/suspended	This account is no longer active. Please contact support.
5	**MSG-AUTH-05**	Toast	Reset link sent	If this email exists, a password reset link has been sent.
6	**MSG-AUTH-06**	In line	Reset link invalid/expired	This reset link is invalid or has expired. Please request a new one.
7	**MSG-AUTH-07**	In line	Too many reset requests	You've requested a reset too many times. Please try again later.
8	**MSG-AUTH-08**	Toast	Registration successful	Account created! Please check your email to verify your account.
9	**MSG-AUTH-09**	Toast	Email verified	Your email has been verified.
10	**MSG-CLS-01**	Red, under upload box	Unsupported file format	Unsupported file format. Please upload a JPG or PNG image.
11	**MSG-CLS-02**	Red, under upload box	File exceeds max size	File exceeds the maximum size of {max_size_mb} MB.
12	**MSG-CLS-03**	Toast	Classification successful	Waste identified successfully.
13	**MSG-CLS-04**	Toast (warning)	Low-confidence result	We're not fully sure about this item — please confirm or select the category manually.
14	**MSG-CLS-05**	In line	AI processing failed	We couldn't process this image. Please try again with a clearer photo.
15	**MSG-CLS-06**	Toast	Misclassification report submitted	Thanks! Your feedback has been sent for review.
16	**MSG-SRT-01**	In line	No search result	No matching waste category found.
17	**MSG-SRT-02**	In line	Invalid QR code	This QR code is invalid or has expired.
18	**MSG-SRT-03**	Toast (Admin)	Catalog entry saved	Waste category saved successfully.
19	**MSG-SRT-04**	Toast (Admin)	Sorting guide published	Sorting guide published successfully.
20	**MSG-MTH-01**	In line	No recyclers found	No recycling facilities found near you.
21	**MSG-MTH-02**	In line	Location permission denied	Location access is required to find nearby facilities. Please enable it in your settings.
22	**MSG-MTH-03**	Toast	Recycler pending verification	Your facility profile is under review and will appear in search once approved.
23	**MSG-MTH-04**	Toast/Email	Recycler approved	Your facility profile has been approved and is now visible to users.
24	**MSG-MTH-05**	In line	Recycler rejected	Your facility profile was not approved: {reason}. Please update and resubmit.

25	**MSG-MTH-06**	Red	Profile incomplete for publishing	Please add at least one accepted waste category and a valid location before publishing.
26	**MSG-NEG-01**	Toast	Handover/negotiation request sent	Your trade request has been sent to {facility_name}.
27	**MSG-NEG-02**	In-app	New message received	New message from {sender_name}.
28	**MSG-NEG-03**	Toast	Offer accepted	{party_name} has accepted your offer.
29	**MSG-NEG-04**	Toast	Recycler receives new handover request with proposed slot	You have a new handover request with a proposed time slot from {user_name}.
30	**MSG-HND-01**	Toast	Handover request sent	Your handover request has been sent.
31	**MSG-HND-02**	Toast	Handover accepted	Your handover request has been accepted.
32	**MSG-HND-03**	Toast	Handover completed	Transaction #{transaction_id} marked as completed.
33	**MSG-HND-04**	In line	Quantity deviates from agreed amount	The confirmed quantity differs from the agreed amount by more than {X}%. Please review before confirming.
34	**MSG-HND-05**	Toast	Handover cancelled	This handover has been cancelled.
35	**MSG-HND-06**	In line	Late cancellation	This cancellation is within {X} hours of the scheduled appointment and will be recorded as a late cancellation.
36	**MSG-PAY-01**	In line	Payment QR generated	Scan this VietQR code to complete your EcoLink service payment.
37	**MSG-PAY-02**	Toast	Payment received successfully	Your service payment has been confirmed successfully.
38	**MSG-PAY-03**	In line	Payment code expired	This payment code has expired. Please generate a new one.
39	**MSG-DSP-01**	Toast	Dispute submitted	Your dispute has been submitted and is under review.
40	**MSG-DSP-02**	Toast	Dispute resolved	Your dispute has been resolved. {outcome_summary}
41	**MSG-DSP-03**	In line	Dispute window expired	You can no longer raise a dispute for this transaction.
42	**MSG-DSP-04**	In line	Duplicate open dispute	There is already an open dispute for this transaction.
43	**MSG-GAM-01**	Toast	Points earned	You earned {points} points!
44	**MSG-GAM-02**	Toast	Voucher redeemed	Voucher redeemed successfully.

45	**MSG-GAM-03**	In line	Insufficient points	You don't have enough points to redeem this reward.
46	**MSG-GAM-04**	In line	Reward out of stock	This reward is no longer available.
47	**MSG-GAM-05**	In line	Daily point cap reached	You've reached your daily point limit for this action.
48	**MSG-GAM-06**	Toast	Virtual Tree stage up	Your Virtual Tree has grown to a new stage!
49	**MSG-GAM-07**	Toast	Eco Fact unlocked	New Eco Fact unlocked!
50	**MSG-GAM-08**	Toast	Daily check-in successful	Check-in successful! You earned {points} points (Streak: {streak_days} days).
51	**MSG-RVW-01**	Toast	Review submitted	Thanks for your review!
52	**MSG-RVW-02**	In line	No completed transaction	You can only review a facility after completing a transaction with them.
53	**MSG-RVW-03**	Toast	Review deleted	Your review has been deleted.
54	**MSG-RVW-04**	Email/notification	Survey invitation	We'd love your feedback — take a quick survey about your recent experience.
55	**MSG-RVW-05**	In line	Survey already completed	You've already completed this survey.
56	**MSG-NTF-01**	Toast	Session expired	Your session has expired. Please sign in again.
57	**MSG-NTF-02**	Internal (Staff/Admin)	Email delivery failed	Email to {email_address} could not be delivered.
58	**MSG-NTF-03**	Red, under field	Input exceeds max length	Exceed max length of {max_length}.
59	**MSG-ADM-01**	Toast (Staff)	Account status updated	Account status updated successfully.
60	**MSG-ADM-02**	Toast (Admin)	Report exported	Report exported successfully.
61	**MSG-ADM-03**	Toast (Admin)	Staff account created	Staff account created successfully.
62	**MSG-ADM-04**	In line	Unauthorized action	You don't have permission to perform this action.
63	**MSG-ADM-05**	Toast (Admin)	Blog post approved	Post '{blog_title}' has been approved and published.
64	**MSG-ADM-06**	Toast (Staff)	Blog post submitted for review	Your post has been submitted to Admin for review.